

Not Learning from Others*

John J. Conlon[†] Malavika Mani Gautam Rao
Matthew Ridley Frank Schilbach

January 1, 2026

Abstract

We study social learning using experiments where two people independently learn relevant information and can share it to make accurate private decisions. Across three experiments, people are substantially less sensitive to information others discover than to equally-relevant information they discovered themselves. This holds when they must learn information from others through discussion; when the experimenter perfectly communicates the information; and even when participants observe others' information with their own eyes. Our results therefore stem not from a failure to elicit information from others but a systematic tendency to underweight it relative to one's own information. Our findings illustrate a powerful barrier to social learning that might underlie many documented cases of failure to learn from others.

Keywords: information aggregation, learning, social learning, experience effects

*We thank the editor and four anonymous referees for their helpful comments and suggestions. We also thank Sandro Ambuehl, Nava Ashraf, Ned Augenblick, Oriana Bandiera, Abhijit Banerjee, Rafael Batista, Leonardo Bursztyn, Arun Chandrasekhar, Katie Coffman, Stefano DellaVigna, Esther Duflo, Ben Enke, Christine Exley, Ben Golub, Emir Kamenica, David Laibson, Shengwu Li, Ulrike Malmendier, Madeline McKelway, Michel Maréchal, Sendhil Mullainathan, Tommy O'Donnell, Ricardo Perez-Truglia, Matthew Rabin, Chris Roth, Dmitry Taubinsky, David Yanagizawa-Drott, and numerous seminar and conference participants for helpful suggestions and comments. We thank Sangeetha Ramanathan and the entire team at the Behavioral Development Lab in Chennai for excellent research assistance. We thank all our study participants for their time and patience. This project was funded by the Wellspring Philanthropic Fund. We received IRB approval from MIT, protocol #1810538700. Each experiment was pre-registered on the AEA registry, number AEARCTR-0004253.

[†]Conlon: Carnegie Mellon University (jconlon@andrew.cmu.edu); Mani: University of Warwick (Malavika.Mani@warwick.ac.uk); Rao: UC Berkeley and NBER (grao@berkeley.edu); Ridley: University of Warwick (Matthew.Ridley@warwick.ac.uk); Schilbach: MIT and NBER (fschilb@mit.edu)

1 Introduction

We often learn new information through our own actions or experiences: experimenting with a new technology, trying out a new restaurant, or personally visiting several schools before choosing one. But the actions and experiences of *others* also present us with a vast trove of useful data. We can talk to others who have used the technology, read restaurant reviews, or ask other students or parents for their opinions. Efficient social learning requires us to elicit information from others and correctly aggregate it with our own private information. How well do people do this, and what frictions might prevent social learning?

We study these questions experimentally. In our first experiment in Chennai, India, 500 participants face a simple learning task: making incentivized guesses of the fraction of red balls in an urn. To inform their guesses, participants have access to two sets of draws with replacement from the urn. In a control condition, participants make both sets of draws themselves and then make a guess. In the main treatment condition, the information is instead split between two participants. Each gets one set of private draws and the chance to learn each others' draws in an unstructured face-to-face discussion. After the discussion, each participant makes a private guess about the contents of the urn. Participants face aligned incentives to share information since the earnings from one partner's randomly-selected guess are equally split between the two of them.

Absent frictions to social learning, participants should be equally sensitive to information they uncover themselves and information their partner uncovered. In our reduced-form analysis, participants' guesses instead respond 54% less ($p < 0.01$) to signals their partners uncover—and which they can access through the discussion—than to signals they uncover themselves. This lower sensitivity to their partner's information is a mistake that reduces earnings: given the incentives for accurate guesses, an additional draw by one's partner improves one's earnings 82% less than receiving an additional own draw.

What drives this failure of social learning? First, it could reflect imperfect *communication*: participants might be reluctant to elicit or share information, or may mistrust others' competence or memory, preventing them from reliably learning what their partner uncovered. Second, there might be a failure of *information aggregation*: participants might not appropriately combine others' information with their own when forming guesses. To test between these possibilities, we run a treatment in which communication

frictions are shut down by design: the experimenter directly informs each participant about their partner’s draws.

Removing communication frictions does not increase sensitivity to others’ information, implying that our finding is driven by a systematic bias in information aggregation. When the experimenter perfectly communicates each partner’s draws, participants’ guesses are 87% less sensitive to others’ information than to own signals ($p < 0.01$). Discussion with one’s partner after being informed of their draws by the experimenter still leads to 46% less sensitivity to the partner’s information ($p < 0.05$). People thus appear to simply treat others’ signals as inherently less informative, even when communication is clear.

We replicate these findings and rule out several potential mechanisms in a second experiment with 292 adults in the same setting. Most strikingly, participants underweight their partner’s information by 41% relative to their own signal even when they sit beside their partner and can observe them drawing balls from the urn *with their own eyes* ($p = 0.04$). This result eliminates or diminishes any role for (i) distrust of the information communicated by the partner and/or experimenter, since the information is directly observed with one’s own eyes; and (ii) the mode of presentation of the information, including both its visual salience and whether the information is learned draw-by-draw or communicated in summary form.

Our third experiment demonstrates the external validity of our findings among a more educated population. In a simpler between-subjects experiment with 4,489 participants from the UK and US on Prolific, we randomize the order of learning one’s own signals versus a partner’s signals. Again, participants are less sensitive to others’ information than to their own, by 17% ($p < 0.01$), despite it being perfectly communicated to them. Presenting own and others’ signals using visually identical animations does not reduce underweighting, nor does doubling the stakes of the experiment or reducing any sense of competition by barring the partner from making any guess.

Participants appear largely unaware of their underweighting of others’ information, and we provide suggestive evidence that this bias is not fully explained by differential later recall. In a debriefing survey at the end of the third experiment, 77% of participants reported that they treated their own and their partner’s information the same. Yet these same participants are 15% less sensitive to their partner’s information than to their own ($p < 0.01$). In the survey, we also ask participants to recall their own and their partner’s

draws from the last round of the experiment. Among the majority of participants who perfectly recall theirs and their partner’s draws, others’ information is still underweighted by 9% ($p = 0.06$).

In our experiments and, arguably, in many natural environments, generating one’s own information requires taking action or experiencing something oneself, whereas learning from others is often more passive. Our third experiment provides evidence that this difference can at least in part explain the bias against others’ information. Specifically, in our online experiment, we vary whether participants click a button to generate their own draws or passively observe draws appearing on the screen with a label identifying each draw as ‘Your’ or ‘Partner’s’. When clicking to generate their “own” draws, they are 17 to 19% less sensitive to their partner’s information than to their own ($p < 0.01$). In contrast, when they must take no action to generate their own information, this under-sensitivity falls to only 4%, significantly smaller than with active involvement ($p < 0.05$) and not distinguishable from zero ($p = 0.27$). This finding suggests that when people take active efforts to uncover others’ information, social learning might be more effective.

We view the main contribution of this paper as being to the literature on social learning that investigates how agents learn from others (see Mobius and Rosenblat 2014 for a review).¹ We provide evidence for a novel, potentially far-reaching bias in information aggregation that may hinder social learning whenever people have to aggregate their own and others’ information. This phenomenon may underlie other documented cases of incomplete social learning. For example, lab studies of observational learning (largely following Anderson and Holt 1997) find that people put more weight on private information than on what can rationally be inferred from the actions of others (Weizsäcker 2010). Though these findings are consistent with our results, they are also consistent with other (sometimes rational) explanations, such as mistrust of others’ ability (De Filippis et al., 2017), noisy choice (Goeree et al. 2007), overconfidence (Angrisani et al., 2021), altruism (March and Ziegelmeyer, 2020), base-rate neglect (Benjamin et al., 2019), or other behavioral biases (Guarino and Jehiel, 2013). Our finding that agents down-weight others’ *signals*, not just their actions or beliefs, along with other features of our experiment, rules out these explanations.²

¹Existing research finds that people sometimes also react very differently to information depending on the identity of the sender, e.g., they may react more to information coming from celebrities (Alatas et al., 2021) or from people who are socially or economically similar to them or who are of a particular gender (BenYishay and Mobarak, 2019; BenYishay et al., 2020).

²Drehmann et al. (2005) include in their online experiment a treatment similar to Anderson and Holt

Our paper also relates to literatures on experience effects and reinforcement learning. Using observational data, previous work has shown that people’s beliefs and economic decisions are powerfully shaped by their personal experiences, even when much more complete data are readily available (Malmendier and Nagel, 2016; Malmendier et al., 2021; D’Acunto et al., 2021; Malmendier and Shen, 2024). Consistent with this, experiments on reinforcement learning show that people’s beliefs and actions react more to events that personally affect them (Merlo and Schotter 2003, Simonsohn et al. 2008, Miller and Maniadis 2012). Intuitively, burning one’s hand on a hot surface is more impactful than watching someone else do it or being told about it. Our findings echo this idea, with an important difference: even *before* (or without) any consequences or feedback, people are biased against information from others. Our findings might thus apply to decisions where any payoff realization lies in the distant future, such as education or career choices.

Our paper also builds on recent work on “ownership effects” in attention to information. In their experiments, Hartzmark et al. (2021) show that owning an asset draws attention to information about it, even when there is no instrumental reason for this differential attention.³ In our experiments, who uncovers information (oneself or others) affects how much a person reacts to it, even though it bears no relevance for the informativeness of the signals. In both cases, these instrumentally irrelevant factors may affect bottom-up attention to information. To the extent that generating information creates a sense of ownership, we extend the notion of ownership in Hartzmark et al. (2021) to information itself.

More broadly, this paper adds to a growing literature on the drivers of under- and

(1997) but where subjects can see previous decision-makers’ signals (a or b) as well as their choice of urns (A or B). Using their data, we find (analyses not shown) that, even conditional on the total number of a and b signals available to the agent (her signal and all previous players’ signals), she is much more likely to choose urn A if “her” signal was a than if “her” signal was b . These results, while suggestive of and consistent with our finding of intrinsic under-weighting of “others” compared to “own” information, cannot rule out that subjects put more weight on their own information because they receive it last, for example due to base-rate neglect. Though Drehmann et al. (2005) note that people in this treatment choose the urn associated with their own signal more than theory would predict, their paper primarily focuses on the effect (in other experimental treatments) of adding asset prices to the Anderson and Holt (1997) paradigm.

³Hartzmark et al. (2021) ask experimental participants to update their beliefs about experimental assets’ fundamentals after observing their prices change. Incentives for accurate beliefs are identical across owned and non-owned assets, and there are no buy-or-sell decisions to be made. Thus, by design, there is no instrumental reason to pay extra attention to owned assets. Yet, Hartzmark et al. (2021) find strong evidence of greater responsiveness to information about owned assets.

overreaction to information. Recent work mostly focuses on people failing to learn effectively on their own, including neglecting the strength of signals (Augenblick et al., 2024), being influenced by salient features (Bordalo et al., 2025), complexity and cognitive uncertainty (Enke and Graeber, 2023), selective attention (Ba et al., 2025; Schwartzstein, 2014), and the role of associative memory (Enke et al., 2024). We focus on inefficiencies in learning that may arise when some of the signals reach people from others.

Insufficient learning from others could arise in many real-world situations. For example, business analysts can either learn information from co-workers or do the research themselves, educators can impart facts passively or invite students to discover them by trial and error, farmers can use new agricultural technologies themselves or learn from their neighbors about them (Kondylis et al., 2024), and medical experts may supplement their own judgment with information from AI tools (Agarwal et al., 2024). Our results suggest that people will be systematically less sensitive to information originating from others, even when that information is perfectly communicated.

The remainder of this paper is organized as follows. Section 2 presents the broad aspects of the design shared by the different experiments. Section 3 presents the empirical framework. Sections 4, 5, and 6 present the detailed designs and results of the three experiments. Section 7 concludes.

2 Overview of Design

In all three experiments, participants play multiple rounds of the same basic statistical learning exercise: a balls-and-urns task based on a large literature studying individual learning (Benjamin, 2019). Here, we describe the task and features of the design common to all experiments. We defer the discussion of treatment variations and details specific to each experiment to the corresponding sections below. Full experimental scripts are shown in Online Appendix C.

The goal in the experimental task is to guess the number of red balls in an urn containing 20 balls. Participants are informed that the number of red balls is drawn uniformly from 4 to 16 in each round, as explained with the help of the illustration in Appendix Figure A.I(a) in the in-person experiments. In the online experiment, we explain that “the computer will randomly choose the exact number of red marbles [in the urn], where every number between 4 and 16 was equally likely to be chosen.”

In each round, participants receive two independent, noisy signals about the composition of the urn, by privately drawing a number of balls from the urn with replacement.⁴ The number of draws in each ‘signal’ is randomized—either 1, 5, or 9 draws—creating variation in how informed each participant is.⁵ We provided training in the task to participants in Experiments 1 and 2 before the first round. These participants individually played two unincentivized practice rounds with two guesses in each, and during these rounds received two ‘tips’ on making good guesses.⁶ The vast majority understood the tasks, as measured by excellent performance on comprehension checks (Table A.I).

Depending on the treatment condition, participants either play the game entirely on their own—the *Individual* treatment—drawing two sets of balls themselves, or else draw one set of balls themselves and have access to another set of balls that a partner (another participant) drew. The different treatments vary how the information obtained by one’s partner can be learned: via open-ended discussion, directly communicated by the experimenter, and/or observed with one’s own eyes. Guesses are made after making each set of draws (or potentially learning them via one’s partner). We test for frictions in social learning by comparing the sensitivity of guesses to draws across conditions.

Participants have incentives to make accurate guesses. The incentives provided were chosen to be easy for participants to understand: a penalty per ball away from the truth. Formally, each guess is incentivized by a piece-wise linear loss function.⁷ In Experiments 1 and 2, a perfectly accurate guess earns each member of the pair Rs. 105 and the payment decreases by Rs. 15 per ball the guess deviates from the truth. This incentive scheme was explained to participants in Experiments 1 and 2 using the illustration shown in Appendix Figure A.I(b). These incentives are sizable. Rs. 105 is about \$1.50 and Rs. 15 is about \$0.20, while average daily earnings in our Chennai sample are about Rs. 350 (\$5).

These incentives imply that participants’ guesses cannot be interpreted as the mean

⁴In Experiments 1 and 2, participants physically drew balls from an urn in our lab, while in Experiment 3 (the online experiment), the drawing was simulated using an animation of an urn. In each case, participants were informed that both partners were drawing from the same urn.

⁵For each pair, we randomly choose the number of draws in the two signals with uniform probability from $\{(1, 1), (1, 5), (5, 1), (5, 5), (1, 9), (9, 1)\}$. This excludes cases with more than 10 draws total.

⁶The first tip explains that it makes sense to guess there are more red than white balls if you draw more red than white, and vice-versa. The second tip is that “the more balls you draw, the more confident you can be in your guess”.

⁷On top of their participation fee, each person receives a payment equal to $\max\{(A - B \times |g - r|), 0\}$, where g is the guess, r the true number of red balls for the randomly-selected guess, and A and B are constants.

(or median or mode) of their beliefs about the color composition of the urn. We chose this payment rule to be transparent and easily comprehensible to participants, as none of our analyses require identification of any particular statistic of participants’ beliefs (which a more complex proper scoring rule could theoretically elicit). Guesses should therefore be thought of as actions that participants have an incentive to tailor to the signals they receive. Our non-parametric and reduced-form results simply test whether these guesses are equally sensitive to one’s own vs one’s partner’s signals. However, as a benchmark, we also compute what a risk-neutral Bayesian seeking to maximize expected payoffs would guess given the signals and our incentive structure. In addition, our structural model accounts for the incentive structure faced by participants.

Participants make multiple guesses throughout the experiment, and we randomly select one guess to score and pay participants for its accuracy. In Experiments 1 and 2, we select one guess among all the guesses that either partner made (including intermediate guesses). We then pay the two participants equally, irrespective of who made the guess, in separate envelopes at the end of the experiment. Thus, each person has an incentive to increase the accuracy of each guess from their pair. Neglecting to ask one’s partner for information, withholding information from them, or ignoring their information reduces one’s own expected payoff. In Experiment 3, the online experiment, participants never need to (and, indeed, cannot) communicate with each other, and information is shared by design. Each participant is rewarded for a randomly selected one of their own guesses: i.e., we do not split incentives between partners in Experiment 3.

3 Empirical Framework

Our goal is to test whether individuals’ guesses are equally sensitive to signals drawn by themselves versus by others. We further examine how this depends upon the precise mode of social learning, such as whether the partner’s information must be learned through a discussion, is communicated by a third party (the experimenter), and/or is directly observed. We present three types of empirical analyses—non-parametric, reduced form, and structural—to answer these questions. These three approaches impose different assumptions and have different strengths, but ultimately lead to similar conclusions.

3.1 Non-parametric Approach

In the non-parametric approach, we use minimal assumptions and simply plot average guesses in each treatment against the signals drawn. For simplicity, we summarize each signal by the net number of red draws (i.e., the number of red minus the number of white draws). That is, if a participant saw 4 red draws and 1 white draw, we would classify the signal as being 3 net red draws.⁸ To enable a transparent comparison of the sensitivity of guesses to own versus others' signals, we plot the guesses separately against the signals drawn by oneself versus those drawn by one's partner.

3.2 Reduced-form Approach

In our second empirical approach, we impose a linear relationship between signals and the resulting guesses and test for differences in this relationship across treatments. The starting point for our analysis is estimation of the following equation by OLS, separately by treatment:

$$Guess_i = \alpha + \beta^o \cdot Own\ Info_i + \beta^p \cdot Partner's\ Info_i + \epsilon_i, \quad (1)$$

where $Guess_i$ is i 's guess of the number of red balls (after having a chance to learn both signals), and $Own\ Info_i$ and $Partner's\ Info_i$ are the net number of red draws (i.e., red minus white draws) drawn oneself and by one's partner respectively. β^o and β^p capture the sensitivity of participants' guesses to signals drawn themselves and by others. If participants learn their partner's signals and treat them the same as their own signals, then we should expect that $\beta^o = \beta^p$. If instead $\beta^p < \beta^o$, participants in that treatment are less sensitive to their partner's draws than to their own.

When estimating equation (1), we add controls for the order in which participants complete the different treatments. In all three experiments, participants play multiple rounds of the game, doing the different treatments in randomized order. Although they receive no feedback after each round, and thus the scope for learning is limited, we control for treatment order effects by including dummies for round number interacted

⁸This simplification loses some information, e.g., it does not capture the total number of draws. A signal with 1 net red could come from a single draw of a red ball or from 9 draws with 5 red and 4 blue. A Bayesian should react differently to these two signals. The structural model does not share this weakness.

with *Own Info_i* and *Partner’s Info_i*.⁹

We also control for the possibility that differences between β^o and β^p may reflect ‘information order’ effects: i.e., the order in which one receives information may affect the weight placed on it. For a Bayesian, the order of receiving information should not matter. Nonetheless, ex ante it is possible that participants put more weight on signals they saw first (‘first impressions matter’) or on signals they saw last (‘recency effects’). In Experiment 3, the order of learning one’s own and one’s partner’s signals is randomized with equal probabilities, so the comparison between β^o and β^p is unbiased by information order effects. In Experiment 1, instead, participants learn their partner’s signals only after they have received their own signals. Therefore, we compare β^p in different treatments with the coefficient on one’s own *second* set of draws in the *Individual* treatment, estimated as β_2^o in the following regression:

$$Guess_i = \alpha + \beta_1^o \cdot Own\ First\ Info_i + \beta_2^o \cdot Own\ Second\ Info_i + \epsilon_i \quad (2)$$

In practice, we find that participants tend to put more weight on the signals they receive *second*, so treatments that provide partners’ information last would tend to bias us *against* finding under-sensitivity to others’ information.¹⁰

3.3 Structural Approach

In our third empirical approach, we estimate a simple model of quasi-Bayesian updating (Grether, 1980, 1992). This approach has several strengths relative to the reduced-form analysis. First, it exploits the full information content of the signals, including the number of draws, rather than the simplified ‘net red draws’ employed in the reduced form. Second, it accounts for the incentive structure faced by participants, modeling them as risk-neutral agents trying to maximize expected payoffs given their beliefs. Third, by taking the form of a standard learning model, it allows us to estimate interpretable weights placed on one’s own and others’ signals, with a clear Bayesian benchmark. Finally, it also accounts for noisy choice together with censoring in guesses at 4 and 16, which might otherwise cause guesses to appear less sensitive than those of a risk-neutral

⁹To include these controls, we stack the regressions for all treatment conditions in a given experiment and estimate them jointly in one regression, allowing the coefficients α , β^o and β^p to vary by treatment.

¹⁰Experiment 2 has aspects of the design of both Experiment 1 and Experiment 3. Some comparisons involve a randomized order of receiving information, as in Experiment 3. Others are similar to Experiment 1 in that one’s partner’s information is received after one’s own information.

Bayesian. On the other hand, the structural model makes more assumptions than the non-parametric and reduced-form analysis, including imposing risk-neutrality.

Let d_1 be the participant's own signal and let d_2 be her partner's signal, e.g., d_1 might equal {Red, Red, White, Red, White} and d_2 might equal {Red}. We then assume that the participant updates her beliefs about the state of the world s (the number of red balls in the urn) according to a modified version of Bayes' Rule:

$$Posterior(s|d_1, d_2) \propto Prior(s) * P(d_1|s)^{\omega_{1rt}} * P(d_2|s)^{\omega_{2rt}} \quad (3)$$

where $Prior(s)$ is the participant's prior about the probability of state s , and $P(d_i|s)$ is the (objective) probability of observing a set of draws d_i conditional on state s . Recall that participants are told each state is equally likely, and there are 13 possible states $s \in \{4, 5, \dots, 16\}$, so $Prior(s) = \frac{1}{13}$. Next, ω_{1rt} and ω_{2rt} are the weights that the participant puts, respectively, on her own and her partner's signals in treatment t when that round occurs in chronological order r . For $\omega_{1rt} = \omega_{2rt} = 1$, Equation (3) reduces to Bayes' Rule.

We allow ω_{1rt} and ω_{2rt} to differ from the Bayesian benchmark depending on both the treatment condition and the chronological order of the round. In particular, we assume the following functional form to mirror the reduced-form analysis described above:

$$\begin{aligned} \omega_{1rt} &= \beta_{1t} + \mu_{1r} \\ \omega_{2rt} &= \beta_{2t} + \mu_{2r} \end{aligned}$$

where β_{1t} and β_{2t} are, respectively, the weight the participant puts on her own and her partner's signal (or, in Experiment 1's *Individual* treatment, on her own second signal) in treatment t , and μ_{1r} and μ_{2r} are the additional weight she puts on each signal when that treatment occurs in chronological order r .

Just as with the reduced-form analysis, we control for the order in which information arrives in one of two ways. In Experiment 1, the partner's information is always conveyed second, so we use a control condition (the *Individual* treatment) where both signals are drawn by the participant herself. We then compare ω_{2rt} , the weight on the second set of draws across treatments to identify the effect of drawing information oneself net of any information order effects. In Experiment 3, we randomize whether participants' own information or their partner's information comes first, so ω_{1rt} and ω_{2rt} will not be biased by differential treatment of earlier or later signals.

In addition to systematically biased updating, we allow for noisy choice. Doing so allows us to account for heterogeneity in guesses conditional on signals (i.e., not everyone with the same signals makes the same guess). We assume that agents are risk-neutral but calculate the expected payoff of each possible guess with noise. In particular, let $Earnings(g, s)$ be the earnings that a participant would earn if they made guess g and the true state was s . Given the experimental incentives, $Earnings(g, s) = \max\{0, 105 - 15 * |g - s|\}$. We assume that the agent calculates the expected payoff of each guess g using the (potentially biased) updating rule given by Equation 3 plus a random additive error term. That is, we assume the perceived expected payoff from making guess g given draws d_1 and d_2 is given by

$$EP(g|d_1, d_2) = \sum_{s=4}^{16} Posterior(s|d_1, d_2) Earnings(g, s) + \alpha \epsilon_{i,g}. \quad (4)$$

The agent then chooses the guess that maximizes this perceived expected payoff. For simplicity, we assume $\epsilon_{i,g}$ is i.i.d. Type 1 extreme value. The parameter α then governs the extent of noisy choice (Goeree et al., 2007). We estimate the model by maximum likelihood.¹¹

4 Experiment 1: Establishing the Main Results

4.1 Recruitment and Sample

Experiment 1 was conducted in person at the Behavioral Development Lab in Chennai, India, between July and December 2019. Participants were recruited on a rolling basis, with about 4 to 10 individuals completing the experiment on a given day. We recruited individuals—not pairs—residing in low- to middle-income neighborhoods within a reasonable travel time of the lab. Surveyors went door-to-door to advertise an academic

¹¹In particular, given the assumptions above, the probability that an agent with signals d_1 and d_2 will choose guess g is $P(i \text{ guesses } g|d_1, d_2) \propto \exp\left(\frac{1}{\alpha} \left[\sum_{s=4}^{16} Posterior(s|d_1, d_2) Earnings(g, s)\right]\right)$. We then choose parameters that maximize the joint likelihood of observing all the choices in our data. We calculate standard errors by bootstrapping the data, drawing pairs with replacement from the data. Throughout, we report bootstrapped standard errors for legibility but denote significance using bootstrapped confidence intervals (e.g., an estimate is significant at the 5% level if the center 95% of bootstrapped estimates do not include zero).

study on ‘your choices and how you aggregate information’ which would ‘help us understand how you make decisions’. No more specific study details were provided at this stage. Potential participants were informed that they would spend 2 to 3 hours at the study office and could expect to earn Rs. 150 to 280 (\$2 to \$3.90) per person, plus a payment of Rs. 100 (\$1.40) to cover travel expenses. Recruitment stopped when we reached our pre-specified target of 500 individuals. Participants were randomly assigned to pairs within an experimental session.¹²

Column 1 of Table 1 reports demographic characteristics of our sample. 50% of the participants are male. Participants are on average 35 years old and have a bit less than 8 years of education. Participants answered about 80 percent of comprehension questions correctly on the first attempt, indicating fairly high levels of attention and comprehension for a task that was unusual and somewhat complex given the local context.

4.2 Experiment 1: Design

Participants play five rounds of the task, as illustrated in Figure 1, with no feedback between rounds. Participants first play, in randomized order, an *Individual* round and a *Discussion* round. In each round, participants have access to two sets of draws with 1, 5, or 9 draws each.

***Individual* round.** In the *Individual* round, the participant first draws a set of balls from the urn with replacement, then guesses how many red balls are in the urn. Then, they draw a second set of balls from the urn and make a second (and final) guess. All drawing and guessing is done privately, without any need to share information. This round serves as a control condition—a benchmark against which we compare the other treatments.

***Discussion* round.** The *Discussion* round models a common mode of social learning, where we learn from others’ experiences through direct communication with them. Instead of drawing two sets of draws oneself as in the *Individual* round, each participant’s *partner’s* draws—accessible through a discussion—serve as their second set of

¹²Each participant plays four of their five rounds with one randomly-assigned partner of a different gender, and one round with a randomly-assigned partner of the same gender. Participants were introduced to their partner at the start of each round. This variation was induced to study the effect of gender composition on learning and to contrast these findings with a study of learning between spouses. These results are reported in a companion paper (Conlon et al., 2024). Here, we pool results across gender and both types of pairs.

draws. Each person first makes one set of draws followed by a private guess, exactly as in the *Individual* round. Next, the pair are asked to hold a face-to-face discussion and enter a joint guess.¹³ After their discussion, the teammates are separated and each person makes one final, private guess.

Participants can take as long as they like for the unstructured, face-to-face discussion with their partner. While we do not require communication to be truthful, participants have an incentive to share information accurately since one guess per team is randomly chosen to be paid for accuracy at the end of the experiment, with the payment split between the two partners. Participants also have an incentive to help their partner deliberate and make better guesses conditional on information, as in Cooper and Kagel (2005). We record the audio of the discussion (with participants' consent) and present summary statistics of the transcripts in Table A.II. Discussions last an average of 48 seconds and the vast majority (83%) involve some information sharing, with participants about as likely to share their signals as their guesses.

Comparing each participant's final guesses in the *Individual* and *Discussion* rounds reveals whether they learn as much through a discussion with a partner as from information they uncovered themselves. By design, participants have access to the exact same number of draws to inform their final guess in these two rounds, provided they share information. If participants are instead less sensitive to information collected by their partner, this implies either a failure of communication or a failure to aggregate information provided by one's partner.

Participants next play three more rounds, in randomized order, consisting of a second *Discussion* round and two additional treatments in which the experimenter informs the participant of their partner's draws or guesses.

***Informed of Partner's Draws* round.** This round (which we abbreviate as the '*Informed*' round) is designed to shut down any communication frictions between the partners. It is identical to the *Discussion* round except that after participants receive their first set of draws and enter their first guess, they are told their partner's draws (both number and composition) directly by the experimenter, e.g., "Your partner had

¹³The joint guess was included as a comparison to joint guesses made by teams composed of married couples and is not the focus of this paper. Note that having to enter a joint guess might cause teammates to come closer to agreement about the optimal guess, which might be expected to reduce under-sensitivity to each others' information. Experiment 3 and most treatments in Experiment 2 do not include such a joint guess.

five draws, of which three were red and two were white.” Participants then make an additional private guess before moving on to the discussion and their final private guess.

Comparing the guess made after the experimenter informs the participant of their partner’s signal (but *before* discussion) with the second guess in the *Individual* round allows us to directly test whether participants use information they gathered themselves in the same way as information collected by others but perfectly shared with them by a third party. In each case, there is no possibility of joint deliberation.¹⁴ Finally, the *post*-discussion guesses in the *Informed* round reveal how participants use information shared by a third party when additionally given the chance to confirm the information with its source (their partner) and deliberate.

***Informed of Partner’s Guess* round.** This round is the same as the *Informed of Partner’s Draws* round except that the experimenter informs each person of their partner’s private guess (made based on their own draws only), rather than their partner’s draws. The experimenter also shares the number of draws this guess was based on, e.g., “Your partner had 5 draws and, after seeing these draws, they guessed that the urn contains 12 red balls.” Thus, while in the *Informed of Partner’s Draws* round we transmit the signal received by one’s partner, in the *Informed of Partner’s Guess* round we transmit the action (guess) taken based on that signal as well as a measure of the precision of the signal. This round parallels more closely the literature that investigates social learning based on observing others’ actions (Weizsäcker, 2010). In this treatment, less information is transmitted to the participant. Moreover, beliefs about others’ competence might affect how these actions are interpreted and how much is learned about the signals.

4.3 Experiment 1: Results

4.3.1 Non-parametric results

Before comparing participants’ guesses across treatment conditions, we show evidence of good comprehension of the task. Panel A of Figure 2 examines participants’ first guesses, made after drawing the first set of balls by themselves. Reassuringly, the average number of red balls guessed increases in the number of “net red” draws uncovered oneself (pooling

¹⁴Note that this comparison requires controlling for round order effects, since the *Individual* round is always in the first two rounds, while the *Informed* round falls in rounds 3-5.

across all treatments), implying that participants respond sensibly to the information they receive. We can compare this sensitivity to a normative benchmark by computing, for each guess that participants make, what a risk-neutral Bayesian seeking to maximize expected payoffs would guess given the same signals and faced with our incentive structure. On average, participants’ individual guesses (blue dots and lines) are fairly close to this benchmark (pink dashed lines), though somewhat less sensitive to signals than a risk-neutral Bayesian would be. Pooling across all data, the average participant guess increases by 0.61 percentage point for every 1 percentage point increase in what the risk-neutral Bayesian would guess.

Figure 3 contrasts the sensitivity of participants’ guesses to their second set of draws in the *Discussion* and *Informed* rounds, comparing each to the *Individual* round. The blue curve representing the *Discussion* round (Panel A) is distinctly flatter than the grey curve representing the *Individual* round, revealing that participants’ guesses are less sensitive to information gathered by their partner compared to information they collected themselves. This difference is statistically significant: we can reject (F -test, $p = 0.001$) that the differences in average guesses across treatments for each ‘net red’ value are all zero (i.e., that for each pair of dots in Figure 3 the true values lie on top of each other).

Strikingly, the curve is even flatter in the *Informed* round (Panel B), in which we plot participants’ guesses after their partners’ information is *directly* communicated to them by the experimenter (and before any joint deliberation with their partner). Despite having been given *all* decision-relevant information about their partner’s draws directly, participants react to this information much less than they do to information they collected themselves. We can again reject that average guesses conditional on each ‘net red’ value are always equal across treatments (F -test, $p < 0.001$).

Panel C of Figure 3 shows a similar, though somewhat less pronounced, under-sensitivity to partner’s information in the *Informed* round’s post-discussion guesses compared to the pre-discussion guesses shown in Panel B. Since participants can confirm the information given to them by the experimenter directly with its source (their partner), this suggests that low sensitivity to others’ information is not primarily driven by mistrust in the experimenter (though we cannot rule out some role of mistrust in the pre-discussion guesses of the *Informed* rounds).

4.3.2 Reduced-form and structural results

The reduced-form and structural models provide quantitative estimates of sensitivity to own and others' information. Figure 4 plots participants' average sensitivity to the second set of signals, separately within each treatment, using reduced-form estimates of Equation (1). In their final private guesses in the *Discussion* round, participants are less than half as sensitive to their partner's signals (second bar) compared to the corresponding signals in the *Individual* round (first bar, $p < 0.01$).¹⁵ This implies they respond less to information their partner collected than to their 'own' information. Even more starkly, participants put close to zero weight on their partner's information in the *Informed* round, right after it is *directly* shared with them (third bar, $p < 0.01$). Adding a face-to-face discussion with their partner after being informed of their draws somewhat increases participants' sensitivity to their partner's signals. Still, it remains significantly below the sensitivity to their own signals (fourth bar, $p = 0.02$).

The corresponding regression estimates are presented in Table 2 Panel A (columns 1 to 4). Comparing the coefficient β_2 on the second set of information by treatment condition shows a clear result. Participants are 54 percent (0.28/0.52) less sensitive to information collected by their partner in the *Discussion* round relative to information they collected themselves in the *Individual* round ($p < 0.01$). Even more strikingly, they are 86 percent (0.45/0.52) less sensitive to their partner's draws in the pre-discussion *Informed* guess compared to in the *Individual* round ($p < 0.01$). Put differently, participants are seven times more sensitive to their own information than to their partner's, and we cannot reject that participants put no weight on their partner's information at all. The face-to-face discussion increases sensitivity to the partner's information relative to the pre-discussion guess, perhaps through joint deliberation regarding the right answer or increased trust in the information about the partner's signal.¹⁶

The structural estimates in Panel B of Table 2 mirror the reduced-form results.

¹⁵For a risk-neutral Bayesian, this sensitivity would be around 0.72: that is, for every additional net red draw, the expected-payoff maximizing guess on average increases by 0.72. Participants' sensitivity to signals in the *Individual* round is 0.52, so discounting of partners' signals moves participants further away from the risk-neutral rational benchmark.

¹⁶The pattern of results in the *Informed of Partner's Guess* round are similar or more extreme than those that we find in the *Discussion* and *Informed of Partner's Draws* rounds. In the reduced-form estimates, participants are 85% less sensitive to their partner's information in this treatment. Because lower sensitivity to others' information in this treatment can be explained by additional factors such as guesses containing less information than draws or players' beliefs about their partners' ability to make good guesses, we defer the analysis of the *Informed of Partner's Guess* round to Appendix B.1.

Column 1 shows that participants put close to the Bayesian weight ($\beta_1 = 0.92$ vs. the Bayesian benchmark of 1) on their own first signal in the *Individual* treatment, and somewhat greater weight ($\beta_2 = 1.50$) on their second signal in that round. In contrast, participants put much less weight on their partner’s information in the *Discussion* and *Informed* rounds. Most strikingly, participants put no weight at all on their partner’s signals in the (pre-discussion) *Informed* round. The weight on others’ information is somewhat higher in the rounds involving discussion but still 69% to 74% lower than in the *Individual* round.¹⁷

Earnings implications. The expected earnings from guesses are a direct measure of performance in the experiment. Table 3 estimates average expected earnings from guesses as a function of the number of draws in each set of signals. As expected, more draws in the second set of signals in the *Individual* round significantly increases earnings, by Rs. 3.25 per extra draw. However, participants earn only Rs. 0.86, 73% less ($p = 0.02$), for each extra draw their partner makes in the *Discussion* round and Rs. 0.27, 92% less ($p = 0.01$), for each draw their partner makes in the pre-discussion guess in the *Informed* round. In the post-discussion guess in the *Informed* round, additional draws by the partner also earn less than own draws in the *Individual* round (Rs. 1.87 vs Rs 3.25), though this difference is not statistically significant ($p = 0.25$).¹⁸

5 Experiment 2: Exploring Mechanisms and Confounds

Why do participants discount their partner’s information even when it is directly communicated to them? Experiment 2 is designed to isolate potential mechanisms, rule out potential confounds, and evaluate the robustness of our findings.

¹⁷The parameters of the quasi-Bayesian model have a different scale and interpretation than the reduced-form results discussed above. But frictionless social learning implies $\beta_2^o = \beta_2^p$ in both cases, where o and p refer to own and partner’s draws respectively. Appendix B.2 shows that the reduced-form and structural estimates are consistent with each other: data simulated using the structural model produces the same reduced-form results as the empirical data.

¹⁸Appendix Table A.III shows similar regressions but where the dependent variable is the absolute difference between participants’ guesses and the true number of red balls in the urn. Mirroring the results in Table 3, additional draws by the participant reduce this error on average, but this improvement is 70 to 100% smaller for draws that come from participants’ partners and in fact we cannot reject that such draws do not reduce the guessing error at all.

5.1 Recruitment and Sample

Experiment 2 was run at the Behavioral Development Lab in Chennai, India between February and March 2020, after observing the results of Experiment 1. We recruited new participants following a similar procedure as Experiment 1. Data collection ended in March 2020 due to the Covid-19 pandemic, with a sample size of 292 participants (out of an intended sample of 800).¹⁹ Compared to Experiment 1, participants have a similar average age (38 versus 35) and years of education (9 versus 8), but are less likely to be female (31% versus 50%), as reported in column 2 of Table 1.

5.2 Experiment 2: Design

Participants played six rounds corresponding to different treatment conditions, with no feedback between rounds. They first played a *Discussion* round, exactly as in our first experiment, to replicate our earlier findings and provide a baseline to compare other treatments with.²⁰ They then played five rounds in randomized order, consisting of an *Informed* round just as in Experiment 1 and four additional variations of the *Informed* round, described below and in Appendix Figure A.II, Panel A.

***Observe Partner’s Draws* round.** In this round (which we abbreviate to ‘*Observe*’), both participants are in the same booth, so they can each watch their partner drawing balls from the urn with their own eyes.²¹ After both participants have drawn their signals they are separated and each makes a private guess. There is no discussion between partners and no need for the experimenter to share draws. Nor is there any scope for distrust of the experimenter or partner. Both one’s own and one’s partner’s signals are perfectly observable and revealed draw-by-draw in randomized order across individuals. The *only* difference between the two sets of draws is who physically drew the balls from the urn. We designed this to be an extreme treatment, where we anticipated

¹⁹The pre-registered sample size was chosen to be powered to test for gender differences in treatment effects (which we explore in Conlon et al. 2024). Thus, even though the final sample size is smaller than intended, we remain tolerably well-powered to estimate the treatment effects described here. For example, the minimum detectable effect size for the *Informed* treatment is around 50% lower sensitivity to partner’s information relative to own information. This is close to the estimated effect in Experiment 1.

²⁰Participants always played the *Discussion* round first in order to avoid the possibility that our new treatments primed participants to behave differently in the *Discussion* round, and thus provide a cleaner test of whether the *Discussion* results replicated.

²¹We would like to thank Christine Exley for suggesting this treatment.

equal sensitivity to one’s own and others’ information. The remaining treatments are subtler and largely subsumed under this treatment.

***Draw-by-Draw* round.** In the *Informed* treatment, participants draw their own balls one at a time from the urn, while their partner’s information is communicated in summary form (‘2 red and 3 white balls’). Certain updating biases (e.g., base-rate neglect) could cause participants to respond differently to summary information than to learning information draw-by-draw. To test for this channel, the *Draw-by-Draw* round proceeds identically to the *Informed* round, except that the experimenter shares their partner’s draws with each participant one draw at a time, e.g., by saying, ‘Your partner first drew a red ball’, then after a brief pause, ‘Your partner then drew a white ball, ...’ and so on.

***Reverse-Order* round.** In this round, one participant learns their partner’s signal before making any draws themselves. They then make a guess, make their own draws, and make another private guess. Since this treatment is only possible for one person in each pair, we only include guesses from the treated person while analyzing this round.

***No-First-Guess* round.** This round was identical to the *Informed* round except that participants do not make a guess directly after making their own set of draws. We implemented this change to test whether, for example, people are more open to others’ information when they have not yet taken an action or stated a belief based on their own information.

***Higher-Stakes* treatment.** We increased the incentives for accurate guessing by 50% in a randomly-chosen 3 out of 6 rounds. The maximum amount each individual could earn from a guess and their loss in earnings per ball away from the truth were both increased by 50%, to Rs. 158 (\$2.25) and Rs. 22.5, respectively. Participants were informed about the stakes at the beginning of each round.

5.3 Experiment 2: Results

Figure 5 shows the results from Experiment 2. Since this experiment does not include an *Individual* round, we simply compare the sensitivity to own information (β^o) and the partner’s information (β^p) within round, estimating Equation (1) by OLS.²² The

²²We did not include an *Individual* round since the previous experiment established that, if anything, participants are more sensitive to their most recent signals compared to earlier signals. Thus, if the

corresponding regression coefficients are presented in Panel A of Table 4.²³

We first replicate the main finding from Experiment 1: Figure 5 shows that participants are 87% and 58% less sensitive to their partner’s information in the *Discussion* and *Informed* rounds, respectively, and we can reject $\beta^o = \beta^p$ with $p < 0.01$.²⁴ In addition, though not all differences are statistically significant, participants underweight their partner’s information in *every* other treatment.

Most strikingly, participants are less sensitive to their partner’s signals even in the *Observe* treatment, in which they see their partner drawing balls from the urn with their own eyes while sitting beside them. Participants are 41% less sensitive to their partner’s information than to their own in this treatment ($p = 0.04$). This result rules out a large set of confounds, including distrust of information communicated by others, perceived skill differences, order effects, and subtle differences in how information is communicated. Instead, it suggests that the act of producing information (i.e., physically drawing balls from the urn) or associating one piece of information with oneself as opposed to with one’s partner may be driving factors. We explore these mechanisms further in Experiment 3.

The remaining treatments provide further evidence of our core result that participants put less weight on their partner’s information than on their own. When participants learn their partner’s signals before drawing their own signals, in the *Reverse Order* treatment, they are still 53% less sensitive to their partner’s information ($p = 0.04$). While still sizable, the effects in the *Draw-by-Draw* and *No First Guess* treatments are somewhat less pronounced at 38% ($p = 0.18$) and 43% ($p = 0.12$), respectively. The latter two estimates are not statistically significant, perhaps due to the lower-than-intended sample size, but the difference in point estimates is roughly comparable across all six treatments, and we cannot reject that it is the same in all treatments ($p = 0.82$).

The under-sensitivity to others’ information is also not meaningfully affected by the size of the incentives for accurate guessing (Appendix Figure A.V and Table A.IV). In

partner’s information is learned last, this biases us *against* finding less sensitivity to others’ information.

²³Appendix Figure A.III, Panel A shows non-parametric estimates, plotting participants’ guesses first against their own signal and then against their partner’s signal in the *Discussion*, *Informed*, and *Observe* treatments. Like the reduced-form and structural results we discuss below, the non-parametric results indicate underweighting of others’ information: the slope of guesses against one’s own signal is steeper.

²⁴To conserve space, in all rounds except the *Discussion* round, we focus in the main text on only the pre-discussion guesses, after the participant is informed of their partner’s draws. Figure A.IV shows reduced-form results for the post-discussion guesses (except for the *Observe* round in which there was no discussion and thus no post-discussion guess), which look broadly similar to those for the pre-discussion guesses.

particular, we see no significant change in underweighting in rounds that were randomly assigned to have 50% higher stakes, and in fact relative underweighting is somewhat higher with high stakes (52% versus 42%).

The structural estimates (Table 4 Panel B) paint a similar picture, with participants putting significantly less weight on their partner’s information in every round. The weight participants put on their own information tends to be at or above the Bayesian benchmark of $\beta^p = 1$, while they tend to underweight their partner’s information (though, given the imprecision of these estimates, we typically cannot reject equality with the Bayesian benchmark). In particular, in the *Observe* round, participants place 60% less weight (0.91/1.51) on their partner’s information than their own ($p < 0.01$).

6 Experiment 3: External Validity and Mechanisms

Experiment 3 is a large-scale, between-subjects online experiment with three goals. First, it has a simpler, between-subjects design and an online format that other researchers can easily adopt. Second, we further investigate mechanisms, and particularly the importance of taking some action to generate one’s own signals rather than passively receiving information. Third, we test the external validity of our findings with higher-education participants from a different cultural context.

6.1 Recruitment and Sample

We recruited 4,489 participants from the US and UK on the online survey platform Prolific in February 2022, asking participants to complete a “short decision-making experiment” that involved a 15-minute survey. We required participants to have completed at least 50 previous surveys on Prolific with an approval rating above 95%. Participants were paid \$2.50 for completing the survey, plus up to \$2.80 as a bonus for accurate guessing. The resulting sample is similar in age and gender to our Experiment 1 and 2 samples (column 3 of Table 1). A key difference is that the sample is more highly educated, though participants’ task comprehension and performance are comparable across

the three experiments.²⁵

6.2 Experiment 3: Design

Participants recruited on Prolific were directed to a Qualtrics survey that embedded the experiment. Each participant was randomly matched to a partner.²⁶ The experiment had a purely between-subjects design, with participants randomized to one of the treatments—variants of the *Informed* condition—described below. Each participant played five identical rounds of the same treatment without feedback. We randomized across participants whether they drew their own signals first or instead first learned about their partner’s signals.²⁷

Informed of Partner’s Draws treatment ($N = 1,008$). This treatment sought to emulate the *Informed* round from Experiments 1 and 2 as closely as possible in an online format. Participants saw a virtual urn and clicked to draw balls from it one at a time. The drawing and replacement of the balls from the urn was animated. Participants were shown a summary of their partner’s draws, as in the previous *Informed* treatments (e.g., “Your partner got to draw 5 marbles out of **the same** jar. They drew 1 red marble and 4 blue marbles.”).

Observe Partner’s Draws treatment ($N = 1,497$). This treatment (which we abbreviate to ‘*Observe*’) differed from the *Informed* treatment in that participants saw their partner’s draws being revealed using the same ball-by-ball animations as their own draws. The goal was to make the mode of presentation of the two sets of draws as similar as possible. Comparing the *Observe* and *Informed* treatments isolates the role of the presentation of others’ information, including its visual presentation and whether the information is delivered in summary form or signal-by-signal.

Labels Only treatment ($N = 1,487$). This treatment was the same as the *Observe*

²⁵In Experiment 3, we included eight multiple-choice comprehension questions asking participants to explain aspects of the instructions. Participants had to answer each question correctly before they could proceed. The average participant answered 92% of these questions on the first attempt, and more than 80% did so for all eight questions. The results are unchanged if we include only those who answered all questions correctly.

²⁶Since the experiment did not include any direct communication between partners (as there was no *Discussion* round), it was not necessary for partners to be playing the game at the same time. Instead, we pre-generated the signals for each partner from the same ‘urn’.

²⁷The script and a link to the online experiment are provided in Appendix C. Appendix Figure A.II Panel B illustrates the design of the different treatments. Selected screenshots showing how draws were presented to participants are reproduced in Appendix Figure A.VI.

treatment, except that participants no longer had to click a button to generate each of their own draws. The only difference between one’s own and one’s partner’s signals was one word in the text that appeared below the animation (e.g., ‘Your first marble’ versus ‘Partner’s first marble’). If participants are less sensitive to their partner’s draws even in this minimal treatment, it implies that a subtle label is enough to generate a sense of ownership. In turn, comparing this treatment with the *Observe* treatment isolates the effect of taking an action to generate your own information. Taking action to generate information might be necessary to create a sense of ownership or to make that information more salient or vivid.

Non-Rivalry treatment. This treatment aimed to reduce any sense of competition with one’s partner. A randomly-selected half of the participants in the *Informed* treatment were truthfully informed that their partner would not be guessing the contents of the urn. Instead, the partner would only draw signals and be asked to remember them.

Higher-Stakes treatment. We randomized across participants the size of the incentives for accurate guessing. Half of those in each treatment earned a \$1.40 bonus minus \$0.20 cents times the absolute difference between their guess and the true number of red balls in the urn. For the other half of participants, the incentives were doubled.

Survey. After completing the five rounds of the experiment, participants completed a short survey. In the survey, without prior warning, we collected unincentivized measures of recall of their own and their partner’s draws from the last round as a measure of attention and memory. We also elicited participants’ perceptions of whether they used their own and their partner’s signals equally in informing their guesses.

6.3 Experiment 3: Results

To test whether participants are less sensitive to their partner’s information, we simply estimate equation (1) within each treatment condition and test $\beta^o = \beta^p$. Since the order of learning one’s own and one’s partner’s signals was randomized with equal probabilities, order effects do not confound this comparison. Figure 6 and Table 5 Panel A report the reduced-form results.²⁸ In the *Informed* treatment, we see that participants are 17% less

²⁸Appendix Figure A.III, Panel B, reports non-parametric results for Experiment 3, plotting participants’ guesses against their own signal and then against their partner’s. The pattern of results is similar to our reduced-form results, showing a greater responsiveness to own signal in the *Informed* and *Observe* treatments, but a more equal responsiveness in *Labels Only*.

sensitive to their partner’s information than their own ($p < 0.01$). We thus qualitatively replicate our previous findings from Experiments 1 and 2, despite the large differences in presentation, instructions, and samples.

The design of Experiment 3 also permits an even simpler test of sensitivity to own and others’ information. Since participants were randomized to receive their own or their partner’s signals first, we can examine the *first* guess they make—after seeing only the first set of draws—and test whether this guess was less sensitive to draws made by their partner. Appendix Table A.V column 1 reports these results. Once again, we find that participants’ guesses are 17% less sensitive to their partner’s signal than to their own ($p < 0.01$).²⁹ Overall, these results provide strong evidence of lower sensitivity to others’ information even with a very different sample and experimental format.

In the *Observe* treatment, participants continue to be significantly less sensitive to their partner’s information ($p < 0.01$). Indeed, the magnitudes are nearly identical to the *Informed* treatment (19% vs. 17%, $p=0.74$). Consistent with the findings of Experiment 2, this suggests that differences in the presentation of own and others’ information do not explain the lower sensitivity to others’ information. Appendix Table A.V column 2 shows that this also holds for the first guess (10% lower sensitivity to others’ information, $p < 0.01$).

By contrast, participants in the *Labels Only* treatment were only 4% less sensitive to their partner’s information, a difference that was not statistically significant in the reduced-form estimates ($p = 0.27$). The difference in sensitivity to own and partner’s information ($\beta^o - \beta^p$) is significantly lower in the *Labels Only* treatment than in the *Informed* ($p = 0.02$) and *Observe* ($p < 0.01$) treatments. We find a similar pattern in the first guess (Appendix Table A.V column 3). We interpret this result as showing that taking an action to gather information—which, plausibly, increases its salience—plays a role in producing greater sensitivity to it. Merely labeling information as ‘own’ versus ‘partner’s’ when the participant receives the information passively might not create this differential attention.

The structural estimates presented in Panel B of Table 5 again show clear evidence of underweighting of others’ information in the *Informed* and *Observe* treatments, by 31% and 33%, respectively (each $p < 0.01$). Again, the difference in weights is significantly

²⁹Appendix Table A.VI shows that participants’ first guess also underweights others’ signals relative to their own in the first round of the experiment, before participants have experienced any signals of the opposite type.

smaller in the *Labels Only* treatment compared to the *Observe* treatment ($p=0.04$), implying that taking an action to generate one’s own draws increases the weight on own relative to others’ information. However, in contrast to the reduced-form estimates, the structural estimates show significant underweighting of partners’ information even in the *Labels Only* round ($p<0.01$).³⁰

The results of Experiment 3 reinforce our findings from the *Observe* treatment in Experiment 2 that mechanisms such as visual presentation of the information, or distrust of the experimenter, do not drive underweighting. In Experiment 3, both one’s own and one’s partner’s draws are simply displayed on the computer screen, such that it is not clear why one would trust draws assigned to oneself more. The visual presentation of own and others’ draws is also identical in the *Observe* treatment in Experiment 3.

Our additional treatment variations in Experiment 3 enable us to rule out two further possibilities: first, that participants *consciously* underweight others’ information and second, that some form of competitiveness or rivalry drives our main result. Finally, we provide correlational evidence on the role of memory.

Stakes and awareness. As in Experiment 2, the relative sensitivity to own vs. others’ information is not affected by the size of the incentives that participants faced for accurate guesses (Figure A.V, Panel B). For all three treatments, the differential sensitivity to own and partner’s signals is very similar (and statistically indistinguishable) between the low- and high-stakes groups. This finding suggests that participants are either unaware that they are less sensitive to others’ information or that they mistakenly believe it is optimal to discount others’ information. Consistent with the former interpretation, 77% of participants reported in the debriefing survey that they treated both pieces of information the same. Yet these same participants were 15% less sensitive to their partner’s information in the *Informed* treatment, nearly identical to the result in the full sample (Table A.VII Panel C).³¹

³⁰The structural estimates account for the full information content of the signal. For example, a person should place more weight on seeing 9 draws (5 red and 4 white) than on seeing just 1 red draw, whereas the reduced-form analysis treats these as identical (1 net red draw). The structural model also accounts for noisy choices and the fact that guesses are constrained to be between 4 and 16. These differences may explain the discrepancy in the reduced-form and structural analysis of the *Labels Only* treatment.

³¹This also provides evidence against the possibility that participants felt experimenter demand to use their own information more. In all experiments, our instructions emphasized that participants’ goal was just to guess the number of red balls in the urn, and that their and their partner’s draws were coming from the same urn. Among participants who did not report treating both pieces of information equally, 14% reported using their own information more while 8% reported using their partner’s information more.

Competition. Despite the incentives to make accurate guesses, one concern could be that participants underweight their partner’s information out of a sense of competitiveness; e.g., they may enjoy ‘winning’ by making good guesses precisely when their partner guesses poorly. This might plausibly lead to a strategy of ignoring the partner’s draws. However, the *Non-Rivalry* treatment—a sub-treatment of *Informed* in which the partner does not make any guesses—does not increase sensitivity to the partner’s signals (Appendix Figure A.VII and Table A.VIII).

Memory. Our survey at the end of the experiment allows us to provide suggestive evidence on the role of memory (Appendix Figure A.VIII and Table A.IX). Participants were asked to recall their own and their partner’s draws in the final round. Recall of one’s own draws was slightly higher on average (60% vs. 55%, $p < 0.01$).³² However, even when restricting the sample to those who perfectly remember *both* sets of draws, participants still place (marginally) significantly less weight on other’s information, by 9% (Appendix Figure A.IX and Table A.VII Panel C), and we cannot reject that the difference in weights is the same in this subsample as in the overall sample ($p = 0.75$). This suggests that even participants who can recall others’ information (at least when specifically asked to) fail to use it in the same way as their own information. Of course, participants who remember others’ draws are a selected subsample. One limitation of our study is that we did not exogenously manipulate memory of own or others’ information, as would be necessary to quantitatively determine how much of the gap in weights it can explain. Such experiments are a promising direction for future work.

6.4 Experiment 3: Discussion of Mechanisms

What do we learn from Experiment 3 about the potential mechanisms underpinning asymmetric learning from own vs others’ information? We view several aspects of our results as broadly consistent with a bottom-up attention story whereby information gathered oneself simply looms larger in people’s later decisions. First, the asymmetry is

In open-ended responses, participants who reported using their own information more often explained that this was because they (randomly) received more draws than their partners. A 10 percentage point increase in the share of draws received oneself is associated with a 12 percentage point increase in the likelihood of reporting using one’s own information more ($p < 0.01$).

³²The difference in recall is significant in all treatments, although it is smaller in the *Labels Only* treatment. Specifically, recall of own vs. partner’s draws in the different treatments are 64% vs. 56% for the *Informed* treatment, 60% vs. 55% for *Observe*, and 58% vs. 55% for *Labels Only*. We can reject equal memory gaps in the *Labels Only* and the *Informed* treatments ($p=0.01$).

greatly reduced (or even eliminated) when participants need not take action to generate their own signals. Future work could fruitfully explore what types of actions are sufficient to generate these effects, especially in more naturalistic settings. This could also shed light on whether taking similar actions to uncover others’ information would improve social learning.

Second, the relative under-sensitivity to others’ signals in Experiment 3 is notably smaller than in Experiments 1 and 2, largely driven by the fact that participants put a higher weight on their partner’s information in Experiment 3 than in Experiments 1 and 2.³³ One possibility is that this reflects a difference in samples,³⁴ but another is that the online interface reduces the difference between own and partner’s information in a way that may limit the scope for own information to draw more attention. For example, in Experiment 3, participants do not see, learn any details about, or interact with their partner, and thus it might not truly feel like information coming from others. They uncover their own signals only by clicking a button rather than physically drawing from a urn, which might be a less engaging way to discover information. Future work could aim to understand these differences better. For instance, one could experimentally vary participants’ knowledge of the ‘other’, or whether participants interact with their partner in person, via (video) chat, or not at all, or vary information about whether others have seen the same signal. Gathering independent data on what participants attend to or what is salient (Krajbich and Rangel 2011, Li and Camerer 2022, Bohren et al. 2024, Bordalo et al. 2025) would help to further understand the mechanisms involved.

7 Conclusion

This paper presents evidence of a powerful and potentially far-reaching barrier to social learning: people place more weight on information gathered themselves than on information gathered by others. Our results rule out failures of communication as the primary source of this asymmetry, as participants underweight others’ signals even when

³³For ease of comparison, Figure A.X repeats the non-parametric results from the *Informed* round in Experiments 1, 2 and 3 side by side, with the average guesses of a risk-neutral Bayesian displayed as a benchmark.

³⁴For example, it could reflect a documented tendency for lower sensitivity to treatment conditions in online compared to in-person experiments (Gupta et al., 2021). Another potential explanation is that our sample in Experiment 3 is has higher education levels, though we do not find stronger effects among lower-education participants within any of the three experiments (Table A.VII).

they are frictionlessly communicated or even physically observed, and instead point toward failures of information *aggregation* as the primary mechanism: people simply focus more on information that they themselves gathered. This phenomenon appears across three experiments with very different study populations, cultural contexts, experimental formats, and treatment variations.

A limitation of our lab setting is that, while precise and closely controlled, it is fairly abstract and with at best moderate stakes (up to about half a day’s income in Experiments 1 and 2). An open question is to what extent similar findings will appear in field settings and with higher stakes. We speculate that the tendency we identify may underlie documented failures of social learning, whether in information cascade experiments (Weizsäcker, 2010), people reacting to information interventions (Haaland et al., 2023), farmers learning more from their own plots than from neighbors (Duflo et al., 2020; Chandrasekhar et al., 2022), decision-makers under-using AI-based recommendations (Agarwal et al., 2024), or central bankers being sensitive to their own personal economic experiences beyond aggregate data (Malmendier et al., 2021). But underweighting of others’ information could play a role in numerous other settings where social learning is possible.

We document this phenomenon in teams of strangers. In Conlon et al. (2024), we find that the marital context—learning from one’s spouse—appears to counteract the discounting of others’ information for women but not for men. Future work should study the underlying mechanisms behind these differences and, more generally, what types of social or work relationships and contexts shape how effectively people learn from each other. For example, do people learn better from friends or colleagues? How do social status hierarchies affect the weight placed on a person’s independent information? Finally, additional work is required to understand how sensitivity to information coming from others may be increased, where appropriate, particularly in naturalistic settings.

References

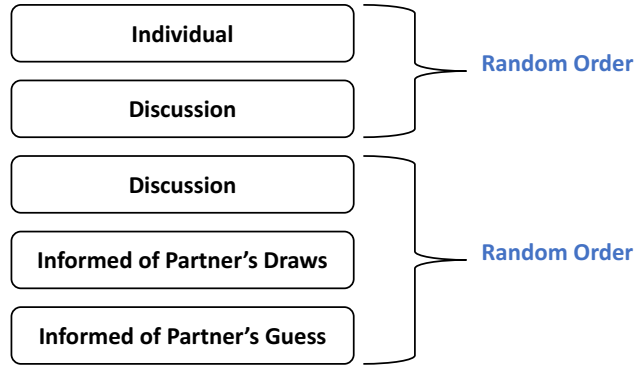
- Agarwal, Nikhil, Alex Moehring, Pranav Rajpurkar, and Tobias Salz, “Combining Human Expertise with Artificial Intelligence: Experimental Evidence from Radiology,” *NBER Working Paper Number 31422*, 2024.
- Alatas, Vivi, Arun G Chandrasekhar, Markus Mobius, Benjamin A Olken, and Cindy Paladines, “Designing Effective Celebrity Public Health Messaging: Results from A Nationwide Twitter Experiment in Indonesia,” 2021.
- Anderson, Lisa R and Charles A Holt, “Information Cascades in the Laboratory,” *American Economic Review*, 1997, 87 (5), 847–862.
- Angrisani, Marco, Antonio Guarino, Philippe Jehiel, and Toru Kitagawa, “Information redundancy neglect versus overconfidence: a social learning experiment,” *American Economic Journal: Microeconomics*, 2021, 13 (3), 163–197.
- Augenblick, Ned, Eben Lazarus, and Michael Thaler, “Overinference from weak signals and underinference from strong signals,” *The Quarterly Journal of Economics*, 2024, p. qjae032.
- Ba, Cuimin, J Aislinn Bohren, and Alex Imas, “Over-and Underreaction to Information: Belief Updating with Cognitive Constraints,” 2025.
- Benjamin, Dan, Aaron Bodoh-Creed, and Matthew Rabin, “Base-Rate Neglect: Foundations and Implications,” *Working Paper*, 2019.
- Benjamin, Daniel J., “Chapter 2 - Errors in Probabilistic Reasoning and Judgment Biases,” *Handbook of Behavioral Economics: Applications and Foundations 1*, 2019, 2, 69–186.
- BenYishay, Ariel and A. Mushfiq Mobarak, “Social learning and incentives for experimentation and communication,” *The Review of Economic Studies*, 2019, 86 (3), 976–1009.
- , Maria Jones, Florence Kondylis, and Ahmed Mushfiq Mobarak, “Gender gaps in technology diffusion,” *Journal of Development Economics*, 2020, 143.
- Bohren, J Aislinn, Josh Hascher, Alex Imas, Michael Ungeheuer, and Martin Weber, “A cognitive foundation for perceiving uncertainty,” Technical Report, National Bureau of Economic Research 2024.
- Bordalo, Pedro, John Conlon, Nicola Gennaioli, Spencer Kwon, and Andrei Shleifer, “How people use statistics,” *Review of Economic Studies*, 2025, p. rdaf022.
- Chandrasekhar, Arun G, Esther Duflo, Michael Kremer, João F Pugliese, Jonathan Robinson, and Frank Schilbach, “Blue spoons: Sparking communication about appropriate technology use,” Technical Report, National Bureau of Economic Research 2022.

- Conlon, John J., Malavika Mani, Gautam Rao, Matthew Ridley, and Frank Schilbach**, “Learning in the Household,” *Working Paper*, 2024.
- Cooper, David J. and John H. Kagel**, “Are two heads better than one? Team versus individual play in signaling games,” *American Economic Review*, 2005, *95* (3), 477–509.
- D’Acunto, Francesco, Ulrike Malmendier, Juan Ospina, and Michael Weber**, “Exposure to Grocery Prices and Inflation Expectations,” *Journal of Political Economy*, 2021, *129* (5), 1615–1639.
- De Filippis, Roberta, Antonio Guarino, Jehiel Philippe, and Toru Kitagawa**, “Updating Ambiguous Beliefs in a Social Learning Experiment,” *Working Paper*, 2017.
- Drehmann, Mathias, Jörg Oechssler, and Andreas Roider**, “Herding and Contrarian Behavior in Financial Markets: An Internet Experiment,” *American Economic Review*, 2005, *95* (5), 1403–1426.
- Duflo, Esther, Daniel Keniston, Tavneet Suri, and Céline Zipfel**, “Chat Over Coffee? Technology Diffusion Through Social and Geographic Networks in Rwanda,” *Working paper*, 2020.
- Enke, Benjamin and Thomas Graeber**, “Cognitive uncertainty,” *The Quarterly Journal of Economics*, 2023, *138* (4), 2021–2067.
- , **Frederik Schwerter, and Florian Zimmermann**, “Associative memory, beliefs and market interactions,” *Journal of Financial Economics*, 2024, *157*, 103853.
- Goeree, Jacob K., Thomas R. Palfrey, Brian W. Rogers, and Richard D. McKelvey**, “Self-Correcting Information Cascades,” *The Review of Economic Studies*, 2007, *74* (3), 733–762.
- Grether, David M**, “Bayes rule as a descriptive model: The representativeness heuristic,” *The Quarterly journal of economics*, 1980, *95* (3), 537–557.
- , “Testing Bayes rule and the representativeness heuristic: Some experimental evidence,” *Journal of Economic Behavior & Organization*, 1992, *17* (1), 31–57.
- Guarino, Antonio and Philippe Jehiel**, “Social Learning with Coarse Inference,” *American Economic Journal: Microeconomics*, 2013, *5* (1), 147–74.
- Gupta, Neeraja, Luca Rigotti, and Alistair Wilson**, “The Experimenters’ Dilemma: Inferential Preferences over Populations,” *arXiv preprint arXiv:2107.05064*, 2021.
- Haaland, Ingar, Christopher Roth, and Johannes Wohlfart**, “Designing information provision experiments,” *Journal of economic literature*, 2023, *61* (1), 3–40.
- Hartzmark, Samuel M, Samuel Hirshman, and Alex Imas**, “Ownership, Learning, and Beliefs,” *The Quarterly Journal of Economics*, 2021, *136* (3), 1665–1717.

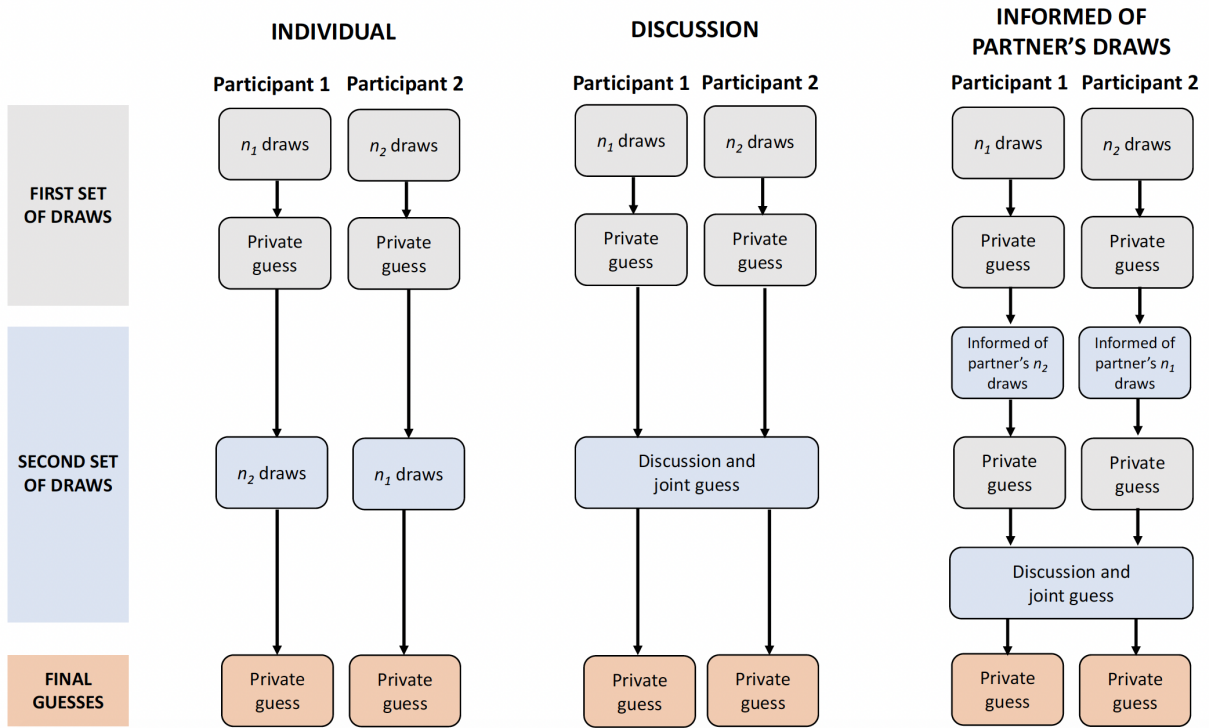
- Kondylis, Florence, John Loeser, Mushfiq Mobarak, Maria Jones, and Daniel Stein**, “Decentralizing Agricultural Demonstration to Accelerate Social Learning,” 2024.
- Krajbich, Ian and Antonio Rangel**, “Multialternative drift-diffusion model predicts the relationship between visual fixations and choice in value-based decisions,” *Proceedings of the National Academy of Sciences*, 2011, 108 (33), 13852–13857.
- Li, Xiaomin and Colin F Camerer**, “Predictable effects of visual salience in experimental decisions and games,” *The Quarterly Journal of Economics*, 2022, 137 (3), 1849–1900.
- Malmendier, Ulrike and Leslie Sheng Shen**, “Scarred consumption,” *American Economic Journal: Macroeconomics*, 2024, 16 (1), 322–355.
- and **Stefan Nagel**, “Learning from inflation experiences,” *The Quarterly Journal of Economics*, 2016, 131 (1), 53–87.
- , — , and **Zhen Yan**, “The making of hawks and doves,” *Journal of Monetary Economics*, 2021, 117, 19–42.
- March, Christoph and Anthony Ziegelmeyer**, “Altruistic Observational Learning,” *Journal of Economic Theory*, 2020, 190, 105–123.
- Merlo, Antonio and Andrew Schotter**, “Learning by not doing: an experimental investigation of observational learning,” *Games and Economic Behavior*, 2003, 42 (1), 116–136.
- Miller, Joshua B and Zacharias Maniadis**, “The weight of personal experience: An experimental measurement,” Technical Report, Working paper 2012.
- Mobius, Markus and Tanya Rosenblat**, “Social Learning in Economics,” *Annual Review of Economics*, 2014, 6 (1), 827–847.
- Schwartzstein, Joshua**, “Selective attention and learning,” *Journal of the European Economic Association*, 2014, 12 (6), 1423–1452.
- Simonsohn, Uri, Niklas Karlsson, George Loewenstein, and Dan Ariely**, “The tree of experience in the forest of information: Overweighing experienced relative to observed information,” *Games and Economic Behavior*, 2008, 62 (1), 263–286.
- Weizsäcker, Georg**, “Do We Follow Others When We Should? A Simple Test of Rational Expectations,” *American Economic Review*, 2010, 100 (5), 2340–60.

Figure 1: Experimental Design

Panel A: Randomization of Rounds



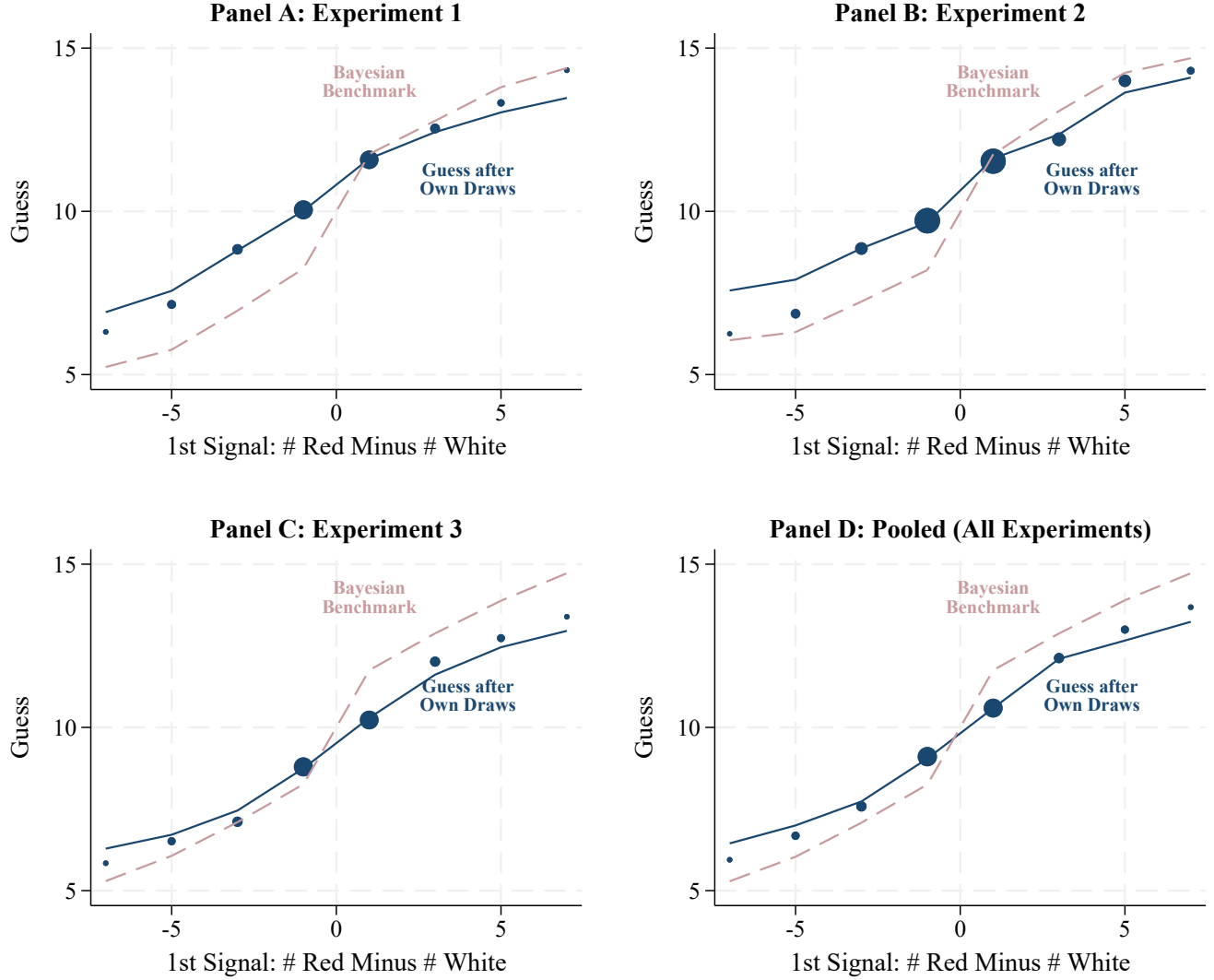
Panel B: Structure of Individual, Discussion, and Informed Rounds



Panel A shows the five rounds of Experiment 1. All participants get matched to a previously unknown partner and complete all five rounds with this partner (with the exception that in one randomly-selected *Discussion* round, participants were re-matched for that round only to generate variation in the relative gender of the partners. We do not exploit this variation in our paper). We randomized the order of the first two rounds (*Individual*, *Discussion*) and the order of the following three rounds: *Discussion*, *Informed of Partner's Guess*, and *Informed of Partner's Draws*.

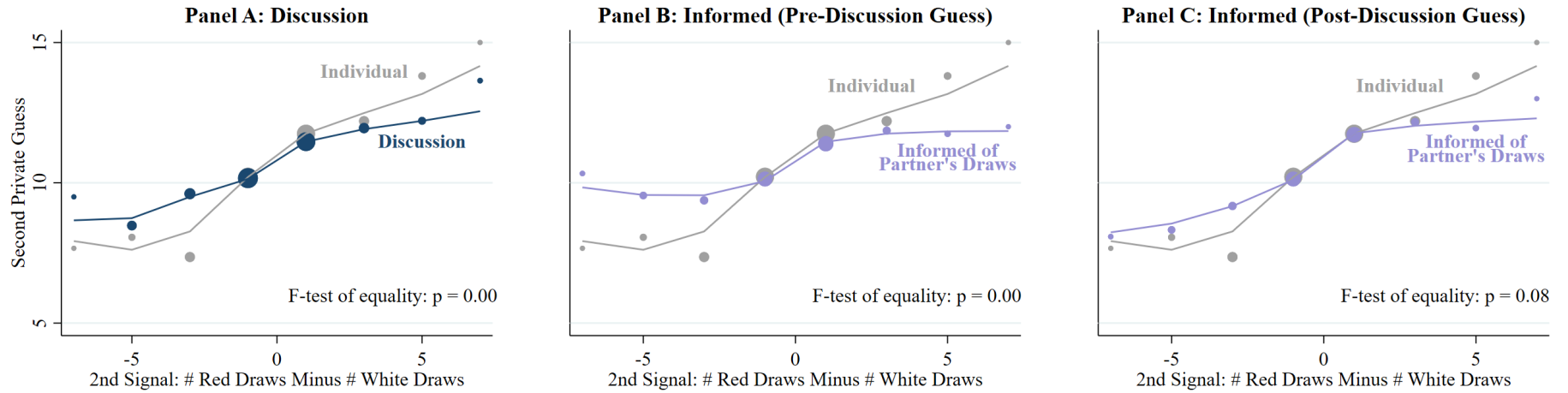
Panel B describes the structure of the different rounds. In the *Individual* round, each participant gets two sets of private draws from the urn and makes a private guess after each set of draws. In the *Discussion* round, each participant makes one set of draws followed by a private guess. The two participants are then asked to discuss and make a joint guess before each makes a final private guess. The *Informed of Partner's Draws* round is identical to the *Discussion* round, except that pre-discussion, each participant is informed about their partner's first set of draws and then asked to make a private guess. In the *Informed of Partner's Guess* round (Appendix B.1), each participant is instead informed pre-discussion about their partner's first private guess and then asked to make a private guess.

Figure 2: Individual Performance vs. Risk-Neutral Bayesian



Notes: This figure plots participants' first private guess against the net number of red draws (red draws minus white draws) in participants' own first (private) signal. We only include observations where participants saw their own signal first (in Experiment 1, this is all observations). The blue solid curve shows locally weighted means (lowess). The pink dotted lines show the average of a risk-neutral Bayesian's guesses given the same signals. Dot size indicates number of observations for each net number of red draws. Panels A through C show data from each of the three experiments separately. Panel D shows pooled data from all three experiments.

Figure 3: Experiment 1: Non-Parametric Results

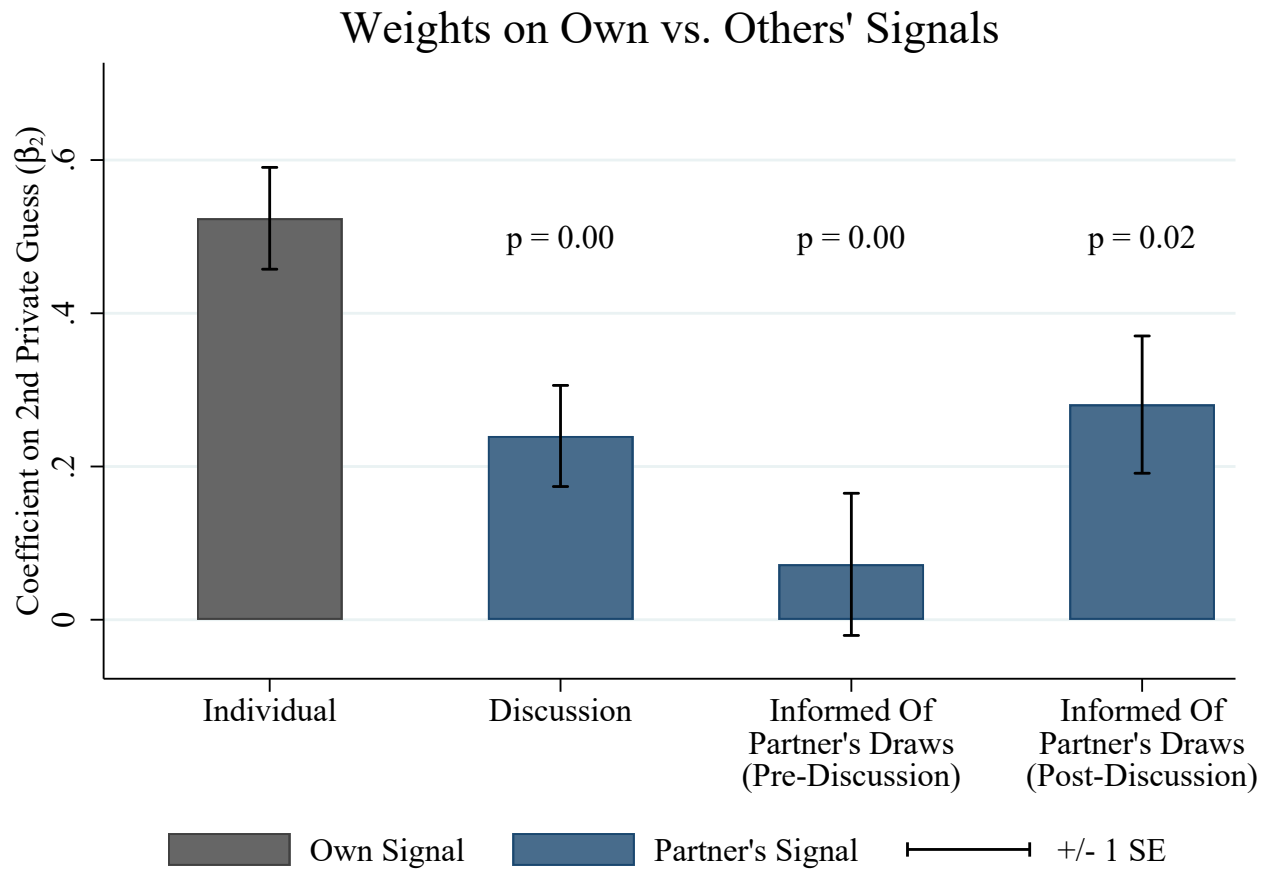


Notes: This figure shows average second private guess of participants in Experiment 1. The x-axis shows the net number of red draws (i.e., red draws minus white draws) in the second signal of the round. Dot size indicates number of observations for each net number of red draws. Lines show locally weighted means (lowess).

- In **Panel A**, the gray dots indicate average guesses in the *Individual* Round, where participants made the second set of draws themselves. The dark-blue dots in the graphs on the left show guesses in the *Discussion* Round, where the second set of draws had to be communicated to the participant via discussion.
- In **Panel B**, the lavender dots show average guesses in the *Informed of Partner's Draws* round, after the respondent is told of his/her partner's draws by the experimenter (but before the joint discussion).
- In **Panel C**, the lavender dots show average guesses in the *Informed of Partner's Draws* round after the joint discussion.

'F-test of equality' in the bottom right shows the p -value of a test of the joint hypothesis that the mean guess is equal across the two rounds at every value of net red draws.

Figure 4: Experiment 1: Reduced-Form Estimates

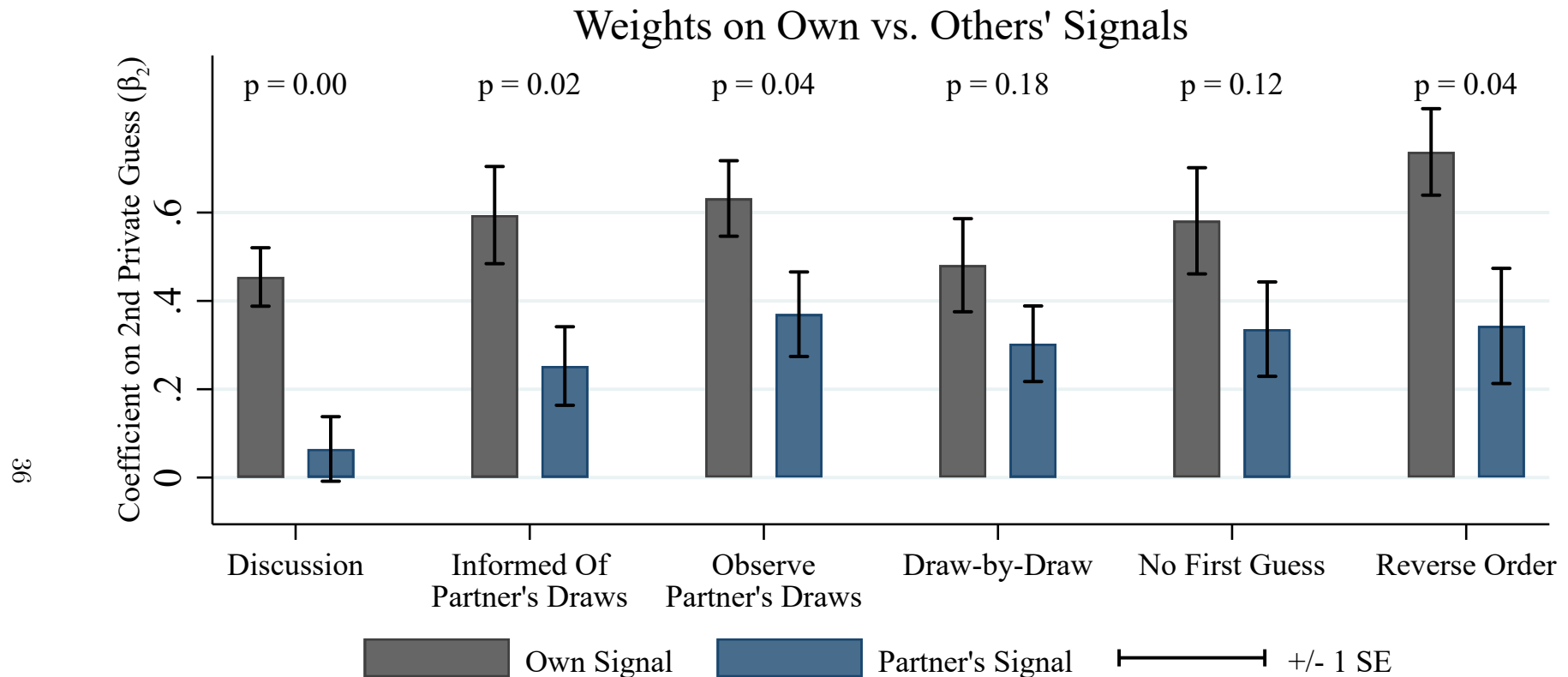


Notes: This figure shows the weights participants put on different signals in Experiment 1. We estimate Equation (1) and then display β_2 for each of the following four types of private guesses:

- (a) *Individual*, in which participants collect all information on their own. For this round, we report the coefficient on the net red draws in the participant's second set of draws, which replaces *Partner's Signal* in Equation (1);
- (b) *Discussion*, in which participants collect the first signal on their own and the second signal (their partner's) is only accessible via discussion;
- (c) *Informed of Partner's Draws (pre-discussion)*, where participants receive the second set of information directly from the experimenter but before any discussion with their partner;
- (d) *Informed of Partner's Draws (post-discussion)*, in which participants receive the second set of information directly *and* have the chance to discuss it with their partner.

For each of the dark-blue bars, we show the p -value of testing whether the weight in that round equals the corresponding weight in the *Individual* round (gray bar).

Figure 5: Experiment 2: Reduced-Form Estimates

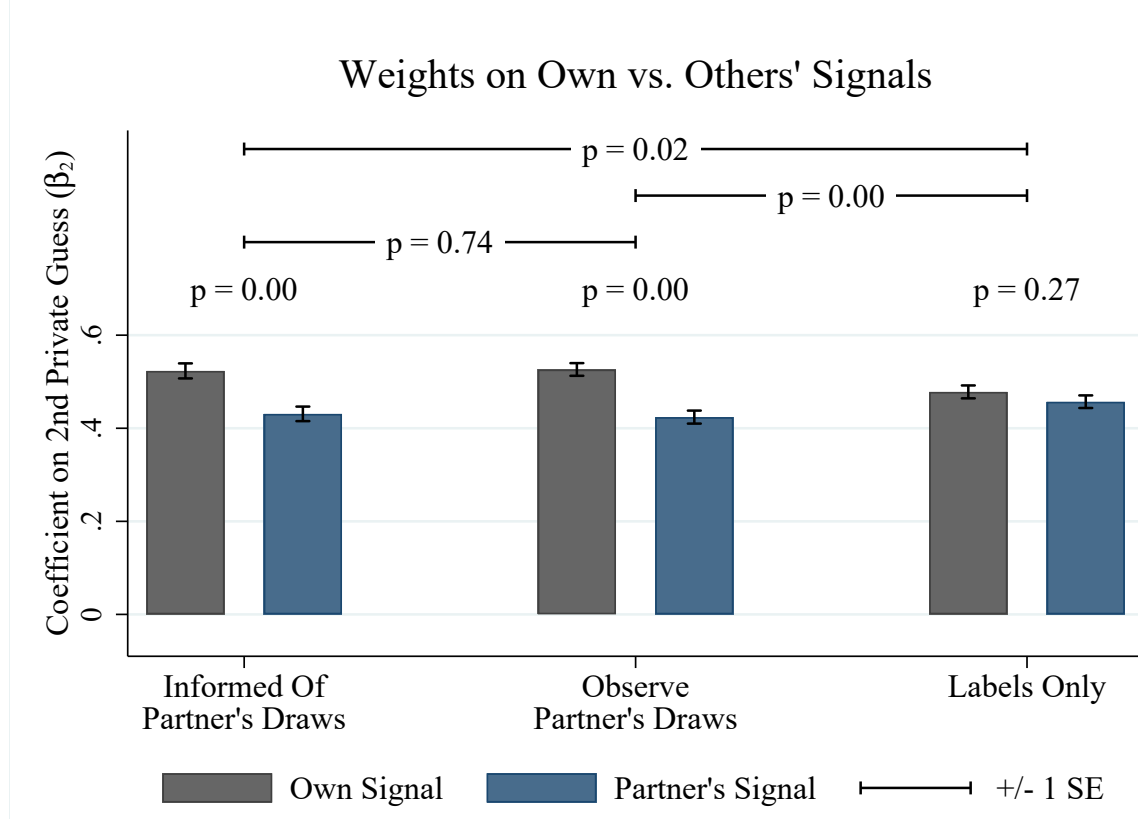


Notes: This figure shows the weights participants put on different signals in Experiment 2. We estimate Equation (1) and then display β_1 in gray and β_2 in dark blue for each treatment. The dependent variable is participants' pre-discussion guess, except in the *Discussion* round. In the *Discussion* round, it is the post-discussion guess as there was no pre-discussion guess. In addition to the *Discussion* and *Informed of Partner's Draws* rounds, we look at the following treatments:

- (a) *Observe Partner's Draws*, in which each participant directly observes their partner's draws (as well as making their own);
- (b) *Draw-by-Draw*, in which participants receive the second set of signals directly one draw at a time;
- (c) *No First Guess*, in which participants receive their partner's signals (and their own) before making their first and only private guess;
- (d) *Reverse Order*, in which one participant receives their partner's signals first and makes their first private guess, and then receives their own signals and makes their second private guess.

For each round, we show the p -value of testing whether the weight on their signal (β_1) equals the corresponding weight on their partner's signal (β_2) in that round.

Figure 6: Experiment 3: Reduced-Form Estimates



Notes: This figure shows the weights participants put on different signals in Experiment 3. We estimate Equation (1) and then display β_1 in gray and β_2 in dark blue for each treatment.

- The first set of bars shows the weights participants put on signals in the *Informed of Partner's Draws* treatment, in which participants clicked to draw their own balls one at a time and were told their partner's number of red and white draws.
- The second set of bars represents the *Observe Partner's Draws* treatment, in which participants clicked to draw their own balls one at a time and directly observed their partner's draws appearing from the urn one at a time.
- The third set of bars corresponds to the *Labels Only* treatment, in which participants did not take any actions and instead passively observed their own and their partner's labeled draws one by one in the exact same format.

For each round, we show the p -value of testing whether the weight on their signal (β_1) equals the corresponding weight on their partner's signal (β_2) in that round.

Table 1: Sample Characteristics

	Experiment 1 (1)	Experiment 2 (2)	Experiment 3 (3)
Female	0.50 (0.50)	0.31 (0.46)	0.57 (0.50)
Age	34.66 (8.58)	38.40 (7.31)	37.70 (13.87)
Years Of Education	7.86 (3.94)	9.02 (3.49)	15.04 (2.03)
Expected Earnings (Relative to Bayesian)	0.82 (0.11)	0.84 (0.12)	0.89 (0.10)
Fraction of Comprehension Questions Correct	0.79 (0.14)	0.79 (0.13)	0.92 (0.13)
Number of Participants	500	293	4489

Notes: This table shows averages of key background characteristics for individuals in each of our three experiments. Standard deviations are in brackets. “Expected Earnings (Relative to Bayesian)” is calculated as the expected payoff of the participant’s guess given the draws they observed, divided by the expected payoff that the Bayesian risk-neutral guess (i.e., expected payoff-maximizing guess) would make given those same draws. “Fraction of Comprehension Questions Correct” shows the proportion of participants who correctly answer questions about the task (summary of questions in Table A.I).

Table 2: Experiment 1: Reduced-Form and Structural Estimates

	Individual (1)	Discussion (2)	Informed (Pre) (3)	Informed (Post) (4)
<i>Panel A: Reduced-Form Estimates</i>				
β_1 : Own First Signal	0.43 (0.06)	0.56 (0.06)	0.56 (0.09)	0.36 (0.09)
β_2^o : Own Second Signal	0.52 (0.07)			
β_2^p : Partner's Signal		0.24 (0.07)	0.07 (0.09)	0.28 (0.09)
Constant	10.71 (0.16)	10.73 (0.16)	10.64 (0.23)	10.66 (0.23)
$\beta_2^p - \beta_2^o$		-0.28*** (0.08)	-0.45*** (0.11)	-0.24** (0.10)
<i>Panel B: Structural Estimates</i>				
β_1 : Own First Signal	0.92 (0.63)	0.87 (0.18)	1.02 (0.41)	0.57 (0.31)
β_2^o : Own Second Signal	1.50 (0.74)			
β_2^p : Partner's Signal		0.40 (0.13)	-0.01 (0.37)	0.46 (0.26)
$\beta_2^p - \beta_2^o$		-1.11*** (0.71)	-1.51*** (0.71)	-1.04** (0.73)
N	500	1000	500	500

Notes: This table shows reduced-form and structural estimates for the weights on signals in Experiment 1. The dependent variable is participants' private guess. "Informed (Pre)" means the second private guess from the *Informed of Partner's Draws* round, after the participant was directly told their partner's signal but before the joint discussion. "Informed (Post)" means the third private guess, after the discussion. All standard errors are clustered at the pair (of two participants) level. Standard errors of the structural estimates are bootstrapped. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_2^o$.

Reduced-form coefficients: Panel A shows reduced-form results, estimating Equation 1 by OLS. "Own First Signal" is the net number of red draws (i.e., red draws minus white draws) in the participant's first set of draws, which they drew themselves in all rounds. "Own Second Signal" is the net number of red draws in the participant's second set of draws in the individual round. "Partner's Signal" is the net number of red draws in the set of draws by the participant's partner, which was the second signal available to the participant in the *Discussion* and *Informed of Partner's Draws* rounds. All regressions include order fixed effects interacted with the participants' first and second signal.

Structural parameters: Panel B shows estimates of the structural model described in Section 3.3. "Own First Signal", "Own Second Signal" and "Partner's Signal" indicate the weights placed on the first set of signals, second set in the *Individual* round, and second (partner's) set in each other round in the agents' quasi-Bayesian updating rule.

Table 3: Experiment 1: Expected Earnings by Type of Guess and Number of Draws

	Individual (1)	Discussion (2)	Informed (Pre) (3)	Informed (Post) (4)
γ_1 : # Own First Draws	1.53 (0.83)	2.52 (0.56)	3.25 (0.76)	2.58 (0.78)
γ_2^o : # Own Second Draws	3.25 (0.86)			
γ_2^p : # Partner's Draws		0.86 (0.65)	0.27 (0.82)	1.87 (0.82)
Constant	102.51 (5.13)	103.36 (4.45)	97.94 (6.18)	97.12 (5.91)
$\gamma_2^p - \gamma_2^o$		-2.39** (1.04)	-2.98*** (1.13)	-1.38 (1.19)
N	500	1000	500	500

Notes: This table compares participants' expected earnings in the *Discussion* and *Informed of Partner's Draws* rounds to their earnings in the *Individual* round. The table shows OLS estimates of the following equation for the *Discussion* and *Informed of Partner's Draws* rounds:

$$Expected\ Earnings_i = \alpha + \gamma_1 \# Own\ First\ Draws_i + \gamma_2^p \# Partner's\ Draws_i + \epsilon_i \quad (5)$$

and OLS estimates of the following equation for the *Individual* round:

$$Expected\ Earnings_i = \alpha + \gamma_1 \# Own\ First\ Draws_i + \gamma_2^o \# Own\ Second\ Draws_i + \epsilon_i \quad (6)$$

where $Expected\ Earnings_{irt}$ is the expected earnings from i 's guess in the round in question, given the signals, and $\# Own\ First\ Draws_i$ indicates the number of draws in the first set of signals, drawn oneself. $\# Own\ Second\ Draws$ is the number of draws in the participant's second set in the *Individual* round and $\# Partner's\ Draws$ is the participant's partner's number of draws, in the *Discussion* and *Informed of Partner's Draws* rounds. In estimation, we stack the estimating equations for all treatments and estimate them jointly, including controls for round order fixed effects. Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\gamma_2^p - \gamma_2^o$.

Table 4: Experiment 2: Reduced-Form and Structural Estimates

	Discussion (1)	Informed (2)	Observe (3)	Draw-by-Draw (4)	No First Guess (5)	Reverse Order (6)
<i>Panel A. Reduced Form Estimates</i>						
β_1 : Own Signal	0.45 (0.07)	0.59 (0.11)	0.63 (0.09)	0.48 (0.11)	0.58 (0.12)	0.74 (0.10)
β_2^p : Partner's Signal	0.06 (0.07)	0.25 (0.09)	0.37 (0.10)	0.30 (0.09)	0.34 (0.11)	0.34 (0.13)
Constant	10.66 (0.19)	10.64 (0.22)	10.63 (0.22)	10.45 (0.21)	10.51 (0.24)	10.38 (0.26)
$\beta_2^p - \beta_1$	-0.39*** (0.09)	-0.34** (0.14)	-0.26** (0.13)	-0.18 (0.13)	-0.25 (0.16)	-0.39** (0.19)
<i>Panel B. Structural Estimates</i>						
β_1 : Own Signal	0.48 (0.11)	1.32 (0.60)	1.51 (0.70)	1.01 (0.69)	1.25 (0.72)	1.29 (0.82)
β_2^p : Partner's Signal	0.07 (0.08)	0.23 (0.36)	0.60 (0.45)	0.33 (0.41)	0.48 (0.55)	0.52 (0.80)
$\beta_2^p - \beta_1$	-0.41*** (0.10)	-1.08*** (0.44)	-0.91*** (0.43)	-0.68** (0.49)	-0.78** (0.41)	-0.77* (0.52)
N	288	292	292	292	292	146

Notes: This table shows reduced-form and structural estimates for rounds in Experiment 2 (our second lab experiment).

Reduced-form coefficients: Panel A shows reduced-form results, estimating Equation 1 by OLS. The dependent variable is participants' private guess. "Informed" refers to the *Informed of Partner's Draws* round and "Observe" to the *Observe Partner's Draws* round. "Own Signal" indicates the net number of red draws (i.e., red draws minus white draws) in the participant's own set of draws. Similarly, "Partner's Signal" indicates the net number of red draws in their partner's set of draws. In estimation, we stack the estimating equations for all treatment and estimate them jointly. The joint regression also includes fixed effects for the order in which participants played treatment conditions, interacted with "Own Signal" and "Partner's Signal." Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Structural parameters: Panel B shows estimates of the structural model described in Section 3.3. "Own Signal" and "Partner's Signal" indicate the weights placed on their own and their partner's set of draws in the agents' quasi-Bayesian updating rule. Bootstrapped standard errors (clustered at the pair level) in parentheses. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table 5: Experiment 3: Reduced-Form and Structural Estimates

	Informed (1)	Observe (2)	Labels Only (3)
<i>Panel A. Reduced Form Estimates</i>			
β_1 : Own Signal	0.52 (0.02)	0.53 (0.01)	0.48 (0.01)
β_2^p : Partner's Signal	0.43 (0.02)	0.42 (0.01)	0.46 (0.01)
Constant	9.56 (0.04)	9.55 (0.03)	9.61 (0.03)
N	5040	7485	7435
$\beta_2^p - \beta_1$	-0.09*** (0.02)	-0.10*** (0.02)	-0.02 (0.02)
p -value: $\beta_2^p - \beta_1$ same as in Informed		0.74	0.02
p -value: $\beta_2^p - \beta_1$ same as in Observe			0.00
<i>Panel B. Structural Estimates</i>			
β_1 : Own Signal	0.49 (0.04)	0.51 (0.04)	0.46 (0.03)
β_2^p : Partner's Signal	0.34 (0.03)	0.34 (0.03)	0.36 (0.03)
$\beta_2^p - \beta_1$	-0.15*** (0.04)	-0.17*** (0.03)	-0.10*** (0.03)
p -value: $\beta_2^p - \beta_1$ same as in Informed		0.52	0.25
p -value: $\beta_2^p - \beta_1$ same as in Observe			0.04
N	5040	7485	7435

Notes: This table shows reduced-form and structural estimates for rounds in Experiment 3 (the online experiment).

Reduced-form coefficients Panel A shows reduced-form results, estimating Equation 1 by OLS. The dependent variable is participants' private guess. "Informed" refers to the *Informed of Partner's Draws* round and "Observe" to the *Observe Partner's Draws* round. "Own Signal" indicates the net number of red draws (i.e., red draws minus white draws) in the participant's own set of draws. Similarly, "Partner's Signal" indicates the net number of red draws in their partner's set of draws. Randomization was between participants in this experiment so we estimate the equation separately for each treatment condition. Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

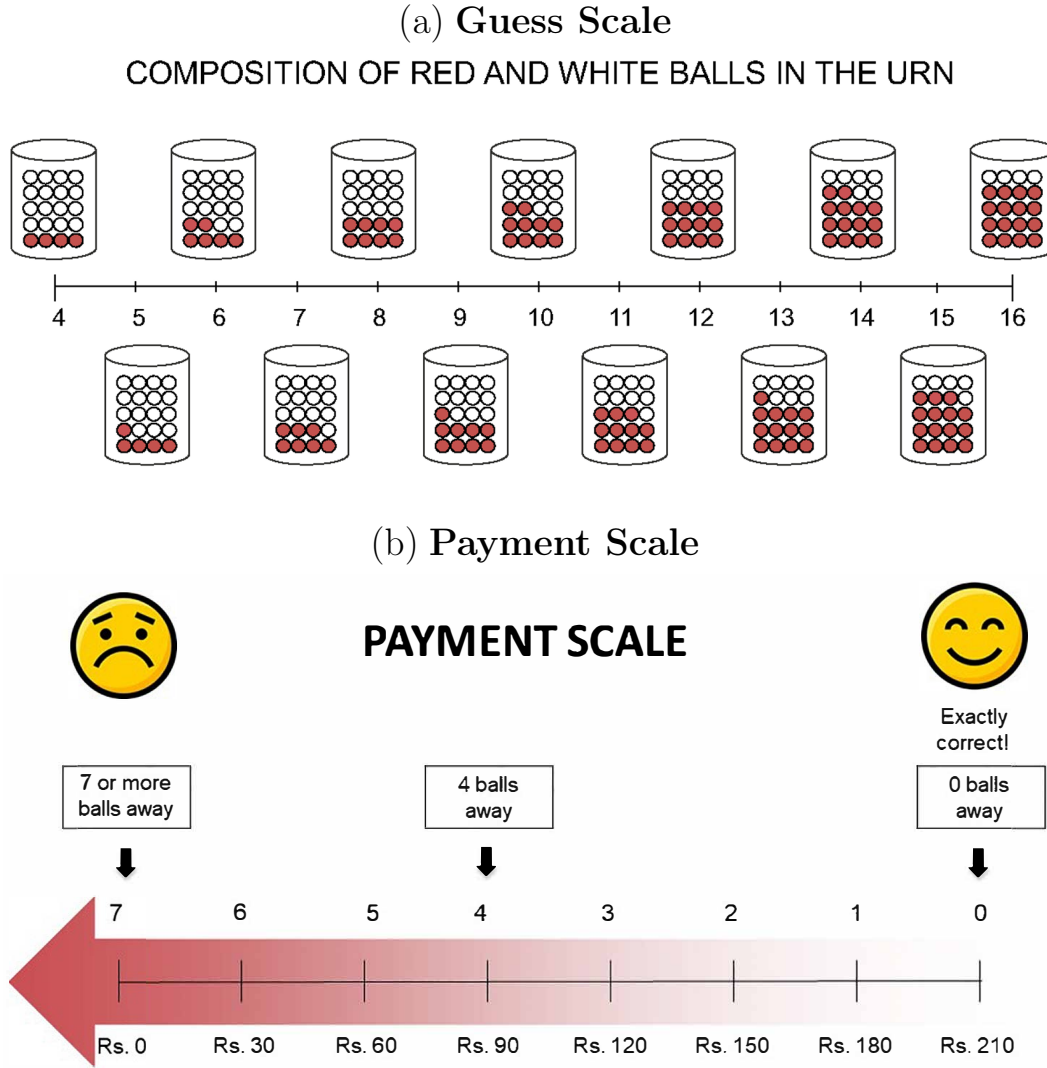
Structural parameters: Panel B shows estimates of the structural model described in Section 3.3. "Own Signal" and "Partner's Signal" indicate the weights placed on their own and their partner's set of draws in the agents' quasi-Bayesian updating rule. Bootstrapped standard errors (clustered at the pair level) in parentheses. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Not Learning from Others: Online Appendix

A Supplementary Figures and Tables

A.1 Supplementary Figures

Figure A.I: Visual Aids



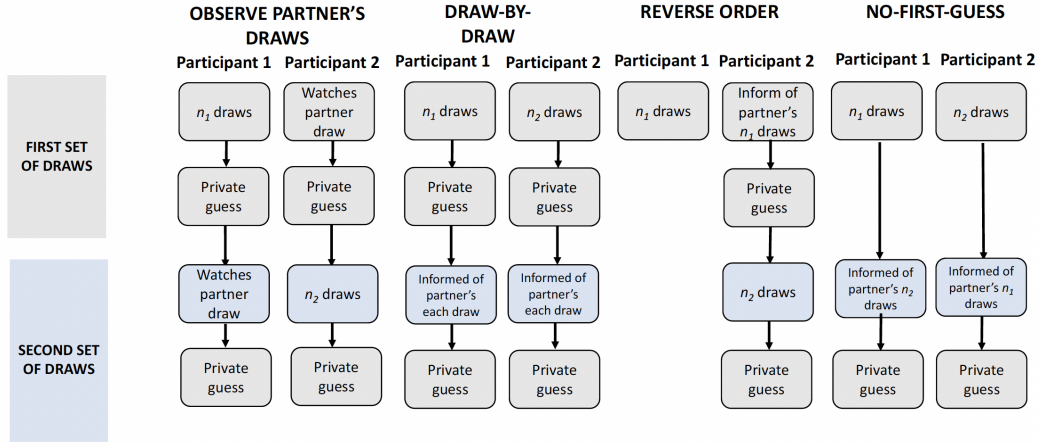
Notes: This figure shows the visual aids used to explain the experiment to study participants in Experiment 1 and Experiment 2.

Panel A: The figure shows the scale which participants used to make their guesses. It shows the 13 possible urn compositions ranging from 4 to 16 red balls (among 20 balls in total). We induced common priors: participants were informed that in each round, each of these compositions was equally likely (probability $1/13$ each). Participants guessed by placing a small token on top of the corresponding number.

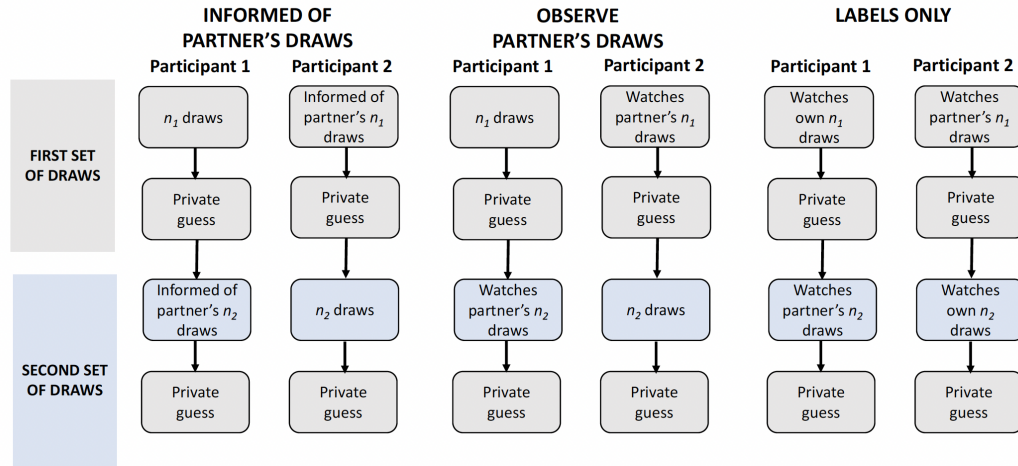
Panel B: The figure shows the scale used to explain the incentives for accurate guessing to participants. For each pair of participants, one of their guesses was randomly selected to determine the pair's payment. In Experiments 1 and 2, on top of their participation fee, each individual receives an amount in Rupees (Rs.) equal to $\max\{(105 - 15 \times |g - r|), 0\}$, where g is the guess and r the true number of red balls for the randomly-selected guess. See more detail in Section 2.

Figure A.II: Experimental Design for Experiment 2 and 3

Panel A: Experiment 2



Panel B: Experiment 3

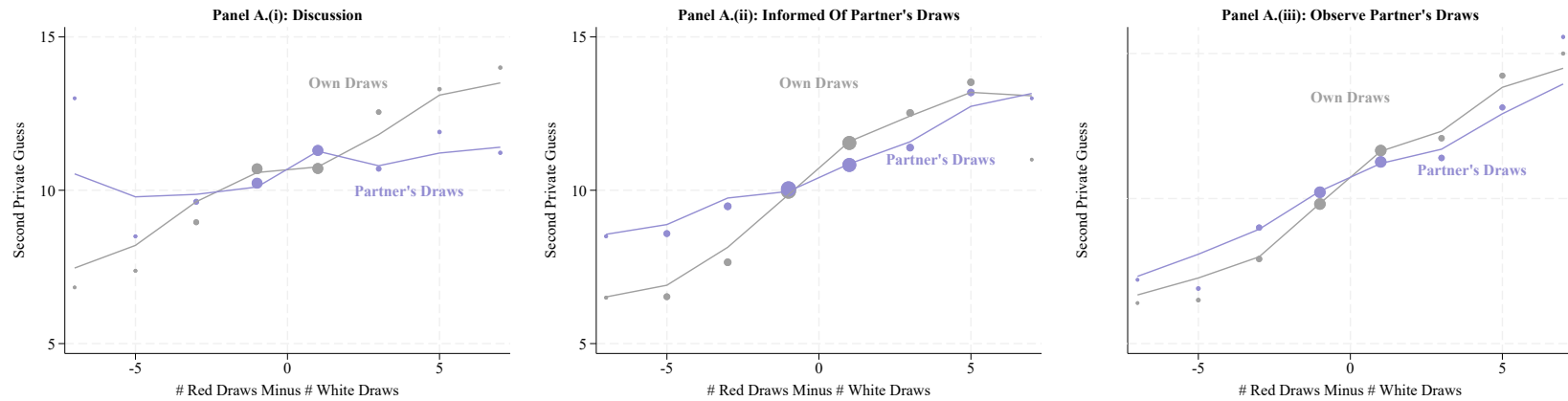


Panel A describes the structure of the different rounds in Experiment 2. In addition to the *Discussion* and *Informed of Partner's Draws* rounds, participants played four variations of the *Informed of Partner's Draws* round. In the *Observe Partner's Draws* round, each participant makes one set of draws while their partner is present, followed by a private guess after each set of draws. The *Draw-by-Draw* round is the same as the *Informed of Partner's Draws* round except each participant is informed about their partner's draws one draw at a time. In the *Reverse-Order* round, one participant learns about their partner's draws first and makes a private guess, and then makes their own set of draws and makes another private guess. In this round, the treatment is only for one participant from the pair. The *No-First-Guess* round is the same as the *Informed of Partner's Draws* round except participants only make one private guess after both sets of draws.

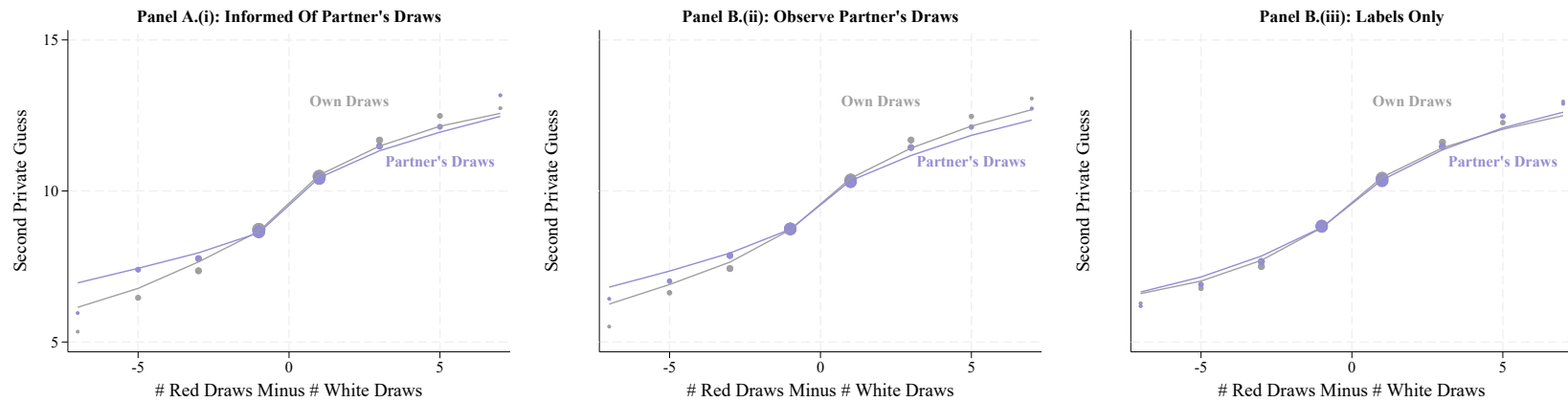
Panel B describes the structure of the different rounds in Experiment 3. In this experiment, the participant's own information and partner's information was presented in the Qualtrics survey using a virtual urn. In the *Informed of Partner's Draws* round, each participant makes one set of draws followed by a private guess. They are informed of their partner's draws and asked to make another private guess. In this experiment, participants played two additional variations of the *Informed of Partner's Draws* round. The *Observe Partner's Draws* round is the same as the *Informed of Partner's Draws*, except each participant watches their partner's draws, followed by a private guess after each set of draws. The *Labels Only* round is identical to the *Informed of Partner's Draws* round, except participants watch both their own draws and their partner's draws, and make a private guess after each set of draws.

Figure A.III: Experiments 2 and 3: Non-Parametric Estimates

Panel A: Experiment 2

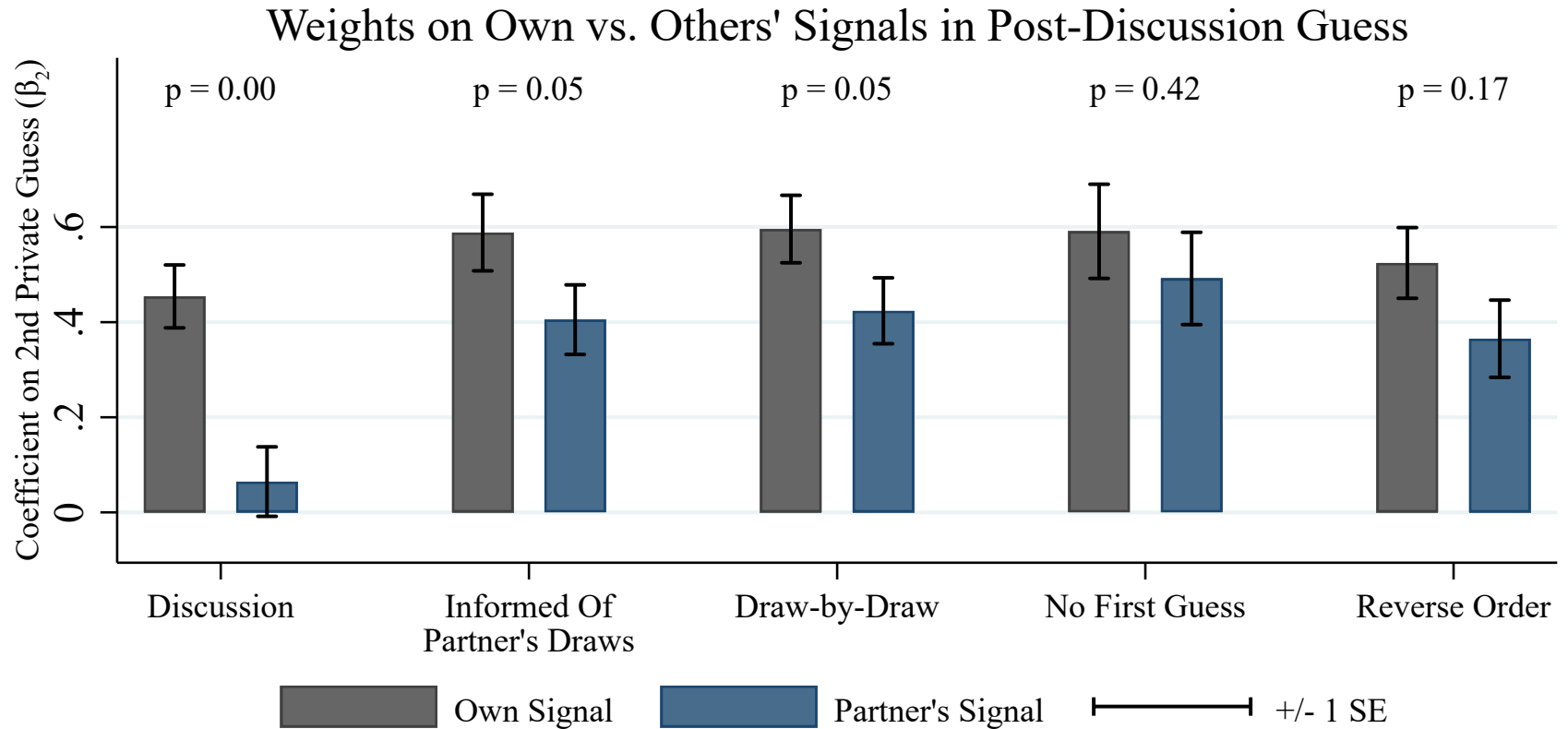


Panel B: Experiment 3



Notes: This figure shows the average second private guesses of participants in Experiment 2 and 3. In each graph, we plot this first against the participant's own signal (unconditional on partner's signal) in gray, and then again against their partner's signal (unconditional on own signal), in blue. The x-axis shows the net number of red draws (i.e. red draws minus white draws) in a given signal. Dots indicate average guesses, with dot size indicating number of observations, while the solid curves show locally weighted means (lowess). Because the signals are symmetrically distributed, equal weighting of own and others' information would imply the two curves should be equally steep. **Panel A** shows the average second private guess in Experiment 2. We show this for A.(i) *Informed of Partner's Draws*, where participants receive the second set of draws directly from the experimenter (and the second guess is before any discussion with their partner); and A.(ii) *Observe Partner's Draws*, where participants watch their partner drawing from the urn. **Panel B** shows the average second guess of participants in Experiment 3. We show results for: B.(i) *Informed of Partner's Draws*, where participants are given a summary of their partner's draws; B.(ii) *Observe Partner's Draws*, where participants watch their partner's draws appear from the urn; and B.(iii) *Labels Only*, where participants passively watch their own as well as their partner's draws appear from the urn.

Figure A.IV: Experiment 2: Reduced-Form Estimates

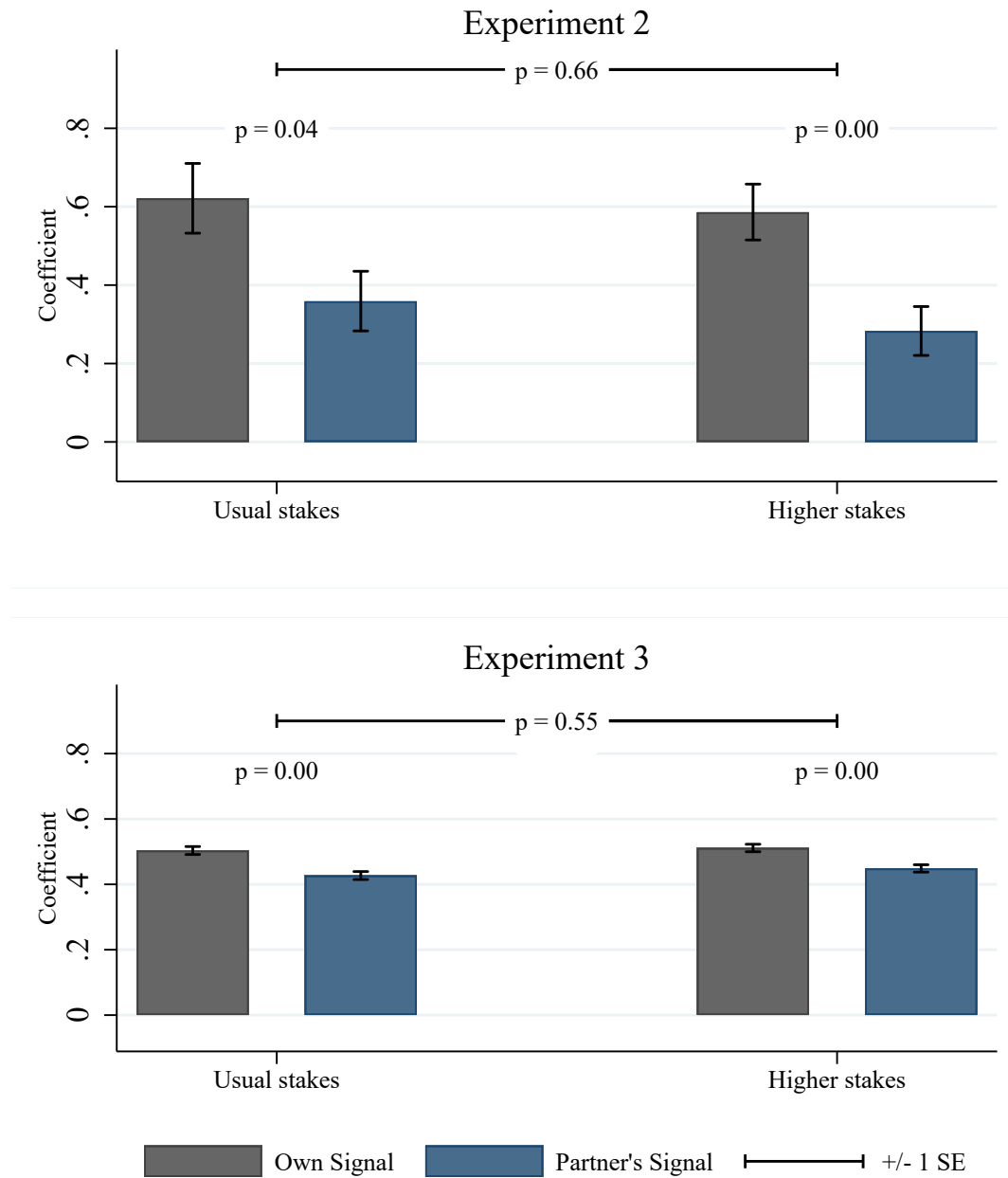


Notes: This figure shows the weights participants put on different signals when making their post-discussion private guess in Experiment 2. We estimate equation (1) and then display β_1 in gray and β_2 in dark blue for each treatment, except the *Observe Partner's Draws* round (in which there was no discussion and thus no post-discussion guess). In addition to the *Discussion* and *Informed of Partner's Draws* rounds, we look at the following treatments:

- (a) *Draw-by-Draw*, in which participants receive the second set of signals directly one draw at a time;
- (b) *No First Guess*, in which participants receive their partner's signals (and their own) before making their first and only private guess;
- (c) *Reverse Order*, in which one participant receives their partner's signals first and makes their first private guess, and then receives their own signals and makes their second private guess.

For each round, we show the p -value of testing whether the weight on their signal (β_1) equals the corresponding weight on their partner's signal (β_2) in that round.

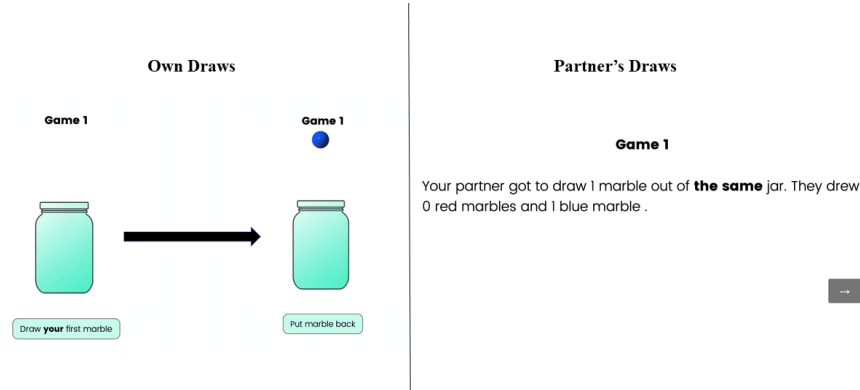
Figure A.V: Weights on Own vs. Others' Signals under Usual vs. Higher Stakes



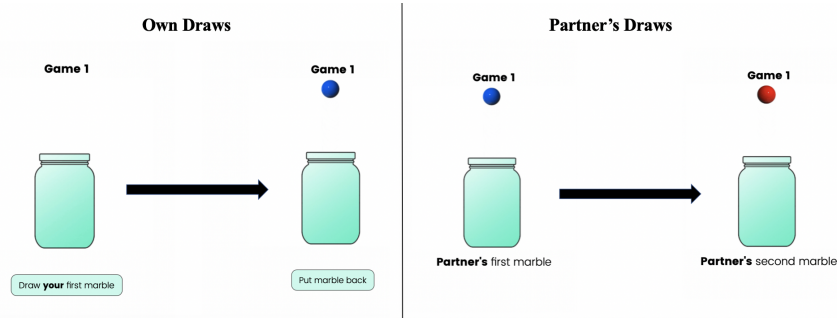
Notes: This figure shows OLS estimates of equation 1 in Experiments 2 and 3, pooling the different treatments, separately by whether participants faced lower or higher stakes (incentives). Above each pair of bars, we show the p -value of testing whether the weight on own information (gray) equals the weight on partner's information (dark blue). The higher, centered p -value in each graph is the p -value of testing whether the difference in weights is the same in the usual and the high stakes condition. In both experiments, we cannot reject that it is.

Figure A.VI: Visual Presentation of Draws in Experiment 3

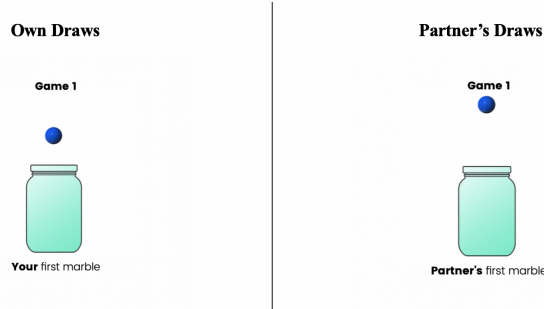
(a) Informed Of Partner's Draws



(b) Observe Partner's Draws

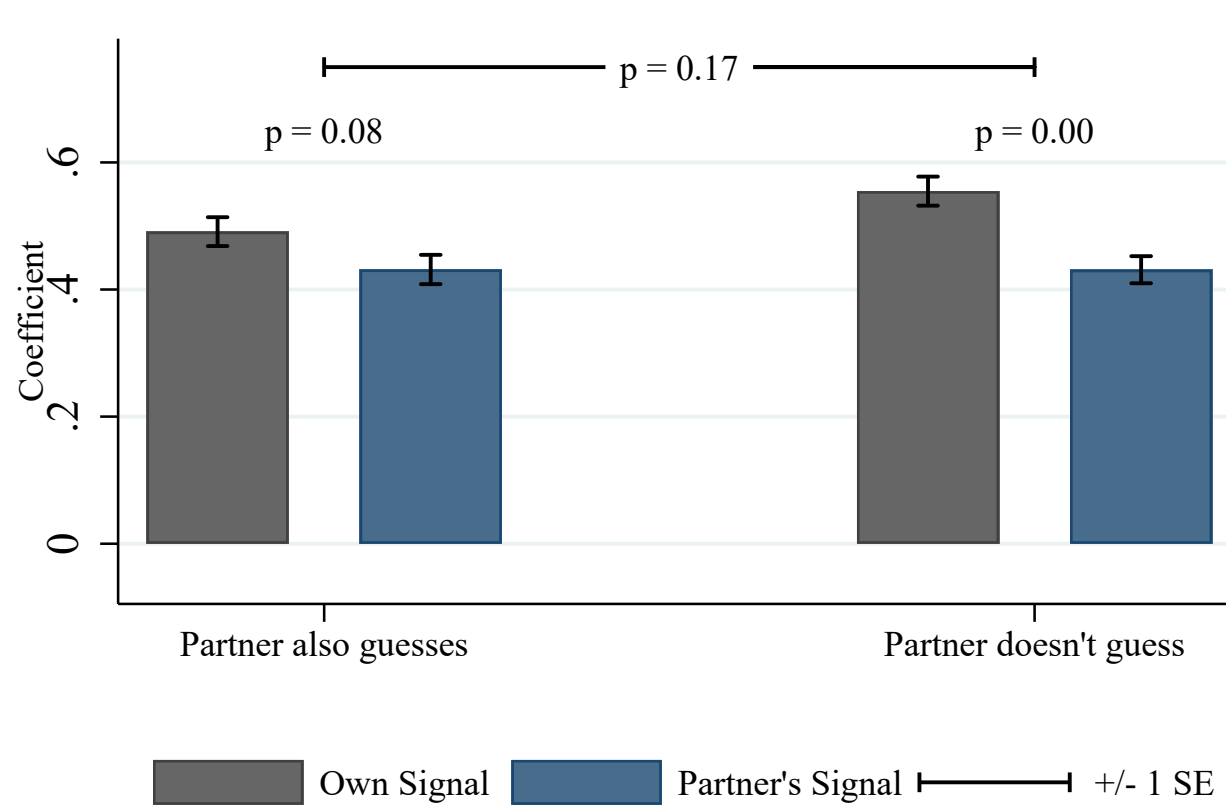


(c) Labels Only



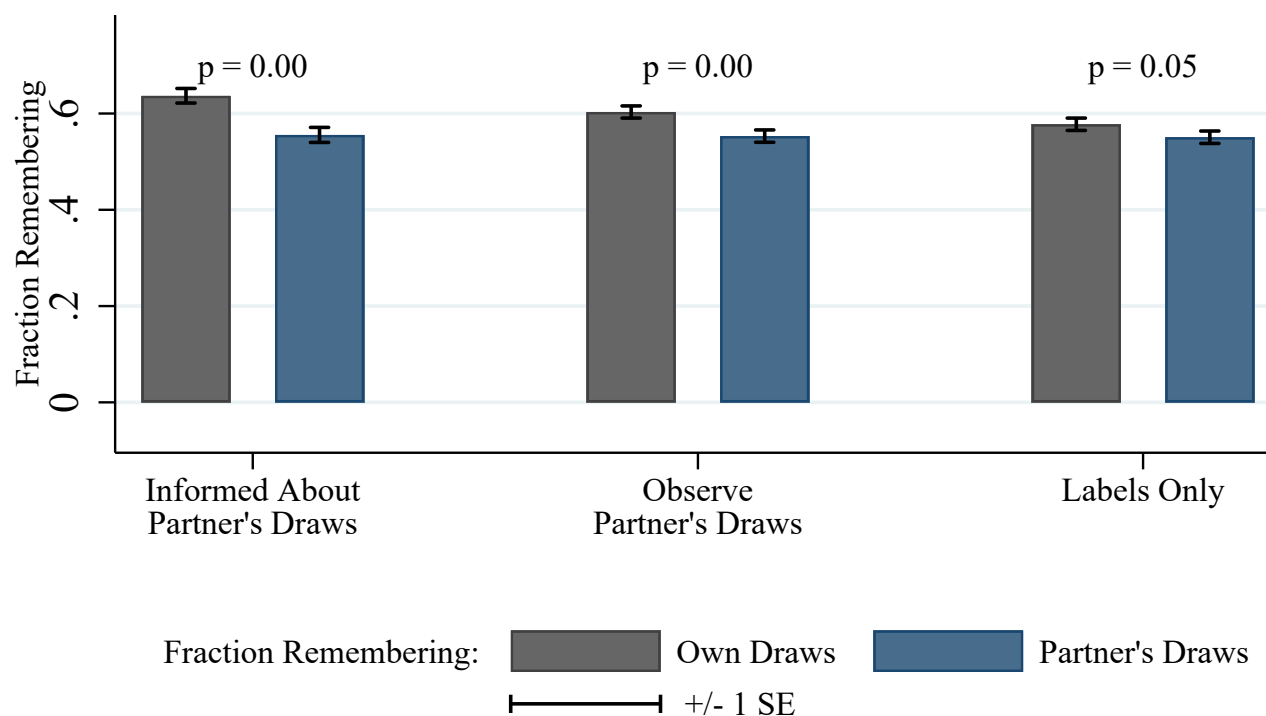
Notes: This figure shows how the participant's own information and partner's information was presented in the Qualtrics survey for the different treatments in Experiment 3. The left panel of the figure shows how their own information was presented, and the right panel shows how their partner's information was presented. The arrows indicate subsequent screens. In all treatments, we emphasized that own and partner's draws were made from the same urn. **Panel A** shows how draws were presented in the *Informed of Partner's Draws* treatment. To obtain their own draws, participants clicked to draw balls one by one from a virtual urn, and after each ball was shown, clicked again to put it back in the urn, which was then animated to shuffle. In contrast, participants learned their partner's draws in summary form as shown in the right part of the panel. **Panel B** shows how draws were presented in the *Observe Partner's Draws* treatment. Participants obtained their own draws in exactly the same way as in the *Informed of Partner's Draws* round. For their partner's draws, participants were shown the same virtual urn and saw their partner's draws being revealed by the same ball-by-ball animation. However, the draws appeared one by one *without* clicking on the urn to obtain them. **Panel C** shows how draws were presented in the *Labels Only* treatment. The participants were shown a virtual urn and saw their own draws revealed by the same ball-by-ball animation, without having to click. Their partner's draws were revealed in exactly the same way.

Figure A.VII: Experiment 3: Reduced-Form Estimates – Guessing vs. Non-guessing Partner in the *Informed* Treatment



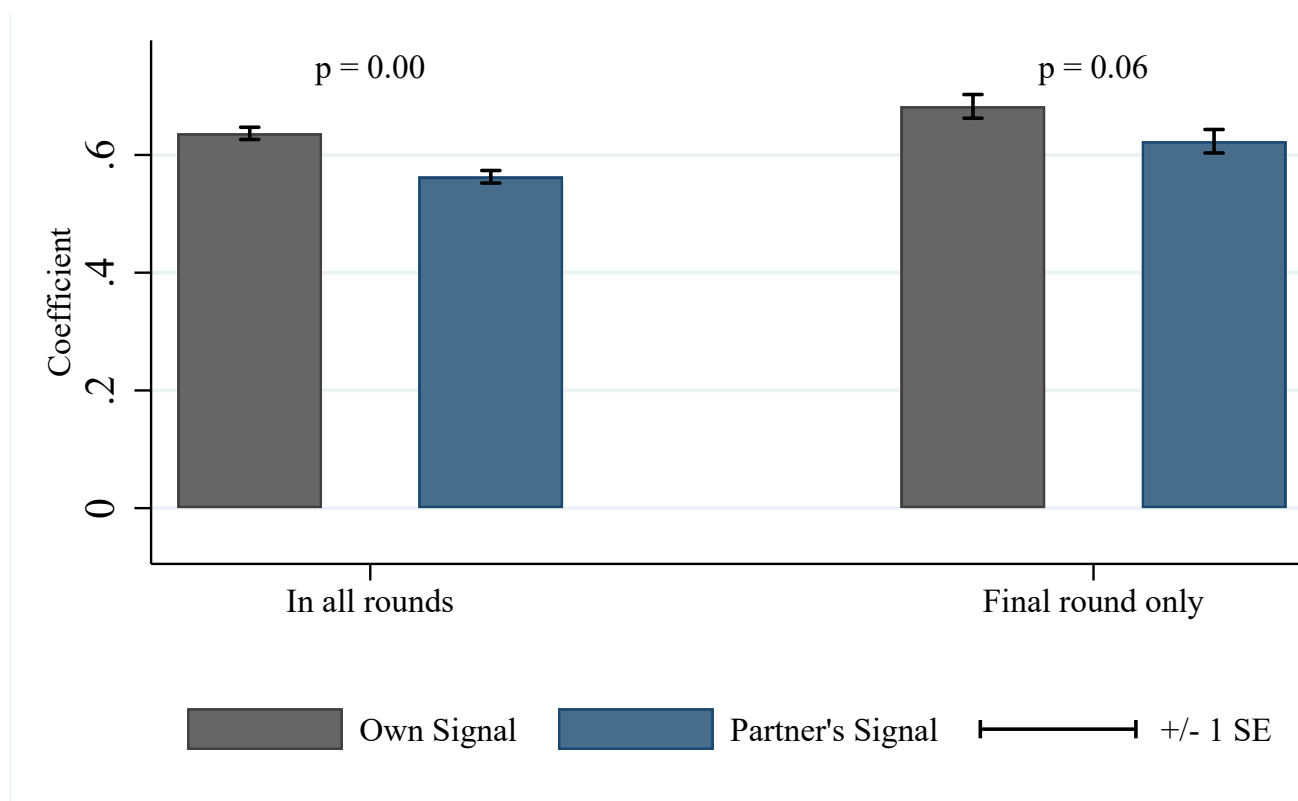
Notes: This figure shows OLS estimates of equation 1 for participants in the *Informed of Partner's Draws* treatment in Experiment 3, separately by whether they were also assigned to the *Non-Rivalry* treatment, i.e., by whether their partner also guessed ('Guessing partner') or not ('Non-guessing partner'). For each of the dark-blue bars, we show the p -value of testing whether the weight on own information (gray bar) equals the weight on partner's information (blue bar). The higher, centered p -value is the p -value of testing whether the difference in weights is the same across the two treatments. We cannot reject that it is. See Table A.VIII for the underlying numbers displayed in this figure.

Figure A.VIII: Memory of Own vs. Others' Signals in Experiment 3

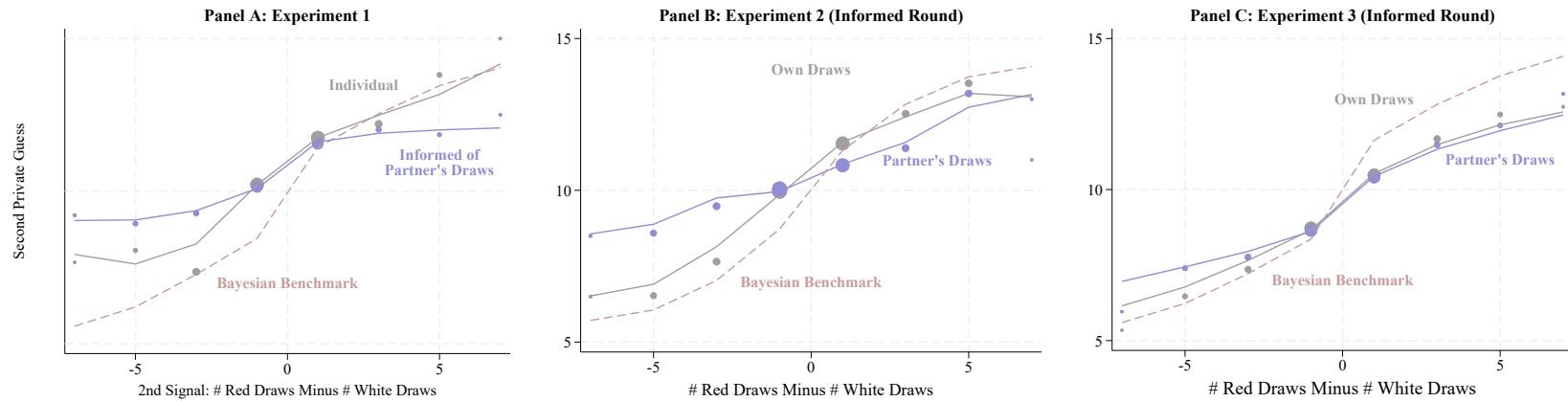


Notes: Participants in Experiment 3 were asked at the end of their final round (out of 5) if they remembered their own and their partner's draws—both number and color composition—from that round. The question was unannounced and unincentivized. This figure shows the fraction of participants correctly remembering their own (gray bar) versus their partner's (blue bar) draws in each treatment in Experiment 3. For each pair of bars, we show the p -value of testing that the same fraction remembered their own draws as remembered their partner's draws. See Table A.IX for the underlying numbers used in this figure.

Figure A.IX: Weights on Own vs. Others' Signals for Participants Who Remember All Draws (Experiment 3)



Notes: Participants in Experiment 3 were asked at the end of their final round (out of 5) if they remembered their own and their partner's draws from that round. This question was unannounced and unincentivized. This figure shows OLS estimates of equation 1 in Experiment 3 for participants who correctly remembered both, pooling across treatments. The left pair of bars shows these participants' weights pooling all five rounds and the right set their weights in the final round only (i.e., the round for which they correctly remembered). Above each pair of bars, we show the p -value of testing whether the weight on own information (gray) equals the weight on partner's information (dark blue).

Figure A.X: Guesses in the *Informed* round vs Bayesian Benchmark: Non-Parametric Estimates

Notes: This figure shows the average second private guesses of participants in the *Individual* and *Informed of Partner's Draws* rounds of Experiment 1 and the *Informed of Partner's Draws* rounds of Experiment 2 and 3. For Experiment 1, we plot the guess in the *Individual* round against the participant's own second signal in gray and the guess in the *Informed of Partner's Draws* round against their partner's signal in blue. For Experiments 2 and 3, we plot the guess against the participant's own signal (unconditional on partner's signal) in gray, and then again against their partner's signal (unconditional on own signal), in blue. The x-axis shows the net number of red draws (i.e. red draws minus white draws) in a given signal. Dots indicate average guesses, with dot size indicating number of observations, while the solid curves show locally weighted means (lowess). Because the signals are symmetrically distributed, equal weighting of own and others' information would imply the two curves should be equally steep. The dotted lines show the average of a risk-neutral Bayesian's guess given the same signals.

A.2 Supplementary Tables

Table A.I: Comprehension and Memory

Question	Experiment 1	Experiment 2
<i>A. Basic Design</i>		
Number of balls	0.97	0.99
Colors of balls	1.00	1.00
<i>B. Common Prior</i>		
Possible < 4 red	0.93	0.96
Possible > 16 red	0.93	0.97
Who chooses number of red balls	0.81	0.84
Likelihood of each number	0.78	0.61
<i>C. Signals</i>		
Learn more from more balls	0.88	0.88
Possible have 4 draws	0.76	0.69
How number draws differs	0.47	0.47
How partner’s draws differ	0.61	0.63
<i>D. Incentives</i>		
Payment if 1 off	0.90	0.95
Payment if way off	0.85	0.92
Payment if 4 off	0.92	0.93
<i>E. Memory</i>		
Correctly remembered own guess	0.92	
Correctly remembered # of own draws	0.97	0.96
Correctly remembered # of own red draws	0.85	0.80
Correctly remembered # of partner’s draws		0.89
Correctly remembered # of partner’s red draws		0.70

Notes: This table shows summary statistics of participants’ comprehension of the task and their memory of previous draws and guesses. Column 1 shows the sample of 500 individuals of Experiment 1; and column 2 shows the sample of 292 individuals of Experiment 2. Panels A through D show the fraction of participants who answered each question correctly. For each question, we corrected the participant if they gave a wrong answer. Panel E shows the fraction of people who correctly remembered their own and their partner’s in some of the rounds.

- **Panel A** shows answers to questions “How many balls are in the urn?” (correct answer: 20), and “What colors are the balls?” (red and white).
- **Panel B** “Is it possible to have less than 4 /more than 16 red balls?” (no); “Who chooses how many balls are red?” (the computer), and “Are some numbers more likely than others?” (no).
- **Panel C** “Do you learn more from one draw or five draws?” (five); “Can you get exactly 4 draws in any round?” (no); “Will you have the same or different numbers of draws across rounds?” (could be same or different); “Will your partner have the same or different number to you?” (could be same or different).
- **Panel D** shows the fraction of people who could correctly indicate their payment on the scale if their guess was 1, 11, or 4 balls off.
- **Panel E** shows the proportion of participants who correctly remember their own guess and draws. “Correctly remembered own guess” correspond to the fraction of people who correctly remember their own guess in the *Informed of Partner’s Guess* round of Experiment 1. “Correctly remembered # of own draws” and “Correctly remembered # of own red draws” correspond to the fraction of people who correctly remember their own draws in the in the *Informed of Partner’s Draws* round in Experiment 1, and correspond to results pooled across 4 rounds, including the *Observe Partner’s Draws* round in Experiment 2. “Correctly remember # of partner’s draws” and “Correctly remembered # of partner’s red draws” correspond to the *Observe Partner’s Draws* round in Experiment 2.

Table A.I: (continued) Comprehension - Experiment 3

	Experiment 3
Goal of task	0.81
Number of balls	0.93
Possible numbers of red balls	0.97
Playing with partner	0.88
Drawing with replacement	0.90
Same urn as partner	0.91
Urn re-randomized across rounds	0.93
Incentive scheme	0.99

Notes: This continues Table A.I, showing summary statistics of participants' comprehension of the task in Experiment 3. Participants were asked 8 multiple-choice questions; if they got a question wrong, they had to retry until they got it right (they could re-read the relevant instruction). Shown are the fraction of participants answering each question correctly first time. The questions are shown below, with the correct answer in brackets.

- Goal of task – "What is the goal of the game you are playing today?" (To guess the number of red marbles in a virtual jar)
- Number of balls – "How many marbles are in the jar total?" (20)
- Possible numbers of red balls – "And how many red marbles could possibly be in the jar?" (Between 4 and 16 red marbles)
- Playing with a partner – "Who are you playing this game with?" (a real person who is taking the survey at about the same time with me). Note that in the *Non-rivalry* treatment, the correct answer was 'A real partner who is taking the survey at around the same time as me but doing a different task than what I'm doing'.
- Drawing with replacement – "Which of the following statements is correct: After each draw, the marble is not put back in the jar / After each draw, the marble gets put back and the contents get shuffled" (After each draw, the marble gets put back and the contents get shuffled)
- Same urn as partner – "Which of these statements is correct: My partner and I are drawing marbles from the same jar with the same number of red marbles / ... different number of red marbles / I am drawing marbles from the jar, and my partner is not / My partner is drawing marbles from the jar, and I am not" (My partner and I are drawing marbles from the same jar with the same number of red marbles)
- Urn re-randomized across rounds – "Which of these statements is correct: I will only play this game once / I will play this game 5 times with the contents of the jar always being the same / I will play this game 5 times with the contents of the jar being re-randomized each time" (I will play this game 5 times with the contents of the jar being re-randomized each time)
- Incentive scheme – "How can you affect the outcome of your bonus payment?" (For a randomly chosen guess, the closer I was to the true number of red marbles in the jar, the higher is my bonus)

Table A.II: Transcripts of Joint Discussions: Summary Statistics

	Experiment 1
Any information shared	0.83
Anyone Shared Guess	0.54
Anyone Shared Number of Draws	0.36
Anyone Shared Composition of Draws	0.48
Length of Discussion (seconds)	47.89

Notes: This table shows averages of key characteristics of the discussions in Experiment 1. These variables were constructed using transcripts of the discussions between participants before the joint guesses were made. Variables are measured at the discussion (pair) level.

- We pool the discussions from the two *Discussion* rounds in Experiment 1, excluding those with same-gender pairs. Accurate transcript data were not collected for same-gender discussions due to challenges in identifying the two participants.
- ‘Anyone Shared Guess’ equals one if either person shared their guess and ‘Anyone Shared Number of Draws’ equals one if either person shared their number of draws. ‘Anyone Shared Composition of Draws’ equals one if either person shared the specific color composition of their draws (e.g. “I drew 4 red balls and 1 white ball”) or mentioned that they drew more of one color (“I drew more red balls than white”). “Any information shared” equals one if either person shared their guess, number of draws, or composition of draws.
- For each variable, we drop observations for which the variable could not be recorded due to an inaudible or unclear recording. This affected 14% of the sample for Length of Discussion and 46% of the sample for each of the other variables in the table.

Table A.III: Experiment 1: Error in Guess by Type of Guess and Number of Draws

	Individual	Discussion	Informed (Pre)	Informed (Post)
	(1)	(2)	(3)	(4)
γ_1 : # Own First Draws	-0.13 (0.06)	-0.08 (0.04)	-0.08 (0.05)	-0.02 (0.05)
γ_2^o : # Own Second Draws	-0.16 (0.06)			
γ_2^p : # Partner's Draws		-0.05 (0.05)	0.03 (0.06)	0.00 (0.06)
Constant	4.12 (0.46)	3.55 (0.33)	3.37 (0.46)	3.14 (0.46)
$\gamma_2^p - \gamma_2^o$		0.11 (0.08)	0.19** (0.08)	0.16* (0.08)
N	500	1000	500	500

Notes: This table compares the error in participants' guesses (the absolute difference between their guess and the true number of red balls in the urn) in the *Discussion*, *Informed of Partner's Draws*, and *Individual* round. The table shows OLS estimates of the following equation for the *Discussion* and *Informed of Partner's Draws* rounds:

$$|Guess - Truth|_{irt} = \alpha + \gamma_1 \# Own First Draws_i + \gamma_2^p \# Partner's Draws_i + \epsilon_i \quad (7)$$

and OLS estimates of the following equation for the *Individual* round:

$$|Guess - Truth|_{irt} = \alpha + \gamma_1 \# Own First Draws_i + \gamma_2^o \# Own Second Draws_i + \epsilon_i \quad (8)$$

where $|Guess - Truth|_{irt}$ is the absolute value of difference between i 's guess and the true number of red balls in the urn in the round in question, and $\# Own First Draws_i$ indicates the number of draws in the first set of signals, drawn oneself. $\# Own Second Draws$ is the number of draws in the participant's second set in the Individual round and $\# Partner's Draws$ is the participant's partner's number of draws, in the *Discussion* and *Informed of Partner's Draws* rounds. In estimation, we stack the estimating equations for all treatment and estimate them jointly including controls for round order fixed effects. Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\gamma_2^p - \gamma_2^o$.

Table A.IV: Weights on Own vs. Others' Signals under Usual vs. Higher Stakes

	Usual Stakes (1)	Higher Stakes (2)
<i>Panel A: Experiment 2</i>		
β_1 : Own Signal	0.62 (0.09)	0.59 (0.07)
β_2^p : Partner's Signal	0.36 (0.08)	0.28 (0.06)
Constant	10.47 (0.12)	10.44 (0.11)
$\beta_2^p - \beta_1$	-0.26** (0.13)	-0.30*** (0.11)
<i>p</i> -value: $\beta_2^p - \beta_1$ equal across treatments		0.66
<i>N</i>	1602	1602
<i>Panel B: Experiment 3</i>		
β_1 : Own Signal	0.50 (0.01)	0.51 (0.01)
β_2^p : Partner's Signal	0.43 (0.01)	0.45 (0.01)
Constant	9.56 (0.03)	9.58 (0.03)
$\beta_2^p - \beta_1$	-0.08*** (0.02)	-0.06*** (0.02)
<i>p</i> -value: $\beta_2^p - \beta_1$ equal across treatments		0.55
<i>N</i>	9770	10190

Notes: This table shows OLS estimates of Equation 1 separately by whether participants faced usual or higher stakes (incentives). This table reports the same estimates as Figure A.V. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table A.V: Experiment 3: Sensitivity of First Guesses to Own vs Other’s Signal

	Informed (1)	Observe (2)	Labels Only (3)
β_1 :First Signal $\cdot \mathbb{1}(\text{Own})$	0.66 (0.02)	0.67 (0.02)	0.61 (0.02)
β_2^p :First Signal $\cdot \mathbb{1}(\text{Partner's})$	0.55 (0.02)	0.59 (0.02)	0.63 (0.02)
Constant	9.45 (0.04)	9.52 (0.03)	9.64 (0.03)
$\beta_2^p - \beta_1$	-0.11*** (0.03)	-0.07*** (0.03)	0.02 (0.03)
p -value: $\beta_2^p - \beta_1$ same as in Informed		0.34	0.00
p -value: $\beta_2^p - \beta_1$ same as in Observe			0.02
N	5040	7485	7435

Notes: This table shows reduced-form results, estimating the following equation by OLS:

$$FirstGuess_i = \alpha + \beta_1 \cdot First\ Signal_i \cdot \mathbb{1}(\text{Own}) + \beta_2^p \cdot First\ Signal_i \cdot \mathbb{1}(\text{Partner's}) + \epsilon_i$$

where the dependent variable $FirstGuess_i$ is participant i 's *first* private guess (before seeing the second signal). $First\ Signal_i$ indicates the net number of red draws (i.e., red draws minus white draws) in the first signal that the participant saw, $\mathbb{1}(\text{Own})$ is a dummy variable indicating whether this was i 's own signal, and $\mathbb{1}(\text{Partner's})$ is a dummy variable indicating whether this was i 's partner's signal. "Informed" refers to the *Informed of Partner's Draws* round and "Observe" to the *Observe Partner's Draws* round. Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table A.VI: Experiment 3: Sensitivity of First Guesses to Own vs Other's Signal in Round 1

	Informed (1)	Observe (2)	Labels Only (3)
β_1 :First Signal $\cdot \mathbb{1}(\text{Own})$	0.66 (0.04)	0.65 (0.03)	0.63 (0.04)
β_2^p :First Signal $\cdot \mathbb{1}(\text{Partner's})$	0.42 (0.05)	0.52 (0.04)	0.65 (0.04)
Constant	9.48 (0.07)	9.53 (0.06)	9.73 (0.06)
N	1008	1497	1487
$\beta_2^p - \beta_1$	-0.24*** (0.07)	-0.14** (0.05)	0.02 (0.05)
p -value: $\beta_2^p - \beta_1$ same as in Informed			
p -value: $\beta_2^p - \beta_1$ same as in Observe			
N	1008	1497	1487

Notes: This table shows reduced-form results, estimating the following equation by OLS:

$$FirstGuess_i = \alpha + \beta_1 \cdot First\ Signal_i \cdot \mathbb{1}(\text{Own}) + \beta_2^p \cdot First\ Signal_i \cdot \mathbb{1}(\text{Partner's}) + \epsilon_i$$

where the dependent variable $FirstGuess_i$ is participant i 's *first* private guess (before seeing the second signal) in the first round of the experiment. $First\ Signal_i$ indicates the net number of red draws (i.e., red draws minus white draws) in the first signal that the participant saw, $\mathbb{1}(\text{Own})$ is a dummy variable indicating whether this was i 's own signal, and $\mathbb{1}(\text{Partner's})$ is a dummy variable indicating whether this was i 's partner's signal. "Informed" refers to the *Informed of Partner's Draws* round and "Observe" to the *Observe Partner's Draws* round. Standard errors are clustered at the pair level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table A.VII: Heterogeneity

	Comprehension:		Education:		Performance Belief:	
	Below median (1)	Above median (2)	Below median (3)	Above median (4)	Below median (5)	Above median (6)
Panel A: Experiment 1						
β_1 : Own Info	0.48 (0.04)	0.51 (0.04)	0.41 (0.04)	0.58 (0.04)	0.47 (0.04)	0.44 (0.08)
β_2^p : Partner's Info	0.25 (0.04)	0.33 (0.04)	0.28 (0.04)	0.30 (0.04)	0.28 (0.04)	0.29 (0.06)
$\beta_2^p - \beta_1$	-0.23*** (0.06)	-0.18*** (0.06)	-0.13** (0.06)	-0.28*** (0.06)	-0.18*** (0.06)	-0.16 (0.10)
p -val.: $\beta_2^p - \beta_1$ equal	0.57		0.11		0.84	
Panel B: Experiment 2						
β_1 : Own Info	0.48 (0.06)	0.57 (0.05)	0.48 (0.06)	0.59 (0.05)		
β_2^p : Partner's Info	0.31 (0.04)	0.34 (0.06)	0.24 (0.04)	0.39 (0.05)		
$\beta_2^p - \beta_1$	-0.17** (0.07)	-0.23*** (0.08)	-0.23*** (0.08)	-0.20*** (0.07)		
p -val.: $\beta_2^p - \beta_1$ equal	0.61		0.76			
Panel C: Experiment 3						
					Remember All Draws	Say Treat Same
β_1 : Own Info	0.45 (0.01)	0.54 (0.01)	0.48 (0.01)	0.53 (0.01)	0.68 (0.02)	0.53 (0.01)
β_2^p : Partner's Info	0.36 (0.01)	0.48 (0.01)	0.41 (0.01)	0.46 (0.01)	0.62 (0.02)	0.45 (0.01)
$\beta_2^p - \beta_1$	-0.08*** (0.02)	-0.06*** (0.01)	-0.08*** (0.02)	-0.06*** (0.02)	-0.06* (0.03)	-0.08*** (0.01)
p -val.: $\beta_2^p - \beta_1$ equal	0.37		0.61			

This table shows estimates of Equation 1 estimated on subsets of the data. Columns 1 and 2 show estimates by whether comprehension (the percentage of comprehension questions answered correctly first time) is above or below median. The median was 79% in Experiments 1 and 2, and 100% in Experiment 3 (so 'above median' means everyone who got all questions right). Columns 3 and 4 show estimates by whether years of education is above or below median. Columns 5 and 6 show in Panel A estimates by whether the guesser's belief about their own performance – specifically, how much they expected their guesses to earn on average – is above or below median. This was only asked about in Experiment 1. In Panel C, column 5 restricts the Experiment 3 data to the final round of the experiment and to participants who correctly remember both their own and their partner's draws (asked after the round ended), while column 6 restricts the Experiment 3 data to participants who answered in a debriefing question at the end of the survey that they “treated my draws and my partner's draws the same.” For each pair of columns, “*p*-val.: $\beta_2^p - \beta_1$ equal” is the *p*-value from testing the hypothesis that $\beta_2^p - \beta_1$ is the same in each subsample. The data pools all treatments except the *Individual* round in Experiment 1. Standard errors in parentheses. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table A.VIII: Experiment 3: Reduced-Form Estimates – Rivalry

	Partner also guesses (1)	Partner doesn't guess (2)
β_1 : Own Signal	0.49 (0.02)	0.55 (0.02)
β_2^p : Partner's Signal	0.43 (0.02)	0.43 (0.02)
Constant	9.54 (0.07)	9.59 (0.06)
$\beta_2^p - \beta_1$	-0.06* (0.03)	-0.12*** (0.03)
p-value: $\beta_2^p - \beta_1$ equal across treatments		0.17
N	2525	2515

Notes: This table shows OLS estimates of Equation 1 for participants in the *Informed of Partner's Draws* treatment in Experiment 3, separately by whether their partner also guessed or did not guess (the *Non-Rivalry* treatment). This table reports the same estimates as Figure A.VII. The bottom row shows the p -value of testing whether the difference in weights is the same across the two treatments. Standard errors in parentheses. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_1$.

Table A.IX: Memory of Own vs. Others' Draws in Experiment 3

	Informed (1)	Observe (2)	Labels Only (3)
Fraction Remembering:			
Own Draws	0.64 (0.02)	0.60 (0.01)	0.58 (0.01)
Partner's Draws	0.56 (0.02)	0.55 (0.01)	0.55 (0.01)
p-val.: Equal memory of own and partner's	0.00	0.00	0.05
<i>N</i>	2016	2994	2974

Notes: Participants in Experiment 3 were asked at the end of their final round (out of 5) if they remembered their own and their partner's draws – both number and color composition – from that round. This question was unannounced and unincentivized. This table shows the fraction of participants correctly remembering their own versus their partner's information in each treatment in Experiment 3. In each column we show the p -value of testing whether the fraction remembering own and partner's draws is the same within that treatment. Standard error of the mean in parentheses. This table reports the same estimates as Figure A.VIII.

B Supplementary Information

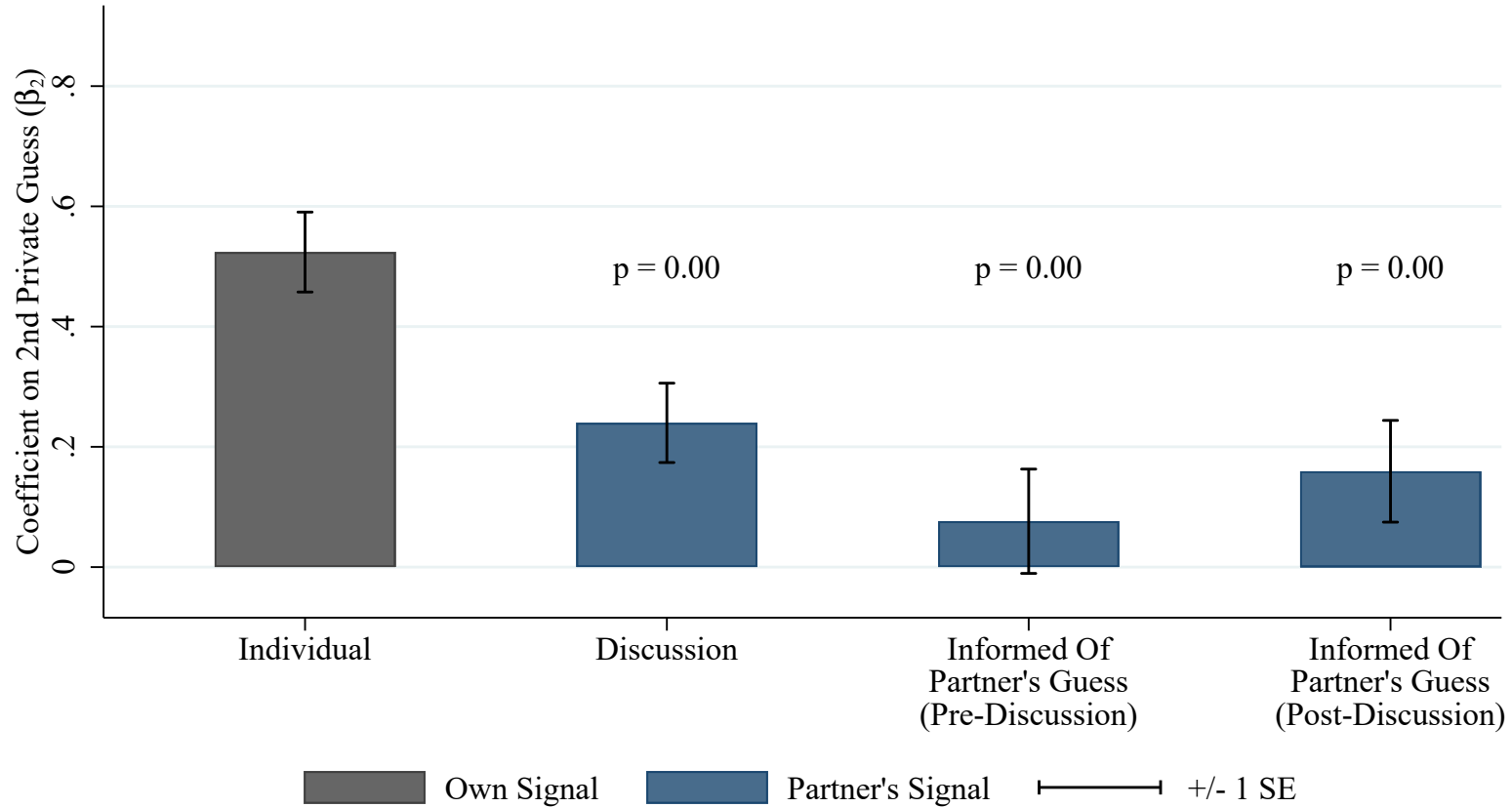
B.1 *Informed of Partner's Guess* Round in Experiment 1

The *Informed of Partner's Guess* round in Experiment 1 is identical to the *Informed of Partner's Draws* round, except that instead of sharing with each person the number of balls of each color their partner drew, the surveyor shares their partner's guess and the total number of draws (1, 5, or 9) on which that guess was based. Figure A.XI shows estimates for the *Individual*, *Discussion*, and *Informed of Partner's Guess* rounds. The results look similar to those for the *Informed of Partner's Draws* round. People strongly discount their partner's information relative to their own in both the pre-discussion and post-discussion guesses. This could be explained by differential processing of own compared to others' information, but also by other (potentially rational) reasons, such as mistrust of partners' guesses or the increased computational difficulty of backing out what the partner's information must have been given their guess. Table A.X shows the corresponding reduced-form and structural estimates, which confirm the visual impressions from Figure A.XI.³⁵

³⁵Note that the structural estimates assume that participants are able to back out from their partner's guess what their information must have been. Less weight on the partner's information could therefore reflect not just intrinsic discounting of others' information but also the extent to which this is a difficult problem for participants to solve (or one they do not attempt to solve).

Figure A.XI: Experiment 1: Reduced Form Estimates – Informed of Partner's Guess Round

Weights on Own vs. Others' Signals



Notes: This figure shows the weights participants put on different signals in Experiment 1. We estimate Equation (1) and then display β_2 for each of the following four types of private guesses:

- Individual*, where participants collect all information on their own. For this round, we replace *Partner's Info* in Equation (1) by the net red draws in the participant's second set of signals;
- Discussion*, in which participants collect the first set of information on their own and the second set (their partner's) is only accessible via discussion;
- Informed of Partner's Guess (pre-discussion)*, where participants have learned their partner's guess and number of draws directly from the experimenter but before any discussion with their partner;
- Informed of Partner's Guess (post-discussion)*, where participants have learned their partner's guess and number of draws *and* had the chance to discuss it with their partner.

For each of the dark-blue bars, we show the *p*-value of testing whether the weight in that round equals the corresponding weight in the *Individual* round (gray bar).

Table A.X: Reduced-Form Estimates in the Informed of Partner's Guess Round

	Individual	Discussion	Informed (Pre)	Informed (Post)
	(1)	(2)	(3)	(4)
β_1 : Own First Signal	0.43 (0.06)	0.56 (0.06)	0.50 (0.10)	0.40 (0.10)
β_2^o : Own Second Signal	0.52 (0.07)			
β_2^p : Partner's Signal		0.24 (0.07)	0.08 (0.09)	0.16 (0.08)
Constant	10.71 (0.16)	10.73 (0.16)	10.58 (0.22)	10.65 (0.22)
$\beta_2^p - \beta_2^o$		-0.28*** (0.08)	-0.45*** (0.10)	-0.36*** (0.10)
N	500	1000	500	500

Notes: This table shows reduced-form estimates of Equation 1 for the *Individual*, *Discussion* and *Informed of Partner's Guess* rounds in Experiment 1. The dependent variable is participants' private guess. 'Informed (Pre)' means the second private guess from the *Informed of Partner's Guess* round, after the participant was directly told their partner's guess but before the joint discussion. "Informed (Post)" means the third private guess, after the discussion. "Own First Signal" is the net number of red draws (i.e., red draws minus white draws) in the participant's first set of draws, which they drew themselves in all rounds. "Own Second Signal" is the net number of red draws in the participant's second set of draws in the individual round. "Partner's Signal" is the net number of red draws in the set of draws by the participant's partner, which was the second signal available to the participant in the *Discussion* and *Informed of Partner's Guess* rounds. All regressions include order fixed effects interacted with the participant's first and second signal. All standard errors are clustered at the pair (of two participants) level. *, **, and *** indicate significance at the $p < 0.10$, 0.05, and 0.01 levels of the difference $\beta_2^p - \beta_2^o$.

B.2 Comparing Structural to Non-Structural Results

In this section we consider whether the estimates of the structural model outlined in Section 3.3 are consistent with the main non-structural results presented elsewhere in Section 4.3. To do so, we simulate, using the estimates of the parameters of the model, what guesses participants in Experiment 1 would make given the signals they had. To eliminate unnecessary noise, instead of simulating just once for each guess of what the participant would choose (which is noisy), we calculate the expected guess. We then produce a version of Panel A of Table 2 using the simulated data. The question these analyses allow us to answer is, “Are the estimated biases from the structural model sufficient to explain the patterns found in the reduced-form and non-structural results?” If the model implied that the non-structural analyses would look very different than in fact they do, this would suggest that the model is not capturing something important about the biases we document.

Panel A of Table A.XI replicates Panel A of Table 2, our main reduced-form results for Experiment 1. Panel B shows the same regressions but using the model-implied expected guesses as the dependent variable rather than participants’ true guesses. Note that these variables, because they are expectations rather than single draws from the distribution of guesses, are mechanically much less noisy than the actual guesses. However, as Table A.XI shows, the *size* of the coefficients are quite similar (i.e., comparing within column across panels). Our interpretation of these results is that the model estimates are sufficient to explain the pattern of results shown in the reduced-form analyses.

Figure A.XII compares the non-parametric results from Figure 3 with similar estimates using the model-simulated data. There are four panels, representing the *Individual* round, *Discussion* round, *Informed of Partner’s Draws* round (pre-discussion), and *Informed of Partner’s Draws* round (post-discussion). Each panel shows the estimates given the actual guesses that participants make (in gray) along with the model-simulated expected guesses (in blue). As expected, actual guesses are noisier, but the slopes of the curves are extremely similar within each panel, suggesting that the non-parametric and structural effect sizes are of comparable magnitude. Note that there is a slight bias in the actual data toward guessing more red balls in the urn, which the structural model by construction cannot deliver (as evidenced by the gray tending to lie above the blue curve).

Table A.XI: Comparing Reduced-Form to Structural Results

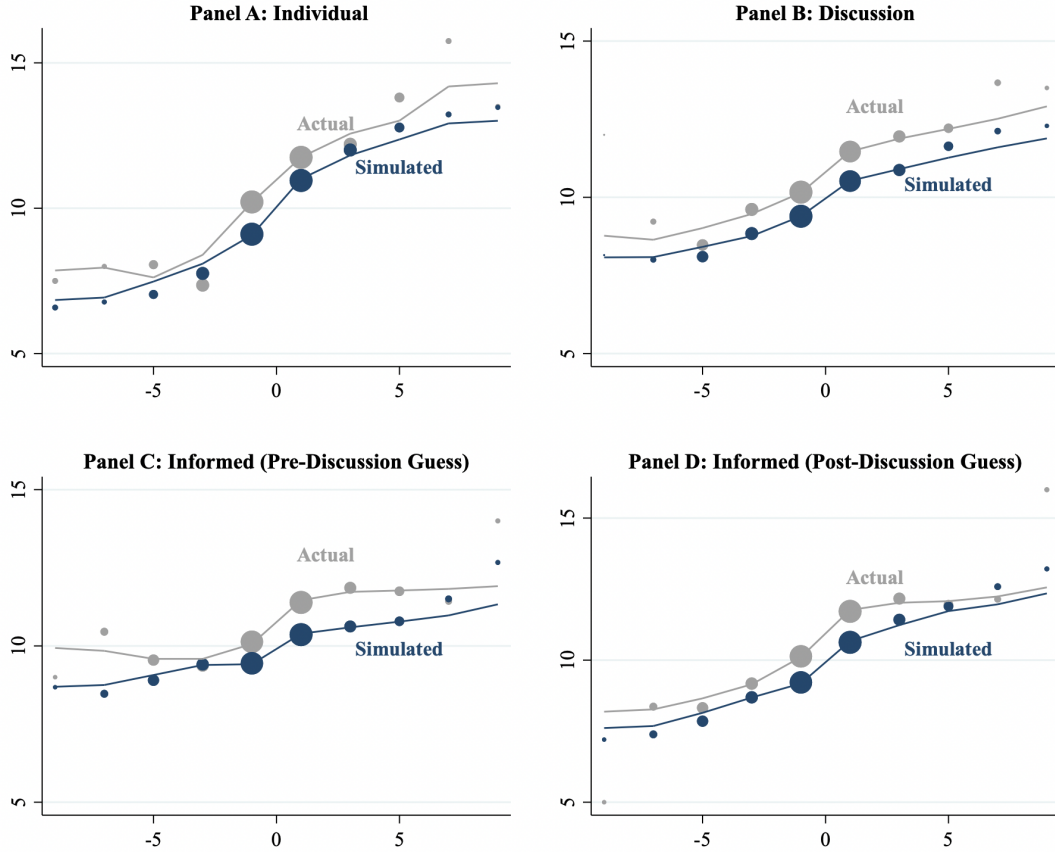
	Individual (1)	Discussion (2)	Informed (Pre) (3)	Informed (Post) (4)
<i>Panel A: Actual Guesses</i>				
β_1 : Own First Signal	0.43 (0.06)	0.56 (0.06)	0.56 (0.09)	0.36 (0.09)
β_2^o : Own Second Signal	0.52 (0.07)			
β_2^p : Partner's Signal		0.24 (0.07)	0.07 (0.09)	0.28 (0.09)
Constant	10.71 (0.16)	10.73 (0.16)	10.64 (0.23)	10.66 (0.23)
$\beta_2^p - \beta_2^o$		-0.28*** (0.08)	-0.45*** (0.11)	-0.24** (0.10)
<i>Panel B: Model-Implied Expected Guesses</i>				
β_1 : Own First Signal	0.41 (0.01)	0.48 (0.01)	0.54 (0.02)	0.38 (0.01)
β_2^o : Own Second Signal	0.58 (0.02)			
β_2^p : Partner's Signal		0.24 (0.01)	0.04 (0.01)	0.27 (0.01)
Constant	9.98 (0.03)	9.98 (0.02)	9.98 (0.03)	9.98 (0.03)
$\beta_2^p - \beta_2^o$		-0.34*** (0.02)	-0.54*** (0.02)	-0.31*** (0.02)
	500	1000	500	500

This table shows reduced-form weights on information in the Individual, Discussion and Informed of Partner's Draws rounds in Experiment 1. "Informed of Partner's Draws (Pre-Discussion)" means the dependent variable is the second private guess from the Informed of Partner's Draws round, after the participant was directly told their partner's information but before discussing it with their partner. "Informed of Partner's Draws (Post-Discussion)" means the dependent variable is the third private guess, after the discussion.

Actual Guesses: Panel A shows reduced-form results, estimating Equation 1 by OLS. The dependent variable is participants' actual private guess. "Own First Signal" is the net number of red draws (i.e., red draws minus white draws) in the participant's first set of signals, which they drew themselves in all rounds. "Own Second Signal" is the net number of red draws in the participant's second set of signals in the individual round. "Partner's Signal" is the net number of red draws in the set of signals drawn by the participant's teammate, which was the second set of signals available to the participant in the Discussion and Informed of Partner's Draws rounds. All regressions include order fixed effects interacted with the participant's first and second info.

Model-Implied Expected Guesses: show the same regressions as Panel A, but use the expected guesses (conditional on actual signals) implied by the structural estimates presented in Panel B of Table 2.

Figure A.XII: Simulated Guesses in Individual, Discussion, and Informed of Partner's Draws Rounds



Notes: This figure compares average actual guesses in Experiment 1 to the average *simulated* guess of participants using the structural model in Section 3.3. The x-axis shows the net number of red draws (i.e. red draws minus white draws) in the second signal of the round. The gray dots indicate average actual guesses, while blue dots indicate average simulated guesses. Panel A includes final private guesses in the *Individual* Round, where participants made the second set of draws themselves. Panel B includes final private guesses in the *Discussion* Round, where the second set of draws had to be communicated to the participant via discussion. Panel C includes the second private guesses in the *Informed of Partner's Draws* round, after the respondent is told of his/her partner's draws by the experimenter but before the joint discussion. Panel D includes final private guesses in the *Informed of Partner's Draws* round, after the joint discussion.

C Experimental Scripts

Contents

C.1	Experimental Script for Experiment 1	71
C.2	Experimental Script for Experiment 2	111
C.3	Experimental Script for Experiment 3	155

C.1 Experimental Script for Experiment 1

SECTION P: PREAMBLE

P0. Group ID
[.>= 1 and . <= 125]

P1. PID:
[.>= 1 and . <= 250]

P1a. Participant 1 ID:
[.>= 1 and . <= 500]

P1b. Participant 2 ID:
[.>= 1 and . <= 500]

P2. Interviewer

P2a. [Prefilled] Randomized interviewer (among present interviewers)

P3. Do you have the following things to proceed with the experiment?
[☐] 1. Sheet with scale for guessing red balls in the urn along with a token
[☐] 2. Sheet with payment mechanism
[☐] 3. Sheet to explain randomization of final guess
[☐] 4. A sample urn with red and white balls (10 each)
[☐] 5. Tray with red and white balls (10 each)
[☐] 6. Lid to cover the urn on the table
[☐] 7. Two practice urns - (16,4) and (9,11)
[☐] 8. Arrange all five urns for the activity at the back of the booth
[☐] 9. Envelope with the guess that counts

P4. Date: YYYY/ MM/ DD

P5. Interview Start Time: : 24HR

General prefills:

P6. [Prefilled] Randomized order of rounds
P7. [Prefilled] Number of red balls in the urn in Round 1.
P8. [Prefilled] Number of red balls in the urn in Round 2.
P9. [Prefilled] Number of red balls in the urn in Round 3.
P10. [Prefilled] Number of red balls in the urn in Round 4.
P11. [Prefilled] Number of red balls in the urn in Round 5.

Round 1 prefills:

P12. [Prefilled] Participant who goes first in Round 1.
P13. [Prefilled] Participant who goes second in Round 1.
P14. [Prefilled] Participant 1's number of draws from Round 1.
P15. [Prefilled] Participant 2's number of draws from Round 1.
P42. [Prefilled] Same-gender/opposite gender.

Round 2 prefills:

- P16. [Prefilled] Participant who goes first in Round 2.
- P17. [Prefilled] Participant who goes second in Round 2.
- P18. [Prefilled] Participant 1's first number of draws from Round 2.
- P19. [Prefilled] Participant 2's first number of draws from Round 2.
- P20. [Prefilled] Participant 1's second number of draws from Round 2.
- P21. [Prefilled] Participant 2's second number of draws from Round 2.

Round 3 prefills:

- P22. [Prefilled] Participant who goes first in Round 3.
- P23. [Prefilled] Participant who goes second in Round 3.
- P24. [Prefilled] Participant 1's number of draws from Round 3.
- P25. [Prefilled] Participant 2's number of draws from Round 3.

Round 4 prefills:

- P26. [Prefilled] Participant who goes first in Round 4.
- P27. [Prefilled] Participant who goes second in Round 4.
- P28. [Prefilled] Participant 1's number of draws from Round 4.
- P29. [Prefilled] Participant 2's number of draws from Round 4.

Round 5 prefills:

- P30. [Prefilled] Participant who goes first in Round 5.
- P31. [Prefilled] Participant who goes second in Round 5.
- P32. [Prefilled] Participant 1's number of draws from Round 5.
- P33. [Prefilled] Participant 2's number of draws from Round 5.
- P43. [Prefilled] Same-gender/opposite gender.

Beliefs about competence prefills:

- P38. [Prefilled] Randomized belief chosen for Participant 2.
- P39. [Prefilled] Randomized belief chosen for Participant 1.

Finalized round and guess prefills:

- P40. [Prefilled] Randomized final round.
- P41. [Prefilled] Randomized final guess.

SECTION A: INTRODUCTION

[Do not read: Talk to participant 1 and ensure that the introduction for participant 2 starts simultaneously.]

Thank you for coming today! My name is $\$p2$. Would you like something to drink before we start the task? May we offer you some water, or tea, or coffee?

Please turn off your mobile phone till the end of this task. We want to ensure that you are attentive throughout the task.

As discussed before, you and another participant will take part in a task in which you can earn some money. Please take this task seriously. Your choices directly affect how much you earn.

This study aims to contribute to the welfare of the people. So, your participation is important to us. Before you make each choice, think well. Please consider these choices to be like decisions you make in real life.

Section A1: General Overview

We will now explain the activity to you. **Similarly, another surveyor is explaining this activity to another participant.** You and another participant will then participate together in a total of five rounds that are all similar. In each round, you will see an urn such as this one.

[Do not read: Show the sample urn with a few red and white balls.]

Each urn contains a total of 20 balls. Some of these balls are white and others are red.

[Do not read: Show the participant a tray with some white balls and some red balls.]

Now we will ask you some questions about this task.

A1. How many balls are inside each urn?

☐ a. 20 → Yes, that is correct.

☐ b. Any other answer → No, there are 20 balls inside each urn.

A2. What are the colors of these balls?

☐ a. Red and white → Yes, that is correct.

☐ b. Any other answer → No, the balls are red and white.

In each round, the number of red and white balls will be unknown to you. You will only know that in total there are 20 balls inside the urn.

Your task will be to guess how many of these balls are red. In each round, you can draw some balls from the urn to help you make this guess.

A3. Now, can you describe to me in your own words what we would like you to do in this task?

[Do not read: Select all points mentioned correctly by the participants.]

☐ a. Draw a few balls from the urn.

☐ b. Based on the draws, guess how many of the balls in the urn are red.

A3a. *[Do not read: Did the participant describe all steps of the task accurately?]*

☐ a. Yes → Proceed with the survey.

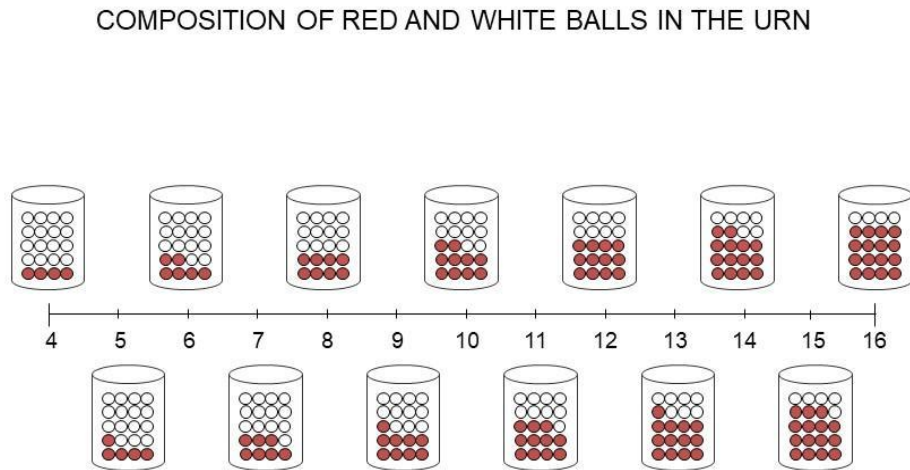
☐ b. No, the participant missed or described incorrectly 1 of the 2 points. → Explain that particular point to the participant.

☐ c. No, the participant missed or described incorrectly both points. → Explain Section A1 to the participant again. Ignore comprehension questions that are part of the text.

Section A2: Randomization of Red and White Balls

[Do not read: Show the scale depicting composition of balls in the urn (Figure 1) to the participant as you explain the below text to them.]

Figure 1: Composition of red and white balls in the urn



Your goal is to guess the number of red balls in the urn, in all five rounds. The number of red balls can be any number ranging from 4 to 16, as you can see on this figure.

- The lowest possible number of red balls is 4. *[Do not read: Point to the urn in the figure containing 4 red balls.]*

A4. Can there be less than 4 red balls in the urn, in any round?

☐ a. Yes → No, the lowest number of red balls is 4.

☐ b. No → Yes, this is correct.

- The highest possible number of red balls is 16. *[Do not read: Point to the urn in the figure containing 16 red balls.]*

A5. Can there be more than 16 red balls in the urn, in any round?

☐ a. Yes → No, the highest number of red balls is 16.

☐ b. No → Yes, this is correct.

- Any other number between 4 and 16 is possible. For example, there could be 6, 7, 11, or 12 red balls in the urn.

We have already used our computer to decide the true number of red balls in the urn in each round. Our computer picked a random number from 4 to 16 for all five rounds. The computer was equally

likely to choose any number from 4 up to 16. So, in each round, any number from 4 to 16 is equally likely to be the true number.

In each round, we will ask you to guess the number of red balls in the urn. When we do, we would like you to use this coin to point to the exact number you would like to guess, on this figure.

[Do not read: Show the participant how to point to the exact number on Figure 1 with the coin.]

Before we proceed, we will ask you some questions.

A6. What is the highest number of red balls that can possibly be in the urn?

☐ a. 16 → Yes, that is correct.

☐ b. Any other answer → No, there can be a maximum of 16 red balls in the urn.

A7. What is the lowest number of red balls that can possibly be in the urn?

☐ a. 4 → Yes, that is correct.

☐ b. Any other answer → No, there can be a minimum of 4 red balls in the urn.

A8. How do we decide the correct number of red balls in the urn? Is it you, me, or the computer?

☐ a. The computer already decided. → Yes, that's correct.

☐ b. Any other answer → No, the computer decides the correct number of red balls in the urn.

A9. Are some numbers of red balls more likely to occur than others?

☐ a. Each number is equally likely to be chosen by the computer. → Yes, that's correct.

☐ b. Any other answer → No, each number is equally likely to be chosen by the computer.

A10. Do you have any questions for us?

☐ a. Yes → *[Do not read: Clarify any doubts or questions they may have.]*

☐ b. No → *[Do not read: Proceed with the survey.]*

A10a. **[if $\$a_{10} = 1$]** *[Do not read]* Specify their question: _____

Let's practice the task a few times. These rounds will not matter for your payment. They are just to help you better understand the task.

Before we begin the example rounds, let me clarify. In the final task, you will be matched with another participant. You will have one set of draws, while the other participant has another set of draws from the same urn. In the example rounds, you will have both sets of draws. Similarly, the participant in the other booth will also have both sets of draws in the example rounds.

[Do not read: Take practice urn 1 that contains 16 red balls and 4 white balls. Use the scale.]

This urn contains 20 balls. Your goal will be to guess how many of them are red. Please draw one ball from this urn. I will shuffle the balls in the urn before you draw from the urn.

A11. Color of the ball : _____

[SurveyCTO check: Limit string length to exactly 1 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A12. How many red balls do you think are in this urn? _____

Please draw an additional five balls from this urn. I will shuffle the balls in the urn before each of your draws.

A13. Order of the draws: _____

[SurveyCTO check: Limit string length to exactly 5 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A14. How many red balls do you think are in this urn? Note that you should use the information from all six draws for your guess. _____

Do you have any questions for us so far?

[Do not read: Clarify any doubts or questions they may have.]

Let's now see how many red balls there are actually in the urn.

[Do not read: Empty the urn in front of the participant and count the number of red balls.]

As you can see, this urn contains mostly red balls. 16 of the balls are red and 4 of them are white.

Let me try to give you a few tips based on this example:

- **Tip 1:** Your draws help you make your guesses.
 - Suppose you draw more red balls than white balls. Then it makes sense to guess that there are more red balls in the urn. For example, if you draw four red balls and one white ball, then there are probably more red than white balls in the urn overall.
 - Suppose you draw more white balls. It makes sense to guess that there are more white balls in the urn.
 - Suppose you draw the same number of red and white balls. It makes sense to guess that the number of red and white balls are the same.
- **Tip 2:** The more balls you draw, the more confident you can be in your guess.
 - For example, suppose you draw only one ball. You do not learn much. Since there are many balls in the urn and you might draw a white ball by chance even if most balls are red.

- Suppose you draw many balls and most of the balls are red. Then, you can be fairly sure that most balls in the urn are red. Suppose you draw many balls and most of the balls are white. Then, you can be fairly sure that most balls in the urn are white.

A15. Can you tell us if you can learn more about the number of red balls from 1 draw or from 5 draws?

☐ a. I will learn more when I draw 5 balls → Yes, that's correct.

☐ b. I will learn more when I draw 1 ball → No, that's incorrect. You can learn more when you draw 5 balls.

☐ c. I will learn the same regardless of how many balls I draw. → No, that's incorrect. You can learn more when you draw 5 balls.

- **Tip 3:** Shuffling the balls in the urn will be helpful. You will be able to learn more about the contents of the urn.
- In this activity, I will ask you to draw a ball from the urn. Once you have looked at the ball, put it back into the urn, and then draw another ball. When you draw the next ball without shuffling the balls properly, there is a higher chance that you will draw the same ball again. If you draw the same ball again, you are not learning anything new. Thus, it is important to shuffle the urn to learn more about the contents of the urn. In each round, I will shuffle the balls in the urn before each of your draws.

Could you please repeat each of these three tips in your own words? I want to be sure I have explained things clearly.

A16. *[Do not read: Select all points mentioned correctly by the participant.]*

☐ a. The more red balls you draw from the urn, the higher the chance that the majority of the balls are red.

☐ b. The more balls you draw, the more confident you can be in your guess.

☐ c. Shuffling the balls in the urn can help you learn more.

[Do not read: Repeat the particular tips that they missed or incorrectly explained just once.]

Let's do another example.

[Do not read: Take practice urn 2 that contains 9 red balls and 11 white balls. Use Figure 1 and the coin.]

Please draw nine balls from this urn.

[Do not read: Shuffle the balls in the urn, before each draw.]

A17. Order of the draws: _____

[SurveyCTO check: Limit string length to exactly 9 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A18. How many red balls do you think are in this urn? _____

Please draw an additional ball from this urn.

[Do not read: Shuffle the balls in the urn, before each draw.]

A19. Color of the draw: _____

[SurveyCTO check: Limit string length to exactly 1 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A20. How many red balls do you think are in this urn? Note that you should use the information from all ten draws for your guess.

Let's now see how many red balls there are in the urn.

[Do not read: Empty the urn in front of the participant and count the number of red balls.]

As you can see, the urn contains more white balls than red balls. There are 9 red balls in the urn and 11 white balls in the urn.

Section A3: Explanation of the Number of Draws

As we said before, you and another participant have to guess the correct number of red balls in the urn. We will pay you according to how close your guess is to the true number of red balls in the urn.

You will make many draws and guesses for five different urns. Similarly, so will the other participant. Now let me tell you more about the number of draws.

The number of draws each of you get will vary between 1, 5, and 9.

The number of draws you and the other participant will get might be different from each other.

The number of draws you and the other participant will get might also be different across sections.

A21. Is it possible that you get exactly 4 draws in any particular round?

☐ a. Yes, I think I can get 4 draws in a round. → No, that is incorrect. You can get either 1, 5 or 9 draws in a round.

☐ b. No, I don't think I can get 4 draws in a round. → Yes, that is correct. You can get either 1, 5 or 9 draws in a round.

A22. How do you think your number of draws will be across rounds? *[Do not read: Read all options and mark one]*

☐ a. You will have the same number of draws across each round. → No, that is incorrect. You may have the same or different number of draws across each round.

☐ b. You will have different number of draws across each round. → No, that is incorrect. You may have the same or different number of draws across each round.

☐ c. You may have the same or different number of draws across each round. → Yes, that is correct.

A23. In any given round, do you think you and the other participant will have the same number of draws?

☐ a. Yes, we will both have the same number of draws. → No, that is incorrect. We might or might not have the same number of draws.

☐ b. No, we will not have the same number of draws. → No, that is incorrect. We might or might not have the same number of draws.

☐ c. We might or might not have the same number of draws. → Yes, that is correct.

Section A4: Randomization of the Guess to be Implemented

Now, let me tell you more about the guesses you and the other participant will make.

Some of these guesses will be on your own. Some others you will make together with the other participant. We will record all your choices on this sheet. *[Do not read: Show the participant the sheet in which you record the choices.]* The computer has chosen only one of these guesses to be the one that counts for your payment. Since you do not know which of those choices will be selected when making your choices, you should take each choice very seriously.

Figure 2: Participant's guesses

	Participant 1 Guess 1	Participant 2 Guess 1	Participant 1 Guess 2	Participant 2 Guess 2	Joint Guess
Round 1					
Round 2					
Round 3					
Round 4					
Round 5					

Section A5. Payment Depending on Performance

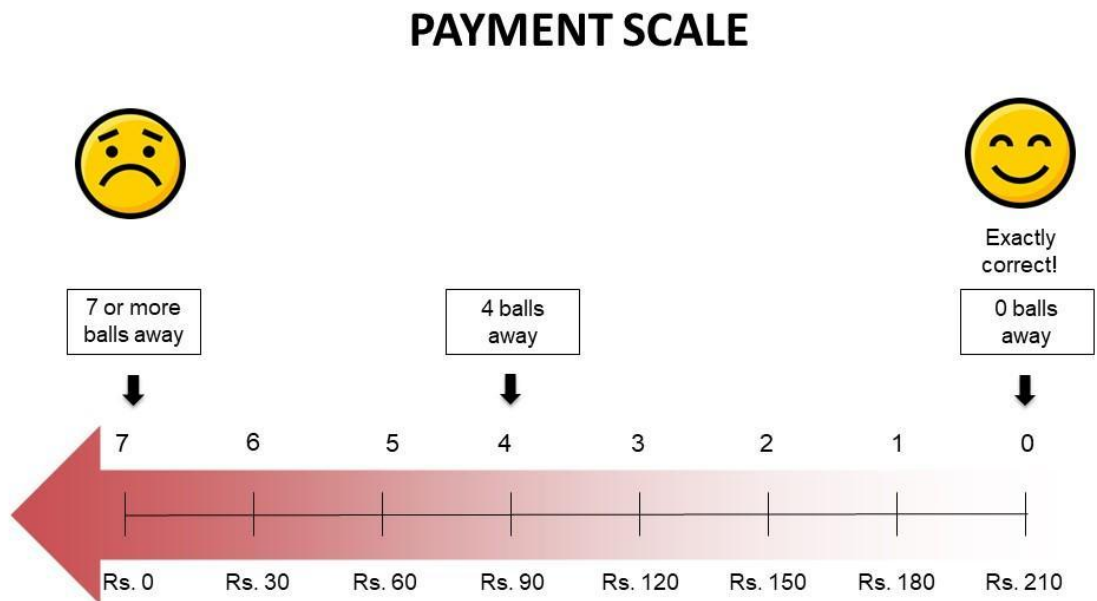
[Do not read: Show them Figure 2 to explain the payment scheme.]

Now, let me explain how the payment scheme works with this figure.

- Regardless of your choices, you will receive a flat payment of Rs. 150. Besides this flat payment, both you and the other participant have a chance to make some more money. How much more money you can earn depends on how close both your guesses are to the true number of balls in the urn. Each of you will be paid half of the total amount you earn separately.
- As we mentioned above, only one of your guesses will be chosen by the computer.
- If the chosen guess is exactly correct, then you and the other participant together could receive, besides the individual flat payment, at the most Rs. 210. So, your total payment could be at most Rs. 255. If your guess is very wrong, then you will make no extra money from the guess. You would only receive the individual flat payment of Rs. 150. If you are in the middle of these two extremes, you get an amount that is between Rs. 150 and Rs. 255.
- Along with this, you and the other participant will have an opportunity to earn an additional Rs. 50 each at the end of this activity.
- Let me give you an example. Suppose the true number of red balls is 5.

[Do not read: Use Figure 2 to explain this entire section.]

Figure 3: Payment Scale



- If your guess is 5, your guess is exactly correct. So, you and the other participant would receive the maximum of Rs. 210 from the guesses (and Rs. 255 in total individually).
- If your guess is 9, it is 4 balls away from the truth since $9 - 5 = 4$. You will lose Rs. 30 for each of the 4 balls. Therefore, you and the other participant would lose Rs. 120 and receive Rs. 90 from the guess (and Rs. 195 in total individually).
- If your guess is 12, it is 7 balls away from the truth since $12 - 5 = 7$. You will lose Rs. 30 each for the 7 balls. Therefore, you and the other participant would lose Rs. 210 and receive nothing from the guesses.
- If your guess is 16, it is 11 balls away from the truth since $16 - 5 = 11$. This guess is quite far away from the truth, so you would not receive any additional payments.

Do you have any questions about how the payment is decided?

Now, we will ask you a few questions before we proceed with the activity.

[Do not read: Ask the participant to use Figure 2 for the following questions.]

A24. Suppose that the true number is 6 red balls for the guess that counts. If you guess that there are 7 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 1 ball away from the true number. Rs. 180 → Yes, that's correct. [if $\$ \{a_{24}\} = 180$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 180. [if $\$ \{a_{24}\} \neq 180$]

A25. Suppose that the true number is 15 red balls for the guess that counts. If you guess that there are 4 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 11 balls away from the true number. Rs. 0 → Yes, that's correct. [if $\$ \{a_{25}\} = 0$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 0. [if $\$ \{a_{25}\} \neq 0$]

A26. Suppose that the true number is 12 red balls for the guess that counts. If you guess that there are 8 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 4 balls away from the true number. Rs. 90 → Yes, that's correct. [if $\$ \{a_{26}\} = 90$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 90. [if $\$ \{a_{26}\} \neq 90$]

Section A6: Overview of the Activity

Before we begin the activity, let me now tell you about the structure of the activity:

- The number of draws for each of you will vary between 1, 5 and 9 over time. Each of you might get a different number of draws. We will inform you about this number in each round.
- Based on your draws, you will each make a few guesses — some on your own, some with the other participant.
- In certain rounds, you will participate with a female participant; in other rounds, you will participate with a male participant.

We are interested in learning how you decide. Thus, we will be present to observe how you make decisions. This procedure has nothing to do with what we think of the decisions you make. We follow it with all the participants.

Do you have any questions before we begin?

[Do not read: Clarify any doubts or questions they may have.]

Comprehension scores: *[In-built CTO Check to calculate their comprehension]*

Randomization of which surveyor will continue the survey: $\$ \{p2a\}$

[Do not read: $\$ \{p2a\}$ will continue with the rest of the survey. The other surveyor will assist.]

SECTION B: ROUNDS

ROUND 1: DISCUSSION

[if $\{p42\}$ = “same gender”]

[Do not read: Introduce the two participants to each other.]

Here is the first urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is green in color — for each round, we will use a different colored urn.

The computer has decided that $\{p12\}$ starts first in this round. In this round, you will draw a few balls and guess the number of red balls in the urn; then $\{p13\}$ will draw a few balls and guess. Following this, we will ask you to make a joint guess.

[Do not read: Ask the $\{p13\}$ to leave and talk to the $\{p12\}$.]

A: Participant 1 Individual Guess

Before we start the activity, can you now please explain to me in your own words what we are asking you to do?

R1_a. *[Do not read: Did the participant mention the following points correctly?]*

☐ a. Draw balls

☐ b. Guess number of red balls

[Do not read: Repeat and correct points that are missing or incorrect.]

This urn contains 20 balls. You can now draw $\{p14\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p14\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p14\}$ times and note down the composition.]

Q1. Order of red/white balls drawn: ____

[SurveyCTO check: Limit string length to exactly $\{p14\}$ and display as field list.]

Q2. *[Do not read]* How many red balls did the $\{p12\}$ draw? ____

Q3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p13\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p12\}$ to wait and ask the $\{p13\}$ to join. Once the $\{p13\}$ joins, ask the $\{p12\}$ to leave.]

B. Participant 2 Individual Guess

Before we start the activity, can you now please explain to me in your own words what we are asking you to do?

R1_b. *[Do not read: Did the participant mention the following points correctly?]*

☐ a. Draw balls

☐ b. Guess number of red balls

[Do not read: Repeat and correct points that are missing or incorrect.]

This urn contains 20 balls and $\{p12\}$ got a chance to draw some balls from it. Now it is your turn. You can draw $\{p15\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p15\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p15\}$ times and note down the composition.]

Q4. Order of red/white balls drawn: _____

[SurveyCTO check: Limit string length to $\{p15\}$ and display as field list.]

Q5. How many red balls did the $\{p13\}$ draw? _____

Q6. How many red balls do you think are in the urn? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p13\}$ to wait and ask the $\{p12\}$ to join.]

C. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

[Do not read: Remain in the booth for the discussion.]

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Our staying here has nothing to do with whether you made ‘good’ or ‘bad’ decisions in previous rounds. This task is designed to be difficult. Most participants find it hard to come up with a good answer. Don't worry about saying anything wrong in front of us. Just try to do your best.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

Q7a. *[Do not read: Discussion start time: |__|__|:|__|__| 24HR]*

Q7b. *[Do not read: Discussion end time: |__|__|:|__|__| 24HR]*

Q7c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if \$q7c = 1]

Q7d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

Q7e. *[Do not read]* What information did the participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

Q7f. *[Do not read]* What information did the participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

Q7g. *[Do not read]* Who made the decision?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

Q7h. *[Do not read]* Notes on interaction: _____

Q8. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Q9. *[Do not read: Which participant moved the coin last on the scale?]*

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: With the other surveyor, ask the participants the following questions simultaneously in separate booths. \$p12\$ will remain in the same booth. Ask \$p13\$ to go to the other booth with the other surveyor.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

Q10. What is the p_{12} 's updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

Q11. What is the p_{13} 's updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 2: INDIVIDUAL

Here is the second urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is orange in color — for each round, we will use a different colored urn.

The computer has decided that $\{p16\}$ starts first in this round. In this round, you will draw a few balls and guess, then your $\{p17\}$ will draw a few balls and guess. You will not get to discuss and guess together in this round. We will also not share your guess with each other.

[Do not read: Ask the $\{p17\}$ to leave and talk to the $\{p16\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. We will now do something slightly different than mentioned before. You can draw $\{p18\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw each ball from the urn, have a look at it, and put it back into the urn. I will shuffle the balls in the urn before each of your draws.

R1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p18\}$ and display as field list.]

R2. *[Do not read]* How many red balls did the $\{p16\}$ draw? ____

R3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p16\}$ to wait and ask the $\{p17\}$ to join. Once the $\{p17\}$ joins, ask the $\{p16\}$ to leave.]

Thank you for your guess. It is now $\{p17\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

B. Participant 2 Individual Guess

This urn contains 20 balls. We will now do something slightly different than mentioned before. You can draw $\{p19\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw each ball from the urn, have a look at it, and put it back into the urn. I will shuffle the balls in the urn before each of your draws.

R4. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p19\}$ and display as field list.]

R5. *[Do not read]* How many red balls did the $\{p17\}$ draw? ____

R6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

You can now draw $\$ \{p21\}$ more balls to learn more about the contents of the urn. Please draw $\$ \{p21\}$ balls in total.

R7. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\$ \{p21\}$ and display as field list.]

R8. *[Do not read]* How many red balls did the $\$ \{p17\}$ draw? _____

R9. Now that you have had the chance to learn more about the content of the urn, can you tell us how many red balls you think are in the urn? [Updated guess] _____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\$ \{p16\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\$ \{p17\}$ to wait and ask the $\$ \{p16\}$ to join. Once the $\$ \{p16\}$ joins, ask the $\$ \{p17\}$ to leave.]

C. Participant 1 Updated Individual Guess

You can now draw $\$ \{p20\}$ more balls to learn more about the contents of the urn. Please draw $\$ \{p20\}$ balls in total.

R10. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\$ \{p20\}$ and display as a field list..]

R11. *[Do not read]* How many red balls did the $\$ \{p16\}$ draw? _____

R12. Now that you have had the chance to learn more about the content of the urn, can you tell us how many red balls you think are in the urn? [Updated guess] _____ *[Do not read: Have them move the coin to make their decision.]*

ROUND 3: INFORMED OF PARTNER'S GUESS

[Do not read: Introduce the two participants to each other.]

Here is the third urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is blue in color — for each round, we will use a different colored urn.

The computer has decided that $\{p22\}$ starts first in this round. In this round, you will draw a few balls and guess, then $\{p23\}$ will draw a few balls and guess. We will share your guesses with each other. Once you learn the other participant's guess, you will make another guess. Following this, we will ask you to make a joint guess.

[Do not read: Ask the $\{p23\}$ to leave and talk to the $\{p22\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p24\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p24\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p24\}$ times and note down the composition.]

S1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p24\}$ and display as field list.]

S2. *[Do not read]* How many red balls did the $\{p22\}$ draw? ____

S3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p23\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p22\}$ to wait and ask the $\{p23\}$ to join. Once the $\{p23\}$ joins, ask the $\{p22\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p22\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p25\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p25\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p25\}$ times.]

S4. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p25\}$ and display as a field list.]

S5. *[Do not read]* How many red balls did the $\{p23\}$ draw? ____

S6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

C. Participant 2 Updated Individual Guess

We will now share $\{p22\}$'s guess with you. Using this information and your own information, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p22\}$ drew $\{p24\}$ balls from this urn and guessed that there are $\{s3\}$ red balls in this urn.

S7a. Can you tell me how many balls you drew from this urn? ____

S7b. Can you tell me what you had guessed? ____

Now that you learned $\{p22\}$'s guess, how many red balls do you think there are in the urn? Please take your time to think about your choice.

S7. What is the $\{p23\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p23\}$ to wait and ask the $\{p22\}$ to join. Once the $\{p22\}$ joins, ask the $\{p23\}$ to leave.]

D. Participant 1 Updated Individual Guess

We will now share $\{p23\}$'s guess with you. Using this information and your own information, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p23\}$ drew $\{p25\}$ balls from this urn and guessed that there are $\{s6\}$ red balls in this urn.

S8a. Can you tell me how many balls you drew from this urn? ____

S8b. Can you tell me what you had guessed? ____

Now that you learned $\{p23\}$'s guess, how many red balls do you think there are in the urn? Please take your time to think about your choice.

S8. What is the $\{p22\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p22\}$ to wait and ask the $\{p23\}$ to join.]

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

[Do not read: Remain in the booth for the discussion.]

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Our staying here has nothing to do with whether you made ‘good’ or ‘bad’ decisions in previous rounds. This task is designed to be difficult. Most participants find it hard to come up with a good answer. Don't worry about saying anything wrong in front of us. Just try to do your best.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

S9a. *[Do not read: Discussion start time: |__|__:|__|__| 24HR]*

S9b. *[Do not read: Discussion end time: |__|__:|__|__| 24HR]*

S9c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $S_{9c} = 1$]

S9d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

S9e. *[Do not read]* What information did the participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

S9f. *[Do not read]* What information did the participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

S9g. *[Do not read]* Who made the decision?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

S9h. *[Do not read]* Notes on interaction: _____

S10. How many red balls do you think there are in the urn? ____ [*Do not read: Have them move the coin to make their decision*]

S11. [*Do not read: Which participant moved the coin last on the scale?*]

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: With the other surveyor, ask the participants the following questions simultaneously in separate booths. \$p22\$ will remain in the same booth. Ask \$p23\$ to go to the other booth with the other surveyor.]

Both of you discussed and made a joint guess. You now have a chance to make another guess based on the information you learnt. We will not share your guess with the other participant and will note it down. This guess may be different from your previous guesses or could be the same. So based on all the information you learnt, how many red balls do you think are in this urn?

S12. What is the \$p22\$'s updated guess? ____ [*Do not read: Have them move the coin to make their decision.*]

S13. What is the \$p23\$'s updated guess? ____ [*Do not read: Have them move the coin to make their decision.*]

We will show you the number of red balls in the urn, if this is chosen at the end. Now, we will proceed to the next round.

ROUND 4: INFORMED OF PARTNER'S DRAWS

Here is the fourth urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is brown in color — for each round, we will use a different colored urn.

The computer has decided that $\{p26\}$ starts first in this round. In this round, you will draw a few balls and guess, then $\{p27\}$ will draw a few balls and guess. After you have made your guesses, we will then share the draws you saw with the other participant. Then you will make another guess. Following this, we will ask you to make a joint guess.

[Do not read: Ask the $\{p27\}$ to leave and talk to the $\{p26\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p28\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p28\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

A1. Which of the following information will we be sharing with your $\{p27\}$?

- ☐ 1. Your guess
- ☐ 2. Draws

[Do not read: Ensure that the participant draws $\{p28\}$ times and note down the composition.]

T1. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\{p28\}$ and display as a field list.]

T2. *[Do not read]* How many red balls did the $\{p26\}$ draw? _____

T3. How many red balls do you think are in the urn? _____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p27\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p26\}$ to wait and ask the $\{p27\}$ to join. Once the $\{p27\}$ joins, ask the $\{p26\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p26\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p29\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p29\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

A2. Which of the following information will we be sharing with your $\{p26\}$?

☐ 1. Your guess

☐ 2. Draws

[Do not read: Ensure that the participant draws $\{p29\}$ times and display as a field list.]

T4. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p29\}$]

T5. *[Do not read]* How many red balls did the $\{p27\}$ draw? ____

T6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

C. Participant 2 Updated Individual Guess

We will now share $\{p26\}$'s draws with you. Using this information and your own draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p26\}$ drew a total of $\{p28\}$ balls from this urn, of which $\{t2\}$ were red balls.

T7a. Can you tell me how many balls you drew from this urn? ____

T7b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

T7. What is the $\{p27\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p27\}$ to wait and ask the $\{p26\}$ to join. Once the $\{p26\}$ joins, ask the $\{p27\}$ to leave.]

D. Participant 1 Updated Individual Guess

We will now share $\{p27\}$'s information with you. Using this information and your own information, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\$ \{p27\}$ drew a total of $\$ \{p29\}$ balls from this urn, of which $\$ \{t5\}$ were red balls.

T8a. Can you tell me how many balls you drew from this urn? ____

T8b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

T8. What is the $\$ \{p26\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\$ \{p26\}$ to wait and ask the $\$ \{p27\}$ to join.]

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

[Do not read: Remain in the booth for the discussion.]

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Our staying here has nothing to do with whether you made 'good' or 'bad' decisions in previous rounds. This task is designed to be difficult. Most participants find it hard to come up with a good answer. Don't worry about saying anything wrong in front of us. Just try to do your best.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

T9a. *[Do not read: Discussion start time: |__|__:|__|__| 24HR]*

T9b. *[Do not read: Discussion end time: |__|__:|__|__| 24HR]*

T9c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\$ \{t9c\} = 1$]

T9d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

T9e. *[Do not read]* What information did participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

T9f. *[Do not read]* What information did participant 2 share?

- ☐ a. Number of draws
- ☐ b. Color composition
- ☐ c. Guess

T9g. *[Do not read]* Who made the decision?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

T9h. *[Do not read]* Notes on interaction: _____

T10. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision]*

T11. *[Do not read: Which participant moved the coin last on the scale?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: With the other surveyor, ask the participants the following questions simultaneously in separate booths. \$p26\$ will remain in the same booth. Ask \$p27\$ to go to the other booth with the other surveyor.]

Both of you discussed and made a joint guess. You now have a chance to make another guess based on the information you learnt. We will not share your guess with the other participant and will note it down. This guess may be different from your previous guesses or could be the same. So based on all the information you learnt, how many red balls do you think are in this urn?

T12. What is the \$p26\$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

T14. What is the \$p27\$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

We will show you the number of red balls in the urn, if this is chosen at the end. Now, we will proceed to the next round.

ROUND 5: DISCUSSION

[if $\{p43\}$ = “opposite gender”]

Here is the fifth urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is black in color — for each round, we will use a different colored urn.

The computer has decided that $\{p30\}$ starts first in this round. In this round, you will draw a few balls and guess the number of red balls in the urn; then $\{p31\}$ will draw a few balls and guess. Following this, we will ask you to make a joint guess.

[Do not read: Ask the $\{p31\}$ to leave and talk to the $\{p30\}$]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p32\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p32\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p32\}$ times and note down the composition.]

U1. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\{p32\}$ and display as a field list.]

U2. *[Do not read]* How many red balls did the $\{p30\}$ draw? _____

U3. How many red balls do you think are in the urn? _____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now your $\{p31\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p30\}$ to wait and ask the $\{p31\}$ to join. Once the $\{p31\}$ joins, ask the $\{p30\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p30\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p33\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p33\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p33\}$ times.]

U4. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\{p33\}$]

U5. *[Do not read]* How many red balls did the $\{p31\}$ draw? ____

U6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p31\}$ to wait and ask the $\{p30\}$ to join.]

C. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Our staying here has nothing to do with whether you made ‘good’ or ‘bad’ decisions in previous rounds. This task is designed to be difficult. Most participants find it hard to come up with a good answer. Don't worry about saying anything wrong in front of us. Just try to do your best.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

U7a. *[Do not read: Discussion start time: |__|__:|__|__| 24HR]*

U7b. *[Do not read: Discussion end time: |__|__:|__|__| 24HR]*

U7c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\{u7c\} = 1]$

U7d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

U7e. *[Do not read]* What information did participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

U7f. *[Do not read]* What information did participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

U7g. *[Do not read]* Who made the decision?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

U7h. *[Do not read]* Notes on interaction: _____

U8. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

U9. *[Do not read: Which participant moves the coin on the scale?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: With the other surveyor, ask the participants the following questions simultaneously in separate booths. $\{p30\}$ will remain in the same booth. Ask $\{p31\}$ to go to the other booth with the other surveyor.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

U10. What is the $\{p30\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

U11. What is the $\{p31\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

We will show you the number of red balls in the urn. if this is chosen at the end. Now, we will proceed to the next round.

SECTION C: Beliefs about Competence

[Do not read: Talk to $\{p30\}$]

Now I would like to ask you a few questions about the game you have been playing today. We will not share your answers with $\{p31\}$.

V1. In general, who, amongst men and women, do you think will be better at this game, i.e. who is making more accurate choices?

- ☐ a. Women are better
- ☐ b. About the same
- ☐ c. Men are better

V2. Who, amongst you and $\{p31\}$, do you think is better at this game, i.e. who is making more accurate choices?

- ☐ a. Participant 1 is better
- ☐ b. About the same
- ☐ c. Participant 2 is better

V3. In one of the rounds, you participated with another participant and not $\{p31\}$. Who, amongst you and the other participant, do you think is better at this game, i.e. who is making more accurate choices?

- ☐ a. Participant 1 is better
- ☐ b. About the same
- ☐ c. Other participant is better

[Do not read: Talk to $\{p31\}$.]

So I would like to ask you a few questions about the game you have been playing today. We will not share your answers with $\{p30\}$.

V4. In general, who, amongst men and women, do you think will be better at this game, i.e. who is making more accurate choices?

- ☐ a. Women are better
- ☐ b. About the same
- ☐ c. Men are better

V5. Who, amongst you and $\{p30\}$, do you think is better at this game, i.e. who is making more accurate choices?

- ☐ a. Participant 1 is better
- ☐ b. About the same
- ☐ c. Participant 2 is better

V6. In one of the rounds, you participated with another participant and not $\{p30\}$. Who, amongst you and the other participant, do you think is better at this game, i.e. who is making more accurate choices?

- ☐ a. Participant 1 is better
- ☐ b. About the same

[] c. Other participant is better

[Do not read: Talk to \$p31].

We're at the end of this activity. Before you can learn how much you and \$p30} will be earning, we have a few more questions for you.

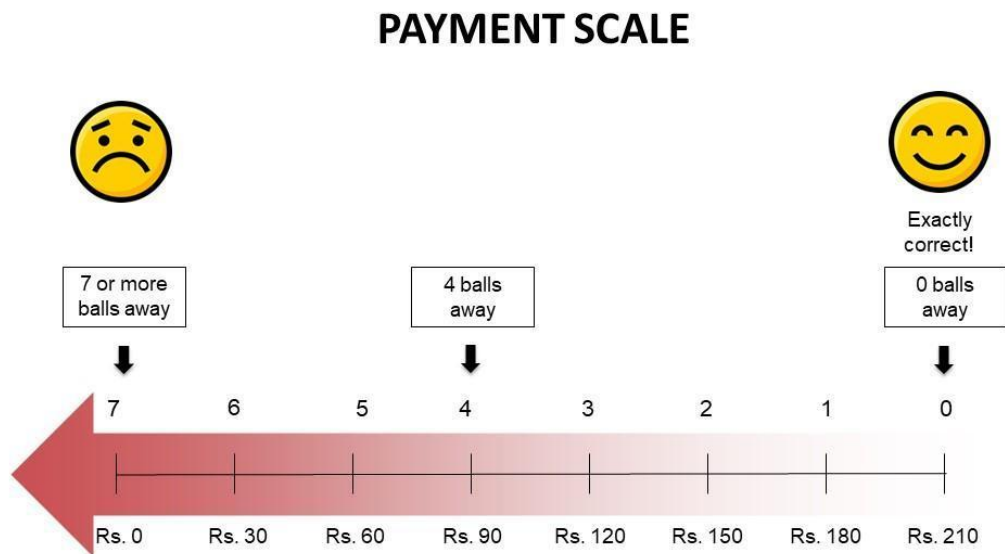
You, \$p30}, and the other participant got to make many guesses after learning some information about the contents of the different urns. Before I proceed, I would like to assure you that anything you tell us will not be shared with any of the participants. I request you to answer the following questions. We will not disclose anything to \$p30}.

Since, you, \$p30}, and the other participant made multiple guesses on your own — how much would each of your guesses earn, on an average?

If your answer is within Rs. 30 of the average of the actual earnings, you can earn Rs. 50 that you will be paid separately. However, if your guess is more than Rs. 30 of the average of the actual earnings, you will not earn anything other than the amount disclosed in the envelope as discussed before. As mentioned before, one of the answers has already been chosen by the computer and you will be paid for that answer. So, think carefully before answering.

[Do not read: Use Figure 3 (the payment scale) for the following questions.]

Figure 3: Payment Scale



W1. How much will \$p30}'s guess earn, on an average? Rs. _____

W2. How much will the other participant's guess earn, on an average? Rs. _____

W3. How much will your guess earn, on an average? Rs. _____

[if $\$ \{p38\}$ = “own guess”]

W3a. [Calculate: $\$ \{p31\}$ ’s average earnings _____]

[if $\$ \{w3a\} - 30 \leq \$ \{w3\} \leq \$ \{w3a\} + 30$]

Your guess about your average earnings is within the Rs. 30 of the average of your actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\$ \{w3\} < \$ \{w3a\} - 30$ or $\$ \{w3\} > \$ \{w3a\} + 30$]

Your guess about your average earnings is not within the Rs. 30 of the average of your actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

[if $\$ \{p38\}$ = “partner’s guess”]

W3b. [Calculate: $\$ \{p30\}$ ’s average earnings _____]

[if $\$ \{w3b\} - 30 \leq \$ \{w1\} \leq \$ \{w3b\} + 30$]

Your guess about $\$ \{p30\}$ ’s average earnings is within the Rs. 30 of the average of their actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\$ \{w1\} < \$ \{w3b\} - 30$ or $\$ \{w1\} > \$ \{w3b\} + 30$]

Your guess about $\$ \{p30\}$ ’s average earnings is not within the Rs. 30 of the average of their actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

[if $\$ \{p38\}$ = “other participant’s guess”]

W3c. [Calculate: the other participant’s average earnings _____]

[if $\$ \{w3c\} - 30 \leq \$ \{w2\} \leq \$ \{w3c\} + 30$]

Your guess about the other participant’s average earnings is within the Rs. 30 of the average of their actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\$ \{w2\} < \$ \{w3c\} - 30$ or $\$ \{w2\} > \$ \{w3c\} + 30$]

Your guess about the other participant’s average earnings is not within the Rs. 30 of the average of their actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

[Do not read: Talk to $\$ \{p30\}$.]

We’re at the end of this activity. Before you can learn how much you and $\$ \{p31\}$ will be earning, we have a few more questions for you.

You, $\{p31\}$, and the other participant got to make many guesses after learning some information about the contents of the different urns. Before I proceed, I would like to assure you that anything you tell us will not be shared with any of the participants. I request you to answer the following questions. We will not disclose anything to $\{p31\}$.

Since, you, $\{p31\}$, and the other participant made multiple guesses on your own — how much would each of your guesses earn, on an average?

If your answer is within Rs. 30 of the average of the actual earnings, you can earn Rs. 50 that you will be paid separately. However, if your guess is more than Rs. 30 of the average of the actual earnings, you will not earn anything other than the amount disclosed in the envelope as discussed before. As mentioned before, one of the answers has already been chosen by the computer and you will be paid for that answer. So, think carefully before answering.

[Do not read: Use Figure 3 (the payment scale) for the following questions.]

W4. How much will $\{p31\}$'s guess earn, on an average? Rs. ____

W5. How much will the other participant's guess earn, on an average? Rs. ____

W6. How much will your guess earn, on an average? Rs. ____

[if $\{p39\} = \text{"own guess"}$]

[if $\{w3b\} - 30 \leq \{w6\} \leq \{w3b\} + 30$]

Your guess about your average earnings is within the Rs. 30 of the average of your actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\{w6\} < \{w3b\} - 30$ or $\{w6\} > \{w3b\} + 30$]

Your guess about your average earnings is not within the Rs. 30 of the average of your actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

[if $\{p39\} = \text{"partner's guess"}$]

[if $\{w3a\} - 30 \leq \{w4\} \leq \{w3a\} + 30$]

Your guess about $\{p31\}$'s average earnings is within the Rs. 30 of the average of their actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\{w4\} < \{w3a\} - 30$ or $\{w4\} > \{w3a\} + 30$]

Your guess about $\{p31\}$'s average earnings is not within the Rs. 30 of the average of their actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

[if $\$p39$ = “other participant’s guess”]

W6a. [Calculate: the other participant’s average earnings _____]

[if $\$w6a - 30 \leq \$w5 \leq \$w6a + 30$]

Your guess about the other participant’s average earnings is within the Rs. 30 of the average of their actual earnings. You will earn an additional Rs. 50 at the end of this survey.

[if $\$w5 < \$w6a - 30$ or $\$w5 > \$w6a + 30$]

Your guess about the other participant’s average earnings is not within the Rs. 30 of the average of their actual earnings. You will not earn an additional Rs. 50 at the end of this survey.

SECTION E: FINAL SECTION

As mentioned before, the envelope in front of you contains the decision made by the computer. Please feel free to open it and have a look at the decision.

F1. *[Do not read: Who opened the envelope?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

F1a. *[Do not read: What guess does the sheet say?]*

- ☐ a. Participant 1's first individual guess
- ☐ b. Participant 2's first individual guess
- ☐ c. Participant 1's second individual guess
- ☐ d. Participant 2's second individual guess
- ☐ e. Participant 1's post-discussion guess
- ☐ f. Participant 2's post-discussion guess
- ☐ g. Joint guess

F1b. *[Do not read: Which urn does the sheet say is the one that determines their payment?]*

- ☐ a. Round 1
- ☐ b. Round 2
- ☐ c. Round 3
- ☐ d. Round 4
- ☐ e. Round 5

[Do not read: Show them the red balls in the urn.]

[if $S_{p4} = 1$]

As you can see, there are S_{p7} red balls in the urn.

[if $S_{p4} = 2$]

As you can see, there are S_{p8} red balls in the urn.

[if $S_{p4} = 3$]

As you can see, there are S_{p9} red balls in the urn.

[if $S_{p4} = 4$]

As you can see, there are S_{p10} red balls in the urn.

[if $S_{p4} = 5$]

As you can see, there are S_{p11} red balls in the urn.

F2. *[Do not read]* Is the finalized guess a post-discussion guess?

- ☐ 0. No
- ☐ 1. Yes

[if $S_{f2} = 1$]

The finalized guess is a private guess, and as mentioned before we will not disclose the guess. However, we will explain the amount you will receive.

[Do not read: With the help of the payment scale, explain their payment to them.]

[if $S_{f2} = 0$]

[Do not read: Show them the grid with responses from each round and point to the one in the envelope. With the help of the payment scale, explain their payment to them.]

SECTION Z: CONCLUSION

Thank you for your time!

Z1. Re-enter PID: _____

Z2. End Time: |_|_|:|_|_| 24HR

C.2 Experimental Script for Experiment 2

SECTION P: PREAMBLE

P1. PID:
[.>= 3000 and .<= 3400]

P1a. Participant 1 ID:
[.>= 1 and .<= 800]

P1b. Participant 2 ID:
[.>= 1 and .<= 800]

P1c. Pair:
☐ 1. Opposite-gender
☐ 2. Same-gender

P2. Interviewer

P2a. [Prefilled] Randomized interviewer (among present interviewers)

P3. Do you have the following things to proceed with the experiment?
☐ 1. Sheet with scale for guessing red balls in the urn along with a token
☐ 2. Sheet with payment mechanism
☐ 3. Sheet to explain randomization of final guess
☐ 4. A sample urn with red and white balls (10 each)
☐ 5. Tray with red and white balls (10 each)
☐ 6. Lid to cover the urn on the table
☐ 7. Two practice urns - (16,4) and (9,11)
☐ 8. Arrange all six urns for the activity at the back of the booth
☐ 9. Envelope with the guess that counts

P4. Date: YYYY/ MM/ DD

P5. Interview Start Time: : 24HR

General prefills:

P6. [Prefilled] Randomized order of rounds
P7. [Prefilled] Number of red balls in the urn in Round 1.
P8. [Prefilled] Number of red balls in the urn in Round 2.
P9. [Prefilled] Number of red balls in the urn in Round 3.
P10. [Prefilled] Number of red balls in the urn in Round 4.
P11. [Prefilled] Number of red balls in the urn in Round 5.
P12. [Prefilled] Number of red balls in the urn in Round 6.

Round 1 prefills:

P14. [Prefilled] Participant who goes first in Round 1.

- P15. [Prefilled] Participant who goes second in Round 1.
- P16. [Prefilled] Participant 1's number of draws from Round 1.
- P17. [Prefilled] Participant 2's number of draws from Round 1.

Round 2 prefills:

- P18. [Prefilled] Participant who goes first in Round 2.
- P19. [Prefilled] Participant who goes second in Round 2.
- P20. [Prefilled] Participant 1's number of draws from Round 2.
- P21. [Prefilled] Participant 2's number of draws from Round 2.
- P22. [Prefilled] Stakes in Round 2.

Round 3 prefills:

- P23. [Prefilled] Participant who goes first in Round 3.
- P24. [Prefilled] Participant who goes second in Round 3.
- P25. [Prefilled] Participant 1's number of draws from Round 3.
- P26. [Prefilled] Participant 2's number of draws from Round 3.
- P27. [Prefilled] Stakes in Round 3.

Round 4 prefills:

- P28. [Prefilled] Participant who goes first in Round 4.
- P29. [Prefilled] Participant who goes second in Round 4.
- P30. [Prefilled] Participant 1's number of draws from Round 4.
- P31. [Prefilled] Participant 2's number of draws from Round 4.
- P32. [Prefilled] Stakes in Round 4.

Round 5 prefills:

- P33. [Prefilled] Participant who goes first in Round 5.
- P34. [Prefilled] Participant who goes second in Round 5.
- P35. [Prefilled] Participant 1's number of draws from Round 5.
- P36. [Prefilled] Participant 2's number of draws from Round 5.
- P37. [Prefilled] Stakes in Round 5.

Round 6 prefills:

- P38. [Prefilled] Participant who goes first in Round 6.
- P39. [Prefilled] Participant who goes second in Round 6.
- P40. [Prefilled] Participant 1's number of draws in Round 6.
- P41. [Prefilled] Participant 2's number of draws in Round 6.
- P42. [Prefilled] Stakes in Round 6.

Finalized round and guess prefills:

- P43. [Prefilled] Randomized final round.
- P44. [Prefilled] Randomized final guess.

SECTION A: INTRODUCTION

[Do not read: Talk to participant 1 and ensure that the introduction for participant 2 starts simultaneously.]

Thank you for coming today! My name is $\$p2$. Would you like something to drink before we start the task? May we offer you some water, or tea, or coffee?

Please turn off your mobile phone till the end of this task. We want to ensure that you are attentive throughout the task.

As discussed before, you and another participant will take part in a task in which you can earn some money. Please take this task seriously. Your choices directly affect how much you earn.

This study aims to contribute to the welfare of the people. So, your participation is important to us. Before you make each choice, think well. Please consider these choices to be like decisions you make in real life.

Section A1: General Overview

We will now explain the activity to you. **Similarly, another surveyor is explaining this activity to another participant.** You and another participant will then participate together in a total of six rounds that are all similar. In each round, you will see an urn such as this one.

[Do not read: Show the sample urn with a few red and white balls.]

Each urn contains a total of 20 balls. Some of these balls are white and others are red.

[Do not read: Show the participant a tray with some white balls and some red balls.]

Now we will ask you some questions about this task.

A1. How many balls are inside each urn?

☐ a. 20 → Yes, that is correct.

☐ b. Any other answer → No, there are 20 balls inside each urn.

A2. What are the colors of these balls?

☐ a. Red and white → Yes, that is correct.

☐ b. Any other answer → No, the balls are red and white.

In each round, the number of red and white balls will be unknown to you. You will only know that in total there are 20 balls inside the urn.

Your task will be to guess how many of these balls are red. In each round, you can draw some balls from the urn to help you make this guess.

A3. Now, can you describe to me in your own words what we would like you to do in this task?

[Do not read: Select all points mentioned correctly by the participants.]

☐ a. Draw a few balls from the urn.

☐ b. Based on the draws, guess how many of the balls in the urn are red.

A3a. *[Do not read: Did the participant describe all steps of the task accurately?]*

☐ a. Yes → Proceed with the survey.

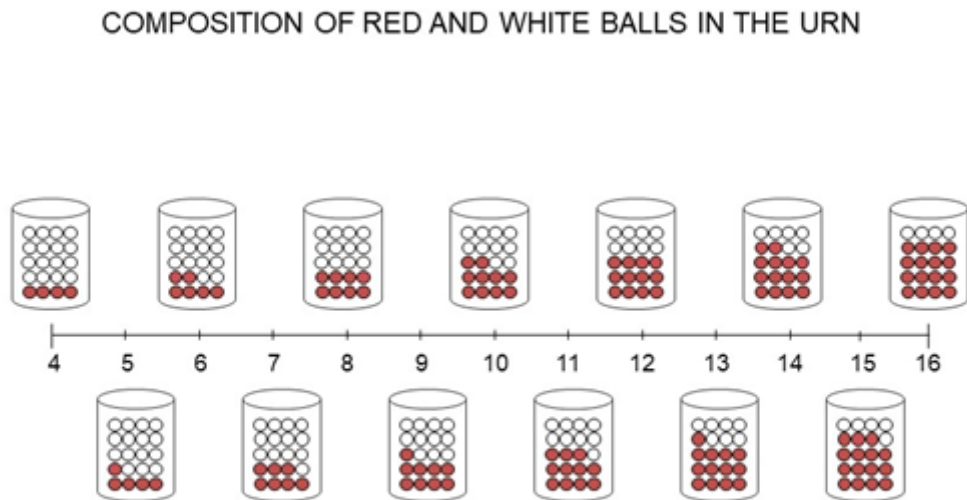
☐ b. No, the participant missed or described incorrectly 1 of the 2 points. → Explain that particular point to the participant.

☐ c. No, the participant missed or described both points incorrectly. → Explain Section A1 to the participant again. Ignore comprehension questions that are part of the text.

Section A2: Randomization of Red and White Balls

[Do not read: Show the scale depicting composition of balls in the urn (Figure 1) to the participant as you explain the below text to them.]

Figure 1: Composition of red and white balls in the urn



Your goal is to guess the number of red balls in the urn, in all six rounds. The number of red balls can be any number ranging from 4 to 16, as you can see on this figure.

- The lowest possible number of red balls is 4. *[Do not read: Point to the urn in the figure containing 4 red balls.]*

A4. Can there be less than 4 red balls in the urn, in any round?

☐ a. Yes → No, the lowest number of red balls is 4.

☐ b. No → Yes, this is correct.

- The highest possible number of red balls is 16. *[Do not read: Point to the urn in the figure containing 16 red balls.]*

A5. Can there be more than 16 red balls in the urn, in any round?

☐ a. Yes → No, the highest number of red balls is 16.

☐ b. No → Yes, this is correct.

- Any other number between 4 and 16 is possible. For example, there could be 6, 7, 11, or 12 red balls in the urn.

We have already used our computer to decide the true number of red balls in the urn in each round. Our computer picked a random number from 4 to 16 for all six rounds. The computer was equally likely to choose any number from 4 up to 16. So, in each round, any number from 4 to 16 is equally likely to be the true number.

In each round, we will ask you to guess the number of red balls in the urn. When we do, we would like you to use this coin to point to the exact number you would like to guess, on this figure.

[Do not read: Show the participant how to point to the exact number on Figure 1 with the coin.]

Before we proceed, we will ask you some questions.

A6. What is the highest number of red balls that can possibly be in the urn?

☐ a. 16 → Yes, that is correct.

☐ b. Any other answer → No, there can be a maximum of 16 red balls in the urn.

A7. What is the lowest number of red balls that can possibly be in the urn?

☐ a. 4 → Yes, that is correct.

☐ b. Any other answer → No, there can be a minimum of 4 red balls in the urn.

A8. How do we decide the correct number of red balls in the urn? Is it you, me, or the computer?

☐ a. The computer already decided. → Yes, that's correct.

☐ b. Any other answer → No, the computer decides the correct number of red balls in the urn.

A9. Are some numbers of red balls more likely to occur than others?

☐ a. Each number is equally likely to be chosen by the computer. → Yes, that's correct.

☐ b. Any other answer → No, each number is equally likely to be chosen by the computer.

A10. Do you have any questions for us?

☐ a. Yes → *[Do not read: Clarify any doubts or questions they may have.]*

☐ b. No → *[Do not read: Proceed with the survey.]*

A10a. **[if $\$a_{10} = 1$]** *[Do not read]* Specify their question: _____

Let's practice the task a few times. These rounds will not matter for your payment. They are just to help you better understand the task.

Before we begin the example rounds, let me clarify. In the final task, you will be matched with another participant. You will have one set of draws, while the other participant has another set of draws from the same urn. In the example rounds, you will have both sets of draws. Similarly, the participant in the other booth will also have both sets of draws in the example rounds.

[Do not read: Take practice urn 1 that contains 16 red balls and 4 white balls. Use the scale.]

This urn contains 20 balls. Your goal will be to guess how many of them are red. Please draw one ball from this urn. I will shuffle the balls in the urn before you draw from the urn.

A11. Color of the ball : _____

[SurveyCTO check: Limit string length to exactly 1 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A12. How many red balls do you think are in this urn? _____

Please draw an additional five balls from this urn. I will shuffle the balls in the urn before each of your draws.

A13. Order of the draws: _____

[SurveyCTO check: Limit string length to exactly 5 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A14. How many red balls do you think are in this urn? Note that you should use the information from all six draws for your guess. _____

Do you have any questions for us so far?

[Do not read: Clarify any doubts or questions they may have.]

Let's now see how many red balls there are actually in the urn.

[Do not read: Empty the urn in front of the participant and count the number of red balls.]

As you can see, this urn contains mostly red balls. 16 of the balls are red and 4 of them are white.

Let me try to give you a few tips based on this example.

- **Tip 1:** Your draws help you make your guesses.
 - Suppose you draw more red balls than white balls. Then it makes sense to guess that there are more red balls in the urn. For example, if you draw four red balls and one white ball, then there are probably more red than white balls in the urn overall.
 - Suppose you draw more white balls. It makes sense to guess that there are more white balls in the urn.
 - Suppose you draw the same number of red and white balls. It makes sense to guess that the number of red and white balls are the same.
- **Tip 2:** The more balls you draw, the more confident you can be in your guess.

- For example, suppose you draw only one ball. You do not learn much. Since there are many balls in the urn and you might draw a white ball by chance even if most balls are red.
- Suppose you draw many balls and most of the balls are red. Then, you can be fairly sure that most balls in the urn are red. Suppose you draw many balls and most of the balls are white. Then, you can be fairly sure that most balls in the urn are white.

A15. Can you tell us if you can learn more about the number of red balls from 1 draw or from 5 draws?

[] a. I will learn more when I draw 5 balls → Yes, that's correct.

[] b. I will learn more when I draw 1 ball → No, that's incorrect. You can learn more when you draw 5 balls.

[] c. I will learn the same regardless of how many balls I draw. → No, that's incorrect. You can learn more when you draw 5 balls.

- **Tip 3:** Shuffling the balls in the urn will be helpful. You will be able to learn more about the contents of the urn.
 - In this activity, I will ask you to draw a ball from the urn. Once you have looked at the ball, put it back into the urn, and then draw another ball. When you draw the next ball without shuffling the balls properly, there is a higher chance that you will draw the same ball again. If you draw the same ball again, you are not learning anything new. Thus, it is important to shuffle the urn to learn more about the contents of the urn. In each round, I will shuffle the balls in the urn before each of your draws.

Could you please repeat each of these three tips in your own words? I want to be sure I have explained things clearly.

A16. *[Do not read: Select all points mentioned correctly by the participant.]*

[] a. The more red balls you draw from the urn, the higher the chance that the majority of the balls are red.

[] b. The more balls you draw, the more confident you can be in your guess.

[] c. Shuffling the balls in the urn can help you learn more.

[Do not read: Repeat the particular tips that they missed or incorrectly explained just once. If they still don't understand the tips after the explanation, inform the monitors immediately.]

Let's do another example.

[Do not read: Take practice urn 2 that contains 9 red balls and 11 white balls. Use Figure 1 and the coin.]

Please draw nine balls from this urn.

[Do not read: Shuffle the balls in the urn, before each draw.]

A17. Order of the draws: _____

[SurveyCTO check: Limit string length to exactly 9 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A18. How many red balls do you think are in this urn? _____

Please draw an additional ball from this urn.

[Do not read: Shuffle the balls in the urn, before each draw.]

A19. Color of the draw: _____

[SurveyCTO check: Limit string length to exactly 1 and display as field list.]

[Do not read: Ask them to use Figure 1 and the coin.]

A20. How many red balls do you think are in this urn? Note that you should use the information from all ten draws for your guess. _____

Let's now see how many red balls there are in the urn.

[Do not read: Empty the urn in front of the participant and count the number of red balls.]

As you can see, the urn contains more white balls than red balls. There are 9 red balls in the urn and 11 white balls in the urn.

Section A3: Explanation of the Number of Draws

As we said before, you and another participant have to guess the correct number of red balls in the urn. We will pay you according to how close your guess is to the true number of red balls in the urn.

You will make many draws and guesses for six different urns. Similarly, so will the other participant. Now let me tell you more about the number of draws.

The number of draws each of you get will vary between 1, 5, and 9.

The number of draws you and the other participant will get might be different from each other.

The number of draws you and the other participant will get might also be different across sections.

A21. Is it possible that you get exactly 4 draws in any particular round?

[] a. Yes, I think I can get 4 draws in a round. → No, that is incorrect. You can get either 1, 5 or 9 draws in a round.

☐ b. No, I don't think I can get 4 draws in a round. → Yes, that is correct. You can get either 1, 5 or 9 draws in a round.

A22. How do you think your number of draws will be across rounds? *[Do not read: Read all options and mark one]*

☐ a. You will have the same number of draws across each round. → No, that is incorrect. You may have the same or different number of draws across each round.

☐ b. You will have a different number of draws across each round. → No, that is incorrect. You may have the same or different number of draws across each round.

☐ c. You may have the same or different number of draws across each round. → Yes, that is correct.

A23. In any given round, do you think you and the other participant will have the same number of draws?

☐ a. Yes, we will both have the same number of draws. → No, that is incorrect. We might or might not have the same number of draws.

☐ b. No, we will not have the same number of draws. → No, that is incorrect. We might or might not have the same number of draws.

☐ c. We might or might not have the same number of draws. → Yes, that is correct.

Section A4: Randomization of the Guess to be Implemented

Now, let me tell you more about the guesses you and the other participant will make.

	Participant 1 Guess 1	Participant 2 Guess 1	Participant 1 Guess 2	Participant 2 Guess 2
Round 1			-	-
Round 2				
Round 3				
Round 4			-	-
Round 5				
Round 6				

	Joint Guess	Participant 1 Guess 3	Participant 2 Guess 3
Round 1			
Round 2			
Round 3			
Round 4			
Round 5			

Some of these guesses will be on your own. Some others you will make together with the other participant. We will record all your choices on this sheet.

[Do not read: Show the participant the sheet in which you record the choices.]

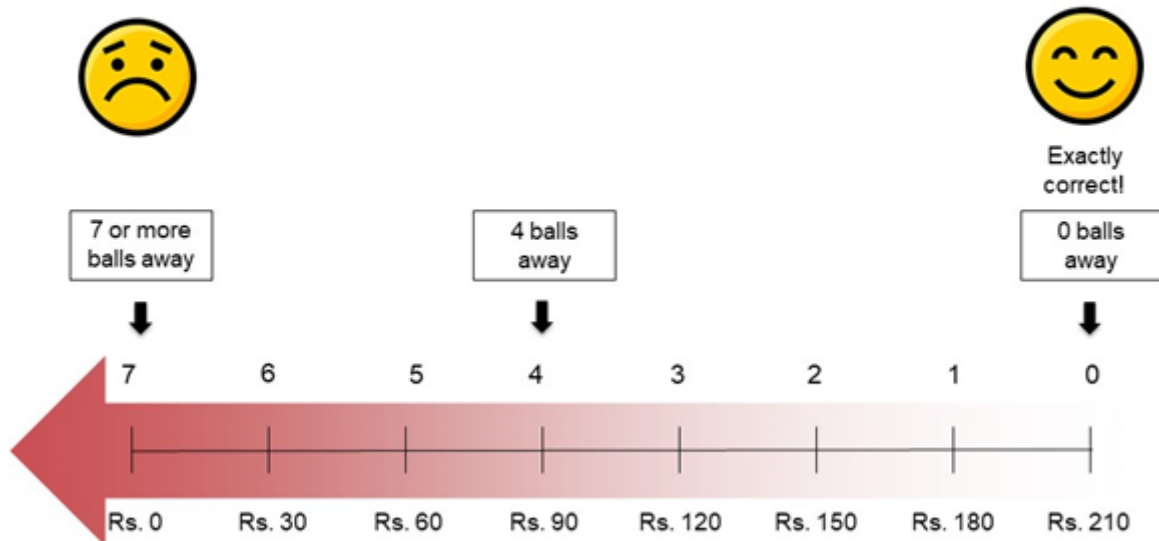
The computer has chosen only one of these guesses to be the one that counts for your payment. Since you do not know which of those choices will be selected when making your choices, you should take each choice very seriously.

Section A5. Payment Depending on Performance

[Do not read: Show them Figure 2 and Figure 3 to explain the payment scheme.]

Figure 2: Payment Scale

PAYMENT SCALE



Now, let me explain how the payment scheme works with this figure.

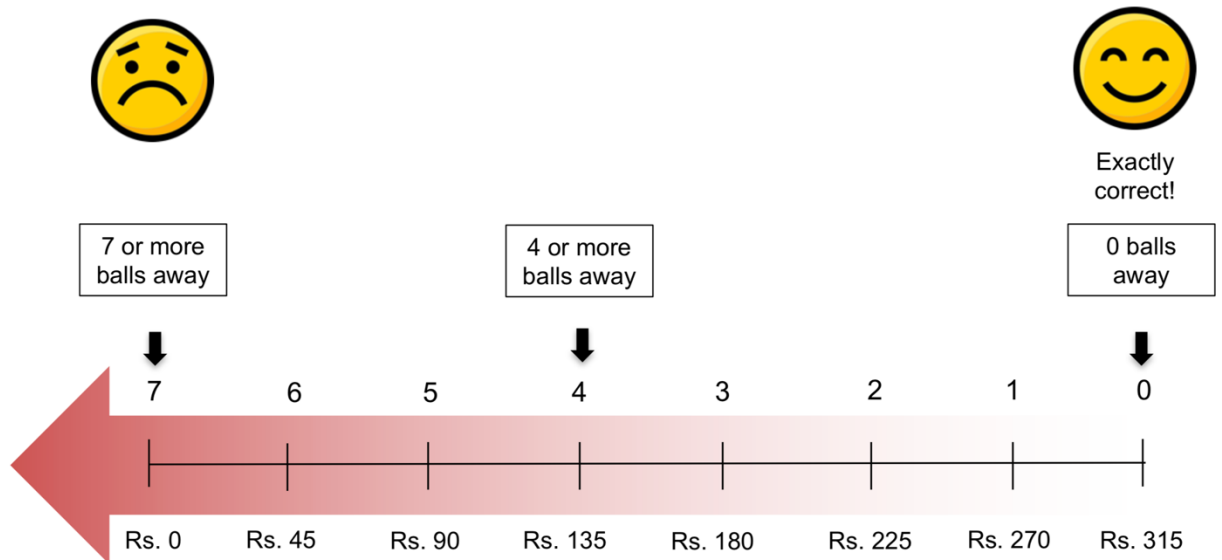
- Regardless of your choices, you will receive a flat payment of Rs. 150. Besides this flat payment, both you and the other participant have a chance to make some more money. How much more money you can earn depends on how close both your guesses are to the true number of balls in the urn. Each of you will be paid half of the total amount you earn separately.
- As we mentioned above, only one of your guesses will be chosen by the computer. The guess and the round chosen by the computer will be mentioned in this envelope *[Do not read: Show the participant the envelope containing the final guess for payment.]*
- If the chosen guess is exactly correct, then you and the other participant together could receive, besides the individual flat payment, an additional amount. This amount varies across 7 rounds. The computer has chosen the final guess and round, so you will be paid accordingly.
- In 4 out of the 7 rounds, you will receive at the most Rs. 210. So, your total payment could be at most Rs. 255. If your guess is very wrong, then you will make no extra money from the guess. You would only receive the individual flat payment of Rs. 150. If you are in the middle of these two extremes, you get an amount that is between Rs. 150 and Rs. 255.
- In 3 out of the 7 rounds, you will receive at the most Rs. 315. So, your total payment could be at most Rs. 308. Similarly if your guess is very wrong, then you will make no extra money from the

guess. You would only receive the individual flat payment of Rs. 150. If you are in the middle of these two extremes, you get an amount that is between Rs. 150 and Rs. 308.

- Let me give you two examples. Suppose the true number of red balls is 5.
- Suppose this is a round where you can earn at most Rs. 210.
[Do not read: Use Figure 2 to explain this entire section.]
 - If your guess is 5, your guess is exactly correct. So, you and the other participant would receive the maximum of Rs. 210 from the guesses (and Rs. 255 in total individually).
 - If your guess is 9, it is 4 balls away from the truth since 9 minus 5 is 4. You will lose Rs. 30 for each of the 4 balls. Therefore, you and the other participant would lose Rs. 120 and receive Rs. 90 from the guess (and Rs. 195 in total individually).
 - If your guess is 12, it is 7 balls away from the truth since 12 minus 5 is 7. You will lose Rs. 30 each for the 7 balls. Therefore, you and the other participant would lose Rs. 210 and receive nothing from the guesses.
 - If your guess is 16, it is 11 balls away from the truth since 16 minus 5 is 11. This guess is quite far away from the truth, so you would not receive any additional payments.
- Suppose this is a round where you can earn at most Rs. 315.
[Do not read: Use Figure 3 to explain this entire section.]

Figure 3: Payment Scale

PAYMENT SCALE



- If your guess is 5, your guess is exactly correct. So, you and the other participant would receive the maximum of Rs. 315 from the guesses (and Rs. 308 in total individually).
- If your guess is 9, it is 4 balls away from the truth since 9 minus 5 is 4. You will lose Rs. 45 for each of the 4 balls. Therefore, you and the other participant would lose Rs. 180 and receive Rs. 135 from the guess (and Rs. 218 in total individually).
- If your guess is 12, it is 7 balls away from the truth since 12 minus 5 is 7. You will lose Rs. 45 each for the 7 balls. Therefore, you and the other participant would lose Rs. 315 and receive nothing from the guesses.

Do you have any questions about how the payment is decided?

Now, we will ask you a few questions before we proceed with the activity.

[Do not read: Ask the participant to use Figure 2 for the following questions.]

Suppose the round that counts is the one where you can at most Rs. 210.

A24. Suppose that the true number is 6 red balls for the guess that counts. If you guess that there are 7 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 1 ball away from the true number. Rs. 180 → Yes, that's correct. [if $\$a_{24} = 180$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 180. [if $\$ \{a_{24}\} \neq 180$]

A25. Suppose that the true number is 15 red balls for the guess that counts. If you guess that there are 4 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 11 balls away from the true number. Rs. 0 → Yes, that's correct. [if $\$ \{a_{25}\} = 0$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 0. [if $\$ \{a_{25}\} \neq 0$]

A26. Suppose that the true number is 12 red balls for the guess that counts. If you guess that there are 8 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 4 balls away from the true number. Rs. 90 → Yes, that's correct. [if $\$ \{a_{26}\} = 90$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 2.]* No, you will be paid Rs. 90. [if $\$ \{a_{26}\} \neq 90$]

[Do not read: Ask the participant to use Figure 3 for the following questions.]

Suppose the round that counts is the one where you can at most Rs. 315.

A27. Suppose that the true number is 6 red balls for the guess that counts. If you guess that there are 7 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 1 ball away from the true number. Rs. 270 → Yes, that's correct. [if $\$ \{a_{27}\} = 270$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 3.]* No, you will be paid Rs. 270. [if $\$ \{a_{27}\} \neq 270$]

A28. Suppose that the true number is 15 red balls for the guess that counts. If you guess that there are 4 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 11 balls away from the true number. Rs. 0 → Yes, that's correct. [if $\$ \{a_{28}\} = 0$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 3.]* No, you will be paid Rs. 0. [if $\$ \{a_{28}\} \neq 0$]

A29. Suppose that the true number is 12 red balls for the guess that counts. If you guess that there are 8 red balls in the urn, how many balls are you away from the true number of red balls? How much would you and the other participant be paid from the guess?

[] a. 4 balls away from the true number. Rs. 135 → Yes, that's correct. [if $\$ \{a_{29}\} = 135$]

[] b. Any other answer → *[Do not read: Explain the payment mechanism again with the help of Figure 3.]* No, you will be paid Rs. 135. [if $\$ \{a_{29}\} \neq 135$]

Section A6: Overview of the Activity

Before we begin the activity, let me tell you about the structure of the activity:

- The number of draws for each of you will vary between 1, 5 and 9 over time. Each of you might get a different number of draws. We will inform you about this number in each round.
- Based on your draws, you will each make a few guesses — some on your own, some with the other participant.
- In all the rounds, you will participate with a female/male participant.

We are interested in learning how you decide. Thus, we will be present to observe how you make decisions. This procedure has nothing to do with what we think of the decisions you make. We follow it with all the participants.

Do you have any questions before we begin?

[Do not read: Clarify any doubts or questions they may have.]

Randomization of which surveyor will continue the survey: $\{p2a\}$

[Do not read: $\{p2a\}$ will continue with the rest of the survey. The other surveyor will assist.]

SECTION B: ROUNDS

ROUND 1: DISCUSSION

[Do not read: Introduce the two participants to each other.]

Here is the first urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is green in color — for each round, we will use a different colored urn.

[if $\{p17a\}$ = “low”]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p17a\}$ = “high”]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

The computer has decided that $\{p14\}$ starts first in this round. In this round, you will draw a few balls and guess the number of red balls in the urn; then $\{p15\}$ will draw a few balls and guess. Following this, we will ask you to make a joint guess.

[Do not read: Ask $\{p15\}$ to leave and talk to $\{p14\}$.]

A: Participant 1 Individual Guess

Before we start the activity, can you now please explain to me in your own words what we are asking you to do?

R1_a. *[Do not read: Did the participant mention the following points correctly?]*

☐ a. Draw balls

☐ b. Guess number of red balls

[Do not read: Repeat and correct points that are missing or incorrect.]

This urn contains 20 balls. You can now draw $\{p16\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back

into the urn. Please draw $\$ \{p16\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\$ \{p16\}$ times and note down the composition.]

Q1. Order of red/white balls drawn: ____

[SurveyCTO check: Limit string length to exactly $\$ \{p16\}$ and display as field list.]

Q2. *[Do not read]* How many red balls did the $\$ \{p14\}$ draw? ____

Q3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\$ \{p15\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask $\$ \{p14\}$ to wait and ask $\$ \{p15\}$ to join. Once the $\$ \{p15\}$ joins, ask the $\$ \{p14\}$ to leave.]

B. Participant 2 Individual Guess

Before we start the activity, can you now please explain to me in your own words what we are asking you to do?

R1_b. *[Do not read: Did the participant mention the following points correctly?]*

☐ a. Draw balls

☐ b. Guess number of red balls

[Do not read: Repeat and correct points that are missing or incorrect.]

This urn contains 20 balls and $\$ \{p14\}$ got a chance to draw some balls from it. Now it is your turn. You can draw $\$ \{p15\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\$ \{p15\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\$ \{p17\}$ times and note down the composition.]

Q4. Order of red/white balls drawn: ____

[SurveyCTO check: Limit string length to $\$ \{p17\}$ and display as field list.]

Q5. How many red balls did the $\$ \{p15\}$ draw? ____

Q6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\$ \{p15\}$ to wait and ask $\$ \{p14\}$ to join.]

C. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

[Do not read: Remain in the booth for the discussion.]

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Our staying here has nothing to do with whether you made ‘good’ or ‘bad’ decisions in previous rounds. This task is designed to be difficult. Most participants find it hard to come up with a good answer. Don't worry about saying anything wrong in front of us. Just try to do your best.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

Q7a. *[Do not read: Discussion start time: |_|_|:|_|_| 24HR]*

Q7b. *[Do not read: Discussion end time: |_|_|:|_|_| 24HR]*

Q7c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\{q7c\} = 1]$

Q7d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

Q7e. *[Do not read]* What information did participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

Q7f. *[Do not read]* What information did participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

Q7g. *[Do not read]* Who made the decision?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

Q7h. *[Do not read]* Notes on interaction: _____

Q8. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Q9. *[Do not read: Which participant moved the coin last on the scale?]*

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: Ask $\{p15\}$ to leave and talk to $\{p14\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

Q10. What is the $\{p14\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p14\}$ to leave and talk to $\{p15\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

Q11. What is the $\{p15\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p14\}$ to join.]

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 2: INFORMED OF PARTNER'S DRAWS

Here is the second urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is brown in color — for each round, we will use a different colored urn.

The computer has decided that $\{p18\}$ starts first in this round. In this round, you will draw a few balls and guess, then $\{p19\}$ will draw a few balls and guess. After you have made your guesses, we will then share the draws you saw with the other participant. Then you will make another guess. Following this, we will ask you to make a joint guess.

[if $\{p22\}$ = “low”]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p22\}$ = “high”]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

[Do not read: Ask the $\{p19\}$ to leave and talk to the $\{p18\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p20\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p20\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

A1. Which of the following information will we be sharing with $\{p19\}$?

☐ 1. Your guess

☐ 2. Draws

[Do not read: Ensure that the participant draws $\{p20\}$ times and note down the composition.]

R1. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\{p20\}$ and display as field list.]

R2.*[Do not read]* How many red balls did the $\{p18\}$ draw? ____

R3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p19\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p18\}$ to wait and ask the $\{p19\}$ to join. Once the $\{p19\}$ joins, ask the $\{p18\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p18\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p21\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p21\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

A2. Which of the following information will we be sharing with $\{p18\}$?

☐ 1. Your guess

☐ 2. Draws

[Do not read: Ensure that the participant draws $\{p21\}$ times.]

R4. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p21\}$ and display as field list.]

R5. *[Do not read]* How many red balls did $\{p19\}$ draw? ____

R6. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

C. Participant 2 Updated Individual Guess

We will now share $\{p18\}$'s draws with you. Using this information and your own draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p18\}$ drew a total of $\{p20\}$ balls from this urn, of which $\{r2\}$ were red balls.

R7a. Can you tell me how many balls you drew from this urn? ____

R7b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

R7. What is the $\{p19\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p19\}$ to wait and ask the $\{p18\}$ to join. Once the $\{p18\}$ joins, ask the $\{p19\}$ to leave.]

D. Participant 1 Updated Individual Guess

We will now share $\{p19\}$'s draws with you. Using this information and your own draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p19\}$ drew a total of $\{p21\}$ balls from this urn, of which $\{r5\}$ were red balls.

R8a. Can you tell me how many balls you drew from this urn? ____

R8b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

R8. What is the $\{p18\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

R9a. *[Do not read: Discussion start time: |__|__|:|__|__| 24HR]*

R9b. *[Do not read: Discussion end time: |__|__|:|__|__| 24HR]*

R9c. *[Do not read]* Did they share information?

[] 0. No

☐ 1. Yes

[if $\{R9c\} = 1]$

R9d. *[Do not read]* Who shared information?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

R9e. *[Do not read]* What information did participant 1 share?

- ☐ a. Number of draws
- ☐ b. Color composition
- ☐ c. Guess

R9f. *[Do not read]* What information did participant 2 share?

- ☐ a. Number of draws
- ☐ b. Color composition
- ☐ c. Guess

R9g. *[Do not read]* Who made the decision?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

R9h. *[Do not read]* Notes on interaction: _____

R10. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

R11. *[Do not read: Which participant moved the coin last on the scale?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

[Do not read: Ask $\{p19\}$ to leave and talk to $\{p18\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

R12. What is the $\{p18\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p18\}$ to leave and talk to $\{p19\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

R13. What is the $\{p19\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p18\}$ to join.]

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 3: DRAW-BY-DRAW

Here is the third urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is blue in color — for each round, we will use a different colored urn.

The computer has decided that $\{p_{23}\}$ starts first in this round. In this round, you will draw a few balls and guess, then $\{p_{24}\}$ will draw a few balls and guess. After you have made your guesses, we will then share the draws you saw with the other participant. Following this, we will ask you to make a joint guess.

[if $\{p_{27}\} = \text{“low”}$]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p_{27}\} = \text{“high”}$]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

[Do not read: Ask the $\{p_{24}\}$ to leave and talk to the $\{p_{23}\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p_{25}\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p_{25}\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

[Do not read: Ensure that the participant draws $\{p_{25}\}$ times and note down the composition.]

S1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p_{25}\}$ and display as a field list.]

S2. *[Do not read]* How many red balls did the $\{p_{23}\}$ draw? ____

S3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p24\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p23\}$ to wait and ask the $\{p24\}$ to join. Once the $\{p24\}$ joins, ask the $\{p23\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p23\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p26\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p26\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

[Do not read: Ensure that the participant draws $\{p26\}$ times and display as a field list.]

S4. *[Do not read]* Order of red/white balls drawn: _____

[Constraint: Limit the number of characters that can be entered to $\{p26\}$.]

S5. *[Do not read]* How many red balls did $\{p24\}$ draw? _____

S6. How many red balls do you think are in the urn? _____ *[Do not read: Have them move the coin to make their decision.]*

C. Participant 2 Updated Individual Guess

We will now share $\{p23\}$'s draws with you. Using this information and your own draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p23\}$ drew a total of $\{p25\}$ balls from this urn. Now I will tell you the order of red/white balls drawn.

Order of red/white balls drawn is $\{s3\}$. *[SurveyCTO: The draws will be displayed as a field list.]*

S7a. Can you tell me how many balls you drew from this urn? _____

S7b. Can you tell me how many red balls you drew from this urn? _____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

S7. What is the $\{p24\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p24\}$ to wait and ask the $\{p23\}$ to join. Once the $\{p23\}$ joins, ask the $\{p24\}$ to leave.]

D. Participant 1 Updated Individual Guess

We will now share $\{p24\}$'s draws with you. Using this information and your own draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

$\{p24\}$ drew a total of $\{p26\}$ balls from this urn.

Order of red/white balls drawn is $\{s5\}$. *[SurveyCTO: The draws will be displayed as a field list.]*

S8a. Can you tell me how many balls you drew from this urn? ____

S8b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

S8. What is the $\{p23\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

S9a. *[Do not read: Discussion start time: |__|__:|__|__| 24HR]*

S9b. *[Do not read: Discussion end time: |__|__:|__|__| 24HR]*

S9c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\{S9c\} = 1]$

S9d. *[Do not read]* Who shared information?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

S9e. *[Do not read]* What information did participant 1 share?

- ☐ a. Number of draws
- ☐ b. Color composition
- ☐ c. Guess

S9f. *[Do not read]* What information did participant 2 share?

- ☐ a. Number of draws
- ☐ b. Color composition
- ☐ c. Guess

S9g. *[Do not read]* Who made the decision?

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

S9h. *[Do not read]* Notes on interaction: _____

S10. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

S11. *[Do not read: Which participant moved the coin last on the scale?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

We will keep aside the urn now. We will show you the number of red balls in the urn, if this urn is chosen at the end.

[Do not read: Ask $\{p24\}$ to leave and talk to $\{p23\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

S12. What is the $\{p23\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p23\}$ to leave and talk to $\{p24\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

S13. What is the $\$p_{24}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\$p_{23}$ to join.]

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 4: NO FIRST GUESS

Here is the fourth urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is green in color — for each round, we will use a different colored urn.

The computer has decided that $\{p28\}$ starts first in this round. In this round, you will draw a few balls and you will not guess, then $\{p29\}$ will draw a few balls and we will then share the draws you saw with the other participant. Following this, we will ask you to make a joint guess.

[if $\{p32\}$ = “low”]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p32\}$ = “high”]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

[Do not read: Ask the $\{p29\}$ to leave and talk to the $\{p28\}$.]

A: Participant 1 Individual Draws

This urn contains 20 balls. Similar to the previous round, you can draw $\{p30\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p30\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant.

[Do not read: Ensure that the participant draws $\{p30\}$ times and note down the composition.]

T1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p30\}$ and display as a field list.]

T2. *[Do not read]* How many red balls did the $\{p28\}$ draw? ____

Thank you. It is now $\{p29\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p28\}$ to wait and ask the $\{p29\}$ to join. Once the $\{p29\}$ joins, ask the $\{p28\}$ to leave.]

B. Participant 2 Individual Draws

This urn contains 20 balls and $\{p28\}$ got a chance to draw some balls from it. Now it is your turn. Similar to the previous round, you can draw $\{p31\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p31\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant.

T3. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p31\}$ and display as a field list.]

T4. *[Do not read]* How many red balls did $\{p29\}$ draw? ____

C. Participant 2 Individual Guess

We will now share $\{p28\}$'s draws with you. Using this information and your own draws, we will ask you to make a guess.

$\{p28\}$ drew a total of $\{p30\}$ balls from this urn, of which $\{t2\}$ were red balls.

T5a. Can you tell me how many balls you drew from this urn? ____

T5b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

T5. What is the $\{p29\}$'s guess? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask the $\{p29\}$ to wait and ask the $\{p28\}$ to join. Once the $\{p28\}$ joins, ask the $\{p29\}$ to leave.]

D. Participant 1 Individual Guess

We will now share $\{p29\}$'s draws with you. Using this information and your own draws, we will ask you to make a guess.

$\{p29\}$ drew a total of $\{p31\}$ balls from this urn, of which $\{t4\}$ were red balls.

T6a. Can you tell me how many balls you drew from this urn? ____

T6b. Can you tell me how many red balls you drew from this urn? ____

How many red balls do you think there are in the urn? Please take your time to think about your choice.

T6. What is the $\$ \{p28\}$'s guess? ____ *[Do not read: Have them move the coin to make their decision.]*

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

T9a. *[Do not read: Discussion start time: |__|__|:|__|__| 24HR]*

T9b. *[Do not read: Discussion end time: |__|__|:|__|__| 24HR]*

T9c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\$ \{T9c\} = 1$]

T9d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

T9e. *[Do not read]* What information did the participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

T9f. *[Do not read]* What information did the participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

T9g. *[Do not read]* Who made the decision?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

T9h. *[Do not read]* Notes on interaction: _____

T10. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

T11. *[Do not read: Which participant moved the coin last on the scale?]*

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

[Do not read: Ask $\{p29\}$ to leave and talk to $\{p28\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

T12. What is the $\{p28\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p28\}$ to leave and talk to $\{p29\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

T13. What is the $\{p29\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p28\}$ to join.]

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 5: REVERSE ORDER

Here is the fifth urn. Both of you will each make draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is orange in color — for each round, we will use a different colored urn.

The computer has decided that $\{p33\}$ starts first in this round. In this round, you will draw a few balls and guess. After you have made your guess, we will share your draws with $\{p34\}$ and then they will draw a few balls and guess. Following this, we will ask you to make a joint guess.

[if $\{p37\}$ = “low”]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p37\}$ = “high”]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

[Do not read: Ask the $\{p34\}$ to leave and talk to the $\{p33\}$.]

A: Participant 1 Individual Guess

This urn contains 20 balls. Similar to the previous round, you can draw $\{p35\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p35\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, we will share your draws with the other participant. However, we will not share your guess.

[Do not read: Ensure that the participant draws $\{p35\}$ times and note down the composition.]

U1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p35\}$ and display as a field list.]

U2. *[Do not read]* How many red balls did the $\{p33\}$ draw? ____

U3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p34\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p33\}$ to wait and ask the $\{p34\}$ to join. Once the $\{p33\}$ joins, ask the $\{p34\}$ to leave.]

B. Participant 2 Individual Guess

This urn contains 20 balls and $\{p33\}$ got a chance to draw some balls from it. Now it is your turn.

We will now share $\{p33\}$'s draws with you. Using this information, we will ask you to make a guess.

$\{p33\}$ drew a total of $\{p35\}$ balls from this urn, of which $\{u2\}$ were red balls.

How many red balls do you think there are in the urn? Please take your time to think about your choice.

U4. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

C. Participant 2 Updated Individual Guess

Similar to the previous round, you can draw $\{p36\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p36\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

[Do not read: Ensure that the participant draws $\{p36\}$ times.]

U5. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p36\}$ and display as field list.]

U6. *[Do not read]* How many red balls did $\{p34\}$ draw? ____

U7. What is the $\{p34\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

We will show you the number of red balls in the urn, if this is chosen at the end. Now, we will proceed to the next round.

E. Joint Guess

Now, we would like you to decide together the number of red balls in the urn.

We are going to remain in the booth while you make your decision together. This is only because we would like to see your decision-making process.

Now both of you will make a decision together. Remember, this does not have to be the same as either of your previous guesses.

U8a. *[Do not read: Discussion start time: |__|__:|__|__| 24HR]*

U8b. *[Do not read: Discussion end time: |__|__:|__|__| 24HR]*

U8c. *[Do not read]* Did they share information?

☐ 0. No

☐ 1. Yes

[if $\{U8c\} = 1$]

U8d. *[Do not read]* Who shared information?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

U8e. *[Do not read]* What information did participant 1 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

U8f. *[Do not read]* What information did participant 2 share?

☐ a. Number of draws

☐ b. Color composition

☐ c. Guess

U8g. *[Do not read]* Who made the decision?

☐ a. Participant 1

☐ b. Participant 2

☐ c. Both

U8h. *[Do not read]* Notes on interaction: _____

U9. How many red balls do you think there are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

U10. *[Do not read: Which participant moved the coin last on the scale?]*

☐ a. Participant 1

☐ b. Participant 2

[] c. Both

[Do not read: Ask $\{p34\}$ to leave and talk to $\{p33\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

U11. What is the $\{p33\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p33\}$ to leave and talk to $\{p34\}$.]

Both of you discussed and made a joint guess. You each now have a chance to make another guess based on the information you have learnt. We will not share your guess with the other participant. This guess may be different from your previous guesses or it may be the same. Based on all the information you learnt, how many red balls do you think are in this urn?

U12. What is the $\{p34\}$'s updated guess? _____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p33\}$ to join.]

We will show you the number of red balls in this urn if this round is chosen at the end to be the round that counts for your payment. Now, we will proceed to the next round.

ROUND 6: OBSERVE PARTNER'S DRAWS

Here is the sixth urn. Both of you will each learn about draws from the same urn. Until the completion of this round, we will be using only this urn. As you can see, this urn is red in color — for each round, we will use a different colored urn.

In this round, you will both be present when you learn about draws from the urn. The computer has decided that $\{p38\}$ starts first in this round and $\{p39\}$ will be present in the booth while you learn about the draws from the urn. Both of you will make a guess. Following this, $\{p39\}$ will learn about the draws and $\{p38\}$ will be present, and then you will both make a guess.

[if $\{p42\}$ = “low”]

[Do not read: Use Figure 2 for this round.]

The computer has decided that you can receive up to Rs. 210 in this round. Remember that you will lose Rs. 30 for each ball that is further away from the true number.

[if $\{p42\}$ = “high”]

[Do not read: Use Figure 3 for this round.]

The computer has decided that you can receive up to Rs. 315 in this round. Remember that you will lose Rs. 45 for each ball that is further away from the true number.

[Do not read: Ask the $\{p38\}$ and $\{p39\}$ to stay in the booth and talk to the $\{p38\}$.]

A: Participant 1 Draws

- This urn contains 20 balls. Similar to the previous round, you can draw $\{p40\}$ balls from the urn to try and learn how many red balls and how many white balls are in the urn. Draw a ball from the urn, have a look at it, and put it back into the urn.

Please draw $\{p40\}$ balls in total. I will shuffle the balls in the urn before each of your draws. As discussed, $\{p39\}$ will be present while you draw balls from the urn.

[Do not read: Ensure that the participant draws $\{p40\}$ times and note down the composition.]

X1. *[Do not read]* Order of red/white balls drawn: ____

[Constraint: Limit the number of characters that can be entered to $\{p40\}$ and display as a field list.]

X2. *[Do not read]* How many red balls did the $\{p38\}$ draw? ____

B: First Individual Guess

[Do not read: Ask $\{p39\}$ to leave and talk to $\{p38\}$.]

X3. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p39\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask $\{p38\}$ to wait and ask the $\{p39\}$ to join. Once $\{p39\}$ joins, ask $\{p38\}$ to leave.]

X4. How many red balls do you think are in the urn? ____ *[Do not read: Have them move the coin to make their decision.]*

[Do not read: Ask $\{p38\}$ to join and talk to $\{p39\}$.]

C: Participant 2 Draws

This urn contains 20 balls and $\{p38\}$ got a chance to learn about the balls in the urn. Now it is your turn. You can draw $\{p41\}$ balls from the urn. Draw a ball from the urn, have a look at it, and put it back into the urn. Please draw $\{p41\}$ balls in total. I will shuffle the balls in the urn before each of your draws.

As discussed $\{p38\}$ will be present while you draw balls from the urn.

[Do not read: Ensure that the participant draws $\{p41\}$ times and note down the composition.]

X5. Order of red/white balls drawn: ____
[SurveyCTO check: Limit string length to $\{p41\}$ and display as a field list.]

D: Individual Updated Guess

[Do not read: Ask $\{p38\}$ to leave and talk to $\{p39\}$.]

Using your draws and $\{p38\}$'s draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

How many red balls do you think there are in the urn? Please take your time to think about your choice.

X6. What is $\{p39\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

Thank you for your guess. It is now $\{p38\}$'s turn. We will not change the contents of the urn and they will remain the same, while you are waiting.

[Do not read: Ask the $\{p39\}$ to wait and ask the $\{p38\}$ to join. Once the $\{p38\}$ joins, ask the $\{p39\}$ to leave.]

Using your draws and $\{p39\}$'s draws, we will ask you to make a new guess. This guess can be the same as your previous guess or it can be different from your previous guess. It is entirely your choice.

How many red balls do you think there are in the urn? Please take your time to think about your choice.

X6. What is $\{p38\}$'s updated guess? ____ *[Do not read: Have them move the coin to make their decision.]*

E. Participant 1 Memory Test

-

[Do not read: Talk to $\{p38\}$.]

X7a. Can you tell me how many balls you drew from this urn? _____

X7b. Can you tell me how many red balls you drew from this urn? _____

X7c. Can you tell me how many balls $\{p39\}$ drew from this urn? _____

X7d. Can you tell me how many red balls $\{p39\}$ drew from this urn? _____

Thank you! We will show you the number of red balls in the urn, if this is chosen at the end.

[Do not read: Ask the $\{p38\}$ to wait and ask the $\{p39\}$ to join. Once the $\{p39\}$ joins, ask the $\{p38\}$ to leave.]

F. Participant 2 Memory Test

[Do not read: Talk to $\{p39\}$.]

X8a. Can you tell me how many balls you drew from this urn? _____

X8b. Can you tell me how many red balls you drew from this urn? _____

X8c. Can you tell me how many balls $\{p38\}$ drew from this urn? _____

X8d. Can you tell me how many red balls $\{p38\}$ drew from this urn? _____

Thank you! We will show you the number of red balls in the urn, if this is chosen at the end. Now, we will proceed to the next round.

SECTION E: FINAL SECTION

As mentioned before, the envelope in front of you contains the decision made by the computer. Please feel free to open it and have a look at the decision.

F1. *[Do not read: Who opened the envelope?]*

- ☐ a. Participant 1
- ☐ b. Participant 2
- ☐ c. Both

F1a. *[Do not read: What guess does the sheet say?]*

- ☐ a. Participant 1's first individual guess
- ☐ b. Participant 2's first individual guess
- ☐ c. Participant 1's second individual guess
- ☐ d. Participant 2's second individual guess
- ☐ g. Joint guess

F1b. *[Do not read: Which urn does the sheet say is the one that determines their payment?]*

- ☐ a. Round 1
- ☐ b. Round 2
- ☐ c. Round 3
- ☐ d. Round 4
- ☐ e. Round 5
- ☐ f. Round 6
- ☐ g. Round 7

[Do not read: Show them the red balls in the urn.]

[if $\{p4\} = 1$]

As you can see, there are $\{p7\}$ red balls in the urn.

[if $\{p4\} = 2$]

As you can see, there are $\{p8\}$ red balls in the urn.

[if $\{p4\} = 3$]

As you can see, there are $\{p9\}$ red balls in the urn.

[if $\{p4\} = 4$]

As you can see, there are $\{p10\}$ red balls in the urn.

[if $\{p4\} = 5$]

As you can see, there are $\{p11\}$ red balls in the urn.

[if $\{p4\} = 6$]

As you can see, there are $\{p12\}$ red balls in the urn.

[if $\{p4\} = 7$]

As you can see, there are $\{p13\}$ red balls in the urn.

F2. *[Do not read]* Is the finalized guess a post-discussion guess?

☐ 0. No

☐ 1. Yes

[if $\{f2\} = 1$]

The finalized guess is a private guess, and as mentioned before we will not disclose the guess. However, we will explain the amount you will receive.

[Do not read: With the help of the payment scale, explain their payment to them.]

[if $\{f2\} = 0$]

[Do not read: Show them the grid with responses from each round and point to the one in the envelope. With the help of the payment scale, explain their payment to them.]

SECTION Z: CONCLUSION

Thank you for your time!

Z1. Re-enter PPID: _____

Z2. End Time: |__|__|:|__|__| 24HR

C.3 Experimental Script for Experiment 3

The online experiment with all treatments for Experiment 3 can be accessed at https://harvard.az1.qualtrics.com/jfe/form/SV_6s2csra9eqCC5pk

Below we reproduce the corresponding scripts.

Thanks for your time! In this survey, you will be playing a guessing game. The goal of the game is to guess the number of red marbles in a virtual jar. You will earn a larger bonus the closer your answer is to the truth (more details on bonuses later).

The jar contains 20 marbles, each of which is either **red** or **blue**. The jar has between 4 and 16 red marbles, and the rest are blue. The computer will randomly choose the exact number of red marbles, where every number between 4 and 16 was equally likely to be chosen.

Q11 To check you understood the previous instructions, we have a few comprehension questions for you.

First, what is the goal of the game you are playing today?

- ☐ To guess the number of red marbles in a virtual jar (1)
- ☐ To guess the number of any marbles in a virtual jar (2)
- ☐ To try to draw red marbles from a virtual jar (3)
- ☐ To solve some word problems (4)

Q14 How many marbles are in the jar total?

- ☐ 10 (1)
- ☐ 15 (2)
- ☐ 20 (3)
- ☐ 25 (4)

Q12 And how many **red** marbles could possibly be in the jar?

- ☐ Between 4 and 16 red marbles (1)
- ☐ Between 2 and 18 red marbles (2)
- ☐ Between 0 and 20 red marbles (3)
- ☐ Between 8 and 12 red marbles (4)

Display This Question:

If instructions_nonguessing_partner = 0

Q16 The goal of the game is to guess how many of the 20 marbles in the jar are **red**.

You will be playing this game with a partner (another participant on Prolific), who is taking this survey at about the same time as you. You and your partner are guessing about the **same jar which has the same number of red marbles in it**.

Display This Question:

If instructions_nonguessing_partner = 1

Q136 The goal of the game is to guess how many of the 20 marbles in the jar are **red**.

You will be playing this game with a partner (another participant on Prolific), who is taking this survey at about the same time as you. You are both using the **same jar which has the same number of red marbles in it**.

However, your partner will be working on a memory task, instead of guessing the number of red marbles.

Display This Question:

If instuctions_nonguessing_partner = 0

Q19

Who are you playing this game with?

- ☐ No one—I am playing the game by myself (1)
 - ☐ An imaginary partner (2)
 - ☐ A real partner who is taking the survey at around the same time as me (3)
-

Display This Question:

If instructions_nonguessing_partner = 1

Q181

Who are you playing this game with?

- ☐ No one—I am playing the game by myself (1)
- ☐ An imaginary partner (2)
- ☐ A real partner who is taking the survey at around the same time as me and guessing the number of red balls (3)
- ☐ A real partner who is taking the survey at around the same time as me but doing a different task than what I'm doing (4)
-

Display This Question:

If own_first = 0

Q87 To help you and your partner make a guess, you will each get to (virtually) draw marbles out of the jar one at a time.

After each draw, the marble is put back into the jar and the contents are shuffled before the next draw. This means it is possible that the same marble will be drawn multiple times.

You will first learn about your partner's draws and make a guess. Then you will learn about your own draws and make another guess.

We will randomly decide whether each of you will draw 1, 5, or 9 marbles from the jar, and you may get a different number of draws than your partner.

Display This Question:

If own_first = 1

Q148 To help you and your partner make a guess, you will each get to (virtually) draw marbles out of the jar one at a time.

After each draw, the marble is put back into the jar and the contents are shuffled before the next draw. This means it is possible that the same marble will be drawn multiple times.

You will first learn about your draws and make a guess. Then you will learn about your partner's draws and make another guess.

We will randomly decide whether each of you will draw 1, 5, or 9 marbles from the jar, and you may get a different number of draws than your partner.

Q89 Which of the following statements is correct?

- ☐ After each draw, the marble is placed back in the jar and the contents get shuffled (1)
- ☐ After each draw, the marble is not put back in the jar (2)

Q49

Which of these statements is correct?

- ☐ My partner and I are drawing marbles from the same jar with the same number of red marbles (1)
- ☐ My partner and I are drawing marbles from different jars with different numbers of red marbles (2)
- ☐ I am drawing marbles from the jar, and my partner is not (3)
- ☐ My partner is drawing marbles from the jar, and I am not (4)

Q52 You will play this game $\{\text{e://Field/num_games_word}\}$ times. Each time, the contents of the jar (that is, how many balls are red vs blue) will be re-randomized. In each game, it is equally likely that the number of red balls in the jar will be any number between 4 and 16.

Q54 Which of these statements is correct?

- ☐ I will only play this game once (1)
- ☐ I will play this game $\{\text{e://Field/num_games_numeral}\}$ times, with the contents of the jar always being the same (2)
- ☐ I will play this game $\{\text{e://Field/num_games_numeral}\}$ times, with the contents of the jar being re-randomized each time (3)

Q21 Great! Now we will explain how your bonus will be calculated.

At the end of the survey, we will randomly choose 1 guess from the $\{\text{e://Field/num_games_numeral}\}$ games you played. Each guess is equally likely to be chosen.

You will then be rewarded with a larger bonus payment the closer your guess in that game was to the truth. In particular, you will earn $\{\text{\$}\{\text{e://Field/max_bonus}\}\}$ if your guess is exactly correct. Your bonus will be $\{\text{\$}\{\text{e://Field/increment}\}\}$ less for each marble your guess is away from the truth (though it cannot

go below zero).

So, for example, if you guessed 10 red marbles, but actually there were 6 red marbles in the jar during that game, you would earn $\$e://Field/example_bonus$ as a bonus.

Q22 How can you affect the amount of your bonus payment?

- ☐ I can't—my answers do not affect my bonus (1)
- ☐ The longer I spend on the survey, the higher my bonus (2)
- ☐ For a randomly chosen guess, the closer I was to the true number of red marbles in the jar, the higher is my bonus (3)

Q24 Great! We are ready to start.

Please proceed to the next page to start Game 1.

End of Block: Instructions

Start of Block: Non-guessing subjects - Instructions

Q288 Thanks for your time! In this survey, you will be playing a memory game. The goal of the game is to recall the number of red marbles and the number of blue marbles that are drawn from a virtual jar. You will earn an additional bonus if your answer is correct (more details on bonuses later).

The jar contains 20 marbles total, each of which is either **red** or **blue**. The computer will randomly choose the exact number of red marbles.

Q290 To check you understood the previous instructions, we have a few comprehension questions for you.

First, what is the goal of the game you are playing today?

- ☐ To guess the number of red marbles in a virtual jar (1)
- ☐ To remember the number of red marbles and number of blue marbles drawn from a virtual jar (2)
- ☐ To try to draw red marbles from a virtual jar (3)
- ☐ To solve some word problems (4)

Q292 How many marbles are in the jar total?

- ☐ 10 (1)
- ☐ 15 (2)
- ☐ 20 (3)
- ☐ 25 (4)

Q297 The goal of the game is to remember how many of the drawn marbles are **red** and how many of the drawn marbles are **blue**.

You will be playing this game with a partner (another participant on Prolific), who is taking this survey at about the same time as you. You are both using the **same jar which has the same number of red marbles in it**. However, your partner will be doing a different task than you: they will be guessing the number of red marbles in the jar, whereas you will be remembering how many marbles of each color are drawn from it.

Q299

Who are you playing this game with?

- ☐ No one—I am playing the game by myself (1)
- ☐ An imaginary partner (2)
- ☐ A real partner who is taking the survey at around the same time as me (3)

Q302 You and your partner will each get to (virtually) draw marbles out of the jar one at a time.

After each draw, the marble is put back into the jar and the contents are shuffled before the next draw.

You will be asked to remember both your own draws from the jar and your partner's draws. We will randomly decide whether each of you will draw 1, 5, or 9 marbles from the jar, and you may get a different number of draws than your partner.

Q308 You will play this game $\{e://Field/num_games_word\}$ times. Each time, the contents of the jar (that is, how many balls are red vs blue) will be re-randomized.

Q310 Which of these statements is correct?

- ☐ I will only play this game once (1)
- ☐ I will play this game $\{e://Field/num_games_numeral\}$ times (3)

Q312 Great! Now we will explain how your bonus will be calculated.

At the end of the survey, the computer will randomly choose 1 of the $\{e://Field/num_games_numeral\}$ games you played. Each game is equally likely to be chosen.

To receive the bonus, you must have correctly recalled **all** of the information from that game:

1. The number of red marbles you drew
2. The number of blue marbles you drew
3. The number of red marbles your partner drew
4. The number of blue marbles your partner drew

You will then be rewarded with \$1 if your answer in that game was correct. Otherwise, you will not receive a bonus.

Q314 How can you affect the amount of your bonus payment?

- ☐ I can't—my answers do not affect my bonus (1)
- ☐ The longer I spend on the survey, the higher my bonus (2)
- ☐ For a randomly chosen answer, if I correctly recalled the number of each colored marble for each player, I get a bonus (3)

Q316 Great! We are ready to start.

Please proceed to the next page to start Game 1.

End of Block: Non-guessing subjects - Instructions

Start of Block: Draw-sharing others' info

Q203 **Game** $\{e://Field/game\}$

Your partner got to draw $\{e://Field/other_draws\}$ marble $\{e://Field/other_s\}$ out of $\{e://Field/jar_language\}$ jar. They drew $\{e://Field/total_other_red\}$ red marble $\{e://Field/other_red_s\}$ and $\{e://Field/other_draws - e://Field/total_other_red\}$ blue marble $\{e://Field/other_blue_s\}$.

End of Block: Draw-sharing others' info

Start of Block: watch_partner_draw

Q205 **Game** $\{e://Field/game\}$

On the next page, you can watch your **partner** draw $\{e://Field/other_draws\}$ marble $\{e://Field/other_s\}$ out of $\{e://Field/jar_language\}$ jar.

Q204 **Game** $\{e://Field/game\}$

Partner's marble

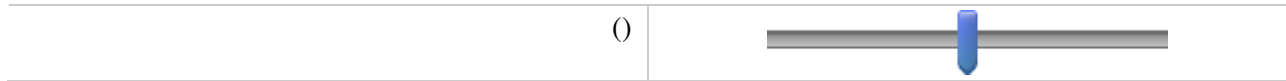
End of Block: watch_partner_draw

Start of Block: Game 1 First Guess

first_guess1 **Game** $\{e://Field/game\}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16



End of Block: Game 1 First Guess

Start of Block: Own draws

Q208 **Game** $\{e://Field/game\}$

On the next page, **you** will get to draw $\{e://Field/own_draws\}$ marble $\{e://Field/own_s\}$ out of $\{e://Field/jar_language\}$ jar.

Q197 **Game** $\{e://Field/game\}$

Draw **your** marble Put marble back

End of Block: Own draws

Start of Block: watch_own_draws

Q212 **Game** $\{e://Field/game\}$

On the next page, you can watch as **your** $\{e://Field/own_draws\}$ marble $\{e://Field/own_s\}$ $\{e://Field/own_to_be\}$ drawn out of $\{e://Field/jar_language\}$ jar.

Q214 **Game** $\{e://Field/game\}$

Your marble

End of Block: watch_own_draws

Start of Block: Individual round—draw other's draws

Q228 Game $\{e://Field/game\}$

On the next page, **you** will get to draw $\{e://Field/other_draws\}$ more marble $\{e://Field/other_s\}$ out of $\{e://Field/jar_language\}$ jar.

Q230 Game $\{e://Field/game\}$

Draw **your** marble Put marble back

End of Block: Individual round—draw other's draws

Start of Block: Game 1 Final Guess

final_guess1 Game $\{e://Field/game\}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16

0	
---	--

End of Block: Game 1 Final Guess

Start of Block: Non-guessing subject Game 1

Q318 How many red and blue marbles did you and your partner draw from this jar (i.e., during game $\{e://Field/game\}$)?

	Number (1)
Red marbles that I drew (1)	
Blue marbles that I drew (2)	
Red marbles that my partner drew (3)	
Blue marbles that my partner drew (4)	

End of Block: Non-guessing subject Game 1

Start of Block: Game Intro

Q201 Please proceed to the next page to start playing **Game** $\{e://Field/game\}$.

Remember, the computer has chosen a new jar for this game and has re-randomized the color of the marbles inside. The number of red marbles is equally likely to be any number between 4 and 16, with the rest of the marbles being blue.

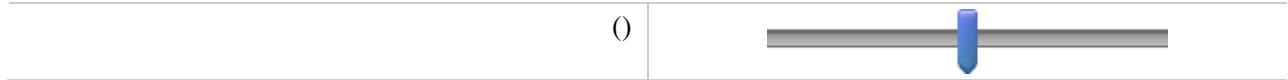
End of Block: Game Intro

Start of Block: Game 2 First Guess

first_guess2 **Game** [\\${e://Field/game}](#)

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16



End of Block: Game 2 First Guess

Start of Block: Game 2 Final Guess

final_guess2 **Game** [\\${e://Field/game}](#)

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16



End of Block: Game 2 Final Guess

Start of Block: Non-guessing subject Game 2

Q320 How many red and blue marbles did you and your partner draw from this jar (i.e., during game [\\${e://Field/game}](#))?

	Number (1)

Red marbles that I drew (1)	
Blue marbles that I drew (2)	
Red marbles that my partner drew (3)	
Blue marbles that my partner drew (4)	

End of Block: Non-guessing subject Game 2

Start of Block: Game 3 First Guess

first_guess3 Game $\{e://Field/game\}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16

	0	
--	---	--

End of Block: Game 3 First Guess

Start of Block: Game 3 Final Guess

final_guess3 Game $\{e://Field/game\}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16

	0	
--	---	--

End of Block: Game 3 Final Guess

Start of Block: Non-guessing subject Game 3

Q322 How many red and blue marbles did you and your partner draw from this jar (i.e., during game $\text{\texttt{\$e://Field/game}}$)?

	Number (1)
Red marbles that I drew (1)	
Blue marbles that I drew (2)	
Red marbles that my partner drew (3)	
Blue marbles that my partner drew (4)	

End of Block: Non-guessing subject Game 3

Start of Block: Game 4 First Guess

first_guess4 Game $\text{\texttt{\$e://Field/game}}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16

0	
---	--

End of Block: Game 4 First Guess

Start of Block: Game 4 Final Guess

final_guess4 **Game** $\{e://Field/game\}$

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16

	()	
--	-----	--

End of Block: Game 4 Final Guess

Start of Block: Non-guessing subject Game 4

Q324 How many red and blue marbles did you and your partner draw from this jar (i.e., during game $\{e://Field/game\}$)?

	Number (1)
Red marbles that I drew (1)	
Blue marbles that I drew (2)	
Red marbles that my partner drew (3)	
Blue marbles that my partner drew (4)	

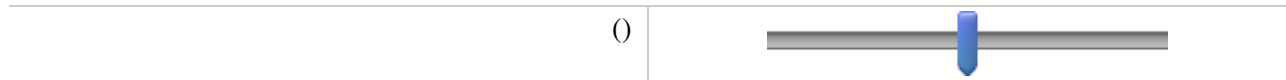
End of Block: Non-guessing subject Game 4

Start of Block: Game 5 First Guess

first_guess5 Game [\\${e://Field/game}](#)

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16



End of Block: Game 5 First Guess

Start of Block: Game 5 Final Guess

final_guess5 Game [\\${e://Field/game}](#)

How many red marbles do you think are in the jar? Remember there are 20 marbles total.

4 5 6 7 8 9 10 11 12 13 14 15 16



End of Block: Game 5 Final Guess

Start of Block: Non-guessing subject Game 5

Q326 How many red and blue marbles did you and your partner draw from this jar (i.e., during game [\\${e://Field/game}](#))?

	Number (1)

Red marbles that I drew (1)	
Blue marbles that I drew (2)	
Red marbles that my partner drew (3)	
Blue marbles that my partner drew (4)	

End of Block: Non-guessing subject Game 5

Start of Block: Memory Check

Q224 Think back to the final game you just played (Game 5). Do you remember how many red and blue marbles you and your partner drew from the jar?

		On a scale from 1 to 5, how sure are you about this?				
	Number (1)	1: Very uncertain (1)	2 (2)	3 (3)	4 (4)	5: Very certain (5)

Red marbles that I drew (1)		☐	☐	☐	☐	☐
Blue marbles that I drew (2)		☐	☐	☐	☐	☐
Red marbles that my partner drew (3)		☐	☐	☐	☐	☐
Blue marbles that my partner drew (4)		☐	☐	☐	☐	☐

End of Block: Memory Check

Start of Block: Final Questions

Q29 Thanks for taking our survey! We have a few final questions for you before we calculate your bonus.

Q217 Did you feel like you were competing with your partner, like you were cooperating with them, or neither?

- ☐ Competing with them (1)
- ☐ Cooperating with them (2)
- ☐ Neither competing not cooperating (3)

Display This Question:

If round_type != 5

Q219 Did you feel like you used the marbles you drew from the jar more than your partner's draws, to come up with your guesses about the contents of the jar?

- ☐ I used my own draws more (1)
- ☐ I treated my draws and my partner's draws the same (2)
- ☐ I used my partner's draws more (3)

Display This Question:

If round_type != 5

Q173 Tell us more about your answer to the previous question. Could you elaborate on how you thought about your partner's draws relative to your own?

Page Break

Q151 Did you feel as though you were playing with a partner, or more like you were playing by yourself?

Though there is an actual partner who got the draws you saw, we wonder whether it feels this way to participants.

- ☐ I completely felt I was playing by myself (1)
- ☐ I mostly felt I was playing by myself (4)
- ☐ I somewhat felt I was playing with a partner (5)
- ☐ I mostly felt I was playing with a partner (6)
- ☐ I completely felt I was playing with a partner (7)

Display This Question:

If Did you feel as though you were playing with a partner, or more like you were playing by yourself... = I completely felt I was playing by myself

Or Did you feel as though you were playing with a partner, or more like you were playing by yourself... = I mostly felt I was playing by myself

JS

Q153 Could you describe more in your own words why you did not feel like you were playing with a partner?

Q232 Were you writing down or recording the color of the marbles you or your partner drew?

- ☐ No, never (1)
- ☐ Yes, but only sometimes (2)
- ☐ Yes, mostly during the earlier games (4)
- ☐ Yes, mostly during the later games (5)
- ☐ Yes, every time (3)

Q149 Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?

- ☐ Most people can be trusted (1)
- ☐ You need to be very careful in dealing with people (2)

Q31 Did you have any technical difficulties with the survey? Or any feedback you'd like to leave?

Q33 How confusing did you find the instructions in the survey?

- ☐ Extremely confusing (1)
- ☐ Moderately confusing (2)
- ☐ Moderately clear (3)
- ☐ Extremely clear (4)

Display This Question:

If How confusing did you find the instructions in the survey? = Extremely confusing

Or How confusing did you find the instructions in the survey? = Moderately confusing

Q34 What did you find confusing? Your feedback is extremely helpful for us to improve the survey in the future.

Q38 What is your gender?

- ☐ Male (1)
- ☐ Female (2)
- ☐ Other (3)
- ☐ Prefer not to answer (4)

Q40 What is your highest level of education?

- ☐ Less than high school degree (1)
- ☐ High school degree or GED (2)
- ☐ Some college, but no degree (3)
- ☐ 2-year college degree (4)
- ☐ 4-year college degree (5)
- ☐ Any post-graduate degree (6)

Q217 How old are you?

Q42 What is your native (first) language?

- ☐ English (1)
- ☐ Other (Please specify) (3) _____

Q46 Great, that's all the questions! Please click the arrow below to see your payment and end the survey.

End of Block: Final Questions

Start of Block: Reveal Payoffs

Display This Question:

If $bonus \geq 0$

And $round_type \neq 5$

Q47

The computer randomly chose $\{e://Field/guess_to_score_language\}$ to calculate your bonus.

You guessed that there were $\{e://Field/answer\}$ red marbles in the jar that game. In fact, there were $\{e://Field/truth\}$ red marbles, meaning that your guess was off by $\{ \text{abs}(e://Field/answer - e://Field/truth) \}$ marbles.

Your bonus payment is therefore $\{\{e://Field/max_bonus\} - \{e://Field/increment\} * \{ \text{abs}(e://Field/answer - e://Field/truth) \} = \{e://Field/bonus\}$. You should receive this bonus within the next few days.

Display This Question:

If $bonus < 0$

And $round_type \neq 5$

Q196

The computer randomly chose $\{e://Field/guess_to_score_language\}$ to calculate your bonus.

Yyou guessed that there were $\{e://Field/answer\}$ red marbles in the jar that game. In fact, there were $\{e://Field/truth\}$ red marbles, meaning that your guess was off by $\{ \text{abs}(e://Field/answer - e://Field/truth) \}$ marbles.

Your bonus payment would therefore be $\{\{e://Field/max_bonus\} - \{e://Field/increment\} * \{ \text{abs}(e://Field/answer - e://Field/truth) \} = \{e://Field/bonus\}$. Since this is negative, you will just not receive a bonus.

End of Block: Reveal Payoffs

Start of Block: Non-guessing subject - Reveal Payoffs

Display This Question:

If $bonus > 0$

And $round_type = 5$

Q327

The computer randomly chose $\{e://Field/guess_to_score_language\}$ to calculate your bonus.

You answered that:

You drew $\{e://Field/answer_red_own\}$ red marbles

You drew $\{e://Field/answer_blue_own\}$ blue marbles

Your partner drew $\{e://Field/answer_red_partner\}$ red marbles

Your partner drew $\{e://Field/answer_blue_partner\}$ blue marbles

This was **correct!**

Your bonus payment is therefore \$1. You should receive this bonus within the next few days.

Display This Question:

If $bonus \leq 0$

And $round_type = 5$

Q328 The computer randomly chose $\{e://Field/guess_to_score_language\}$ to calculate your bonus.

You answered that there were:

$\{e://Field/answer_red_own\}$ red marbles drawn from your jar

$\{e://Field/answer_blue_own\}$ blue marbles drawn from your jar

$\{e://Field/answer_red_partner\}$ red marbles drawn from your partner's jar

$\{e://Field/answer_blue_partner\}$ blue marbles drawn from your partner's jar

The actual answers were:

$\{e://Field/total_own_red\}$ red marbles drawn from your jar

$\{e://Field/own_draws - e://Field/total_own_red\}$ blue marbles drawn from your jar

$\{e://Field/total_other_red\}$ red marbles drawn from your partner's jar

$\{e://Field/other_draws - e://Field/total_other_red\}$ blue marbles drawn from your partner's jar

Since this was not correct, you will not receive a bonus.

End of Block: Non-guessing subject - Reveal Payoffs